Forecasting U.S. Electricity Prices

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Introduction

U.S. electricity prices have been on a sharp upward trend in recent years and this paper sets out to understand why. Using monthly BLS price data spanning November 1978 through early 2026, we apply four time series frameworks: STL decomposition, ETS(A,A,A), ETS(M,A,M), and a suite of ARIMA/ARIMAX models evaluated across forecast horizons of 1, 3, 6, and 12 months.

We find that this recent surge in electricity prices can be explained by 3 catalysts. First, the 2022 Russia-Ukraine war created an energy price shock that passed directly through to electricity prices. Second, explosive data center power demand on a grid that was never designed to handle this capacity. Third, the Iran war has disrupted roughly a quarter of globally traded oil, sending natural gas prices higher and therefore electricity prices with them.

Project I

```
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  message = FALSE
)
# clean up history
rm(list=ls())
# load in packages
library(forecast)
library(ggplot2)    # data visualization
library(kableExtra) # manipulate table styles
library(urca)       # unit root tests
library(tinytex)
library(zoo)
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

Import Data

Read data from a comma-separated values (CSV) file into a data frame. Also checking for NA values.

```
## [1] TRUE
```

View data

observation_date	price
1978-11-01	0.046
1978-12-01	0.046
1979-01-01	0.046
1979-02-01	0.047
1979-03-01	0.047
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1979-06-01	0.052
1979-07-01	0.053
1979-08-01	0.053
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1979-10-01	0.053
1979-11-01	0.051
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1980-04-01	0.056
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1980-07-01	0.064
1980-08-01	0.064
1980-09-01	0.065
1980-10-01	0.063
1980-11-01	0.061
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1981-02-01	0.064
1981-03-01	0.065
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1981-07-01	0.073
1981-08-01	0.075
1981-09-01	0.074
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1981-11-01	0.071
1981-12-01	0.071
1982-01-01	0.073
1982-02-01	0.073
1982-03-01	0.075

1982-04-01	0.075
1982-05-01	0.075
1982-06-01	0.079
1982-07-01	0.079
1982-08-01	0.079
1982-09-01	0.079
1982-10-01	0.077
1982-11-01	0.074
1982-12-01	0.075
1983-01-01	0.075
1983-02-01	0.075
1983-03-01	0.076
1983-04-01	0.075
1983-05-01	0.077
1983-06-01	0.081
1983-07-01	0.082
1983-08-01	0.082
1983-09-01	0.082
1983-10-01	0.080
1983-11-01	0.077
1983-12-01	0.077
1984-01-01	0.078
1984-02-01	0.079
1984-03-01	0.079
1984-04-01	0.080
1984-05-01	0.081
1984-06-01	0.086
1984-07-01	0.087
1984-08-01	0.089
1984-09-01	0.084
1984-10-01	0.081
1984-11-01	0.079
1984-12-01	0.078
1985-01-01	0.079
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1985-03-01	0.079
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1985-05-01	0.080
1985-06-01	0.085
1985-07-01	0.085
1985-08-01	0.085
1985-09-01	NA
1985-10-01	0.082
1985-11-01	0.080

1985-12-01	0.080
1986-01-01	0.081
1986-02-01	0.075
1986-03-01	0.075
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1987-05-01	0.076
1987-06-01	0.084
1987-07-01	0.084
1987-08-01	0.085
1987-09-01	0.084
1987-10-01	0.078
1987-11-01	0.077
1987-12-01	0.077
1988-01-01	0.078
1988-02-01	0.078
1988-03-01	0.078
1988-04-01	0.077
1988-05-01	0.077
1988-06-01	0.084
1988-07-01	0.084
1988-08-01	0.084
1988-09-01	0.084
1988-10-01	0.080
1988-11-01	0.078
1988-12-01	0.078
1989-01-01	0.079
1989-02-01	0.079
1989-03-01	0.079
1989-04-01	0.079
1989-05-01	0.080
1989-06-01	0.086
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1989-08-01	0.086
1989-09-01	0.086
1989-10-01	0.081
1989-11-01	0.080
1989-12-01	0.081
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1990-02-01	0.081
1990-03-01	0.081
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1990-05-01	0.082
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1991-04-01	0.084
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1991-07-01	0.091
1991-08-01	0.090
1991-09-01	0.091
1991-10-01	0.087
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1992-02-01	0.084
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1993-05-01	0.090
1993-06-01	0.094
1993-07-01	0.095
1993-08-01	0.097
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1993-11-01	0.091
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2003-02-01	0.089
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2004-03-01	0.091

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2010-01-01	0.124
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2011-07-01	0.135

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2012-02-01	0.128
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2012-05-01	0.129
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2020-11-01	0.136
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2021-05-01	0.140
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2021-07-01	0.143
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2021-12-01	0.142
2022-01-01	0.147
2022-02-01	0.148
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2022-05-01	0.154
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2025-01-01	0.179
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2025-04-01	0.181
2025-05-01	0.182
2025-06-01	0.190
2025-07-01	0.189
2025-08-01	0.190
2025-09-01	0.188
2025-10-01	NA
2025-11-01	0.189
2025-12-01	0.189
2026-01-01	0.192

Time Series

Preliminary Analysis

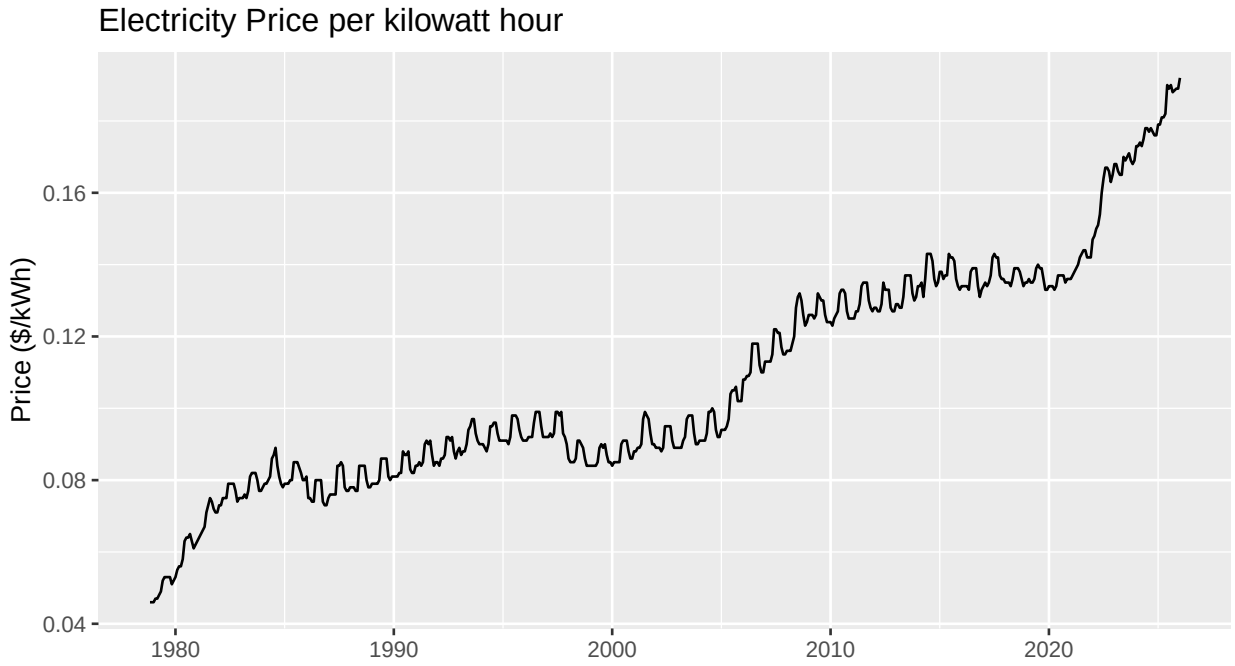
Visualizations

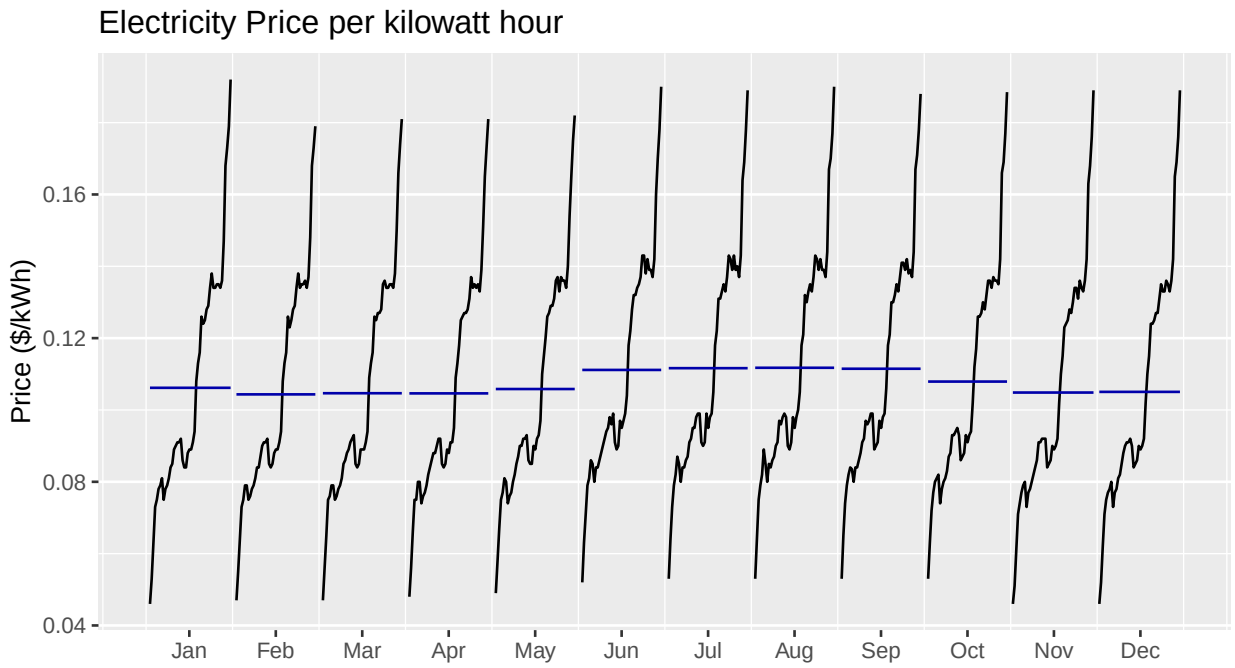
The data is non-stationary.

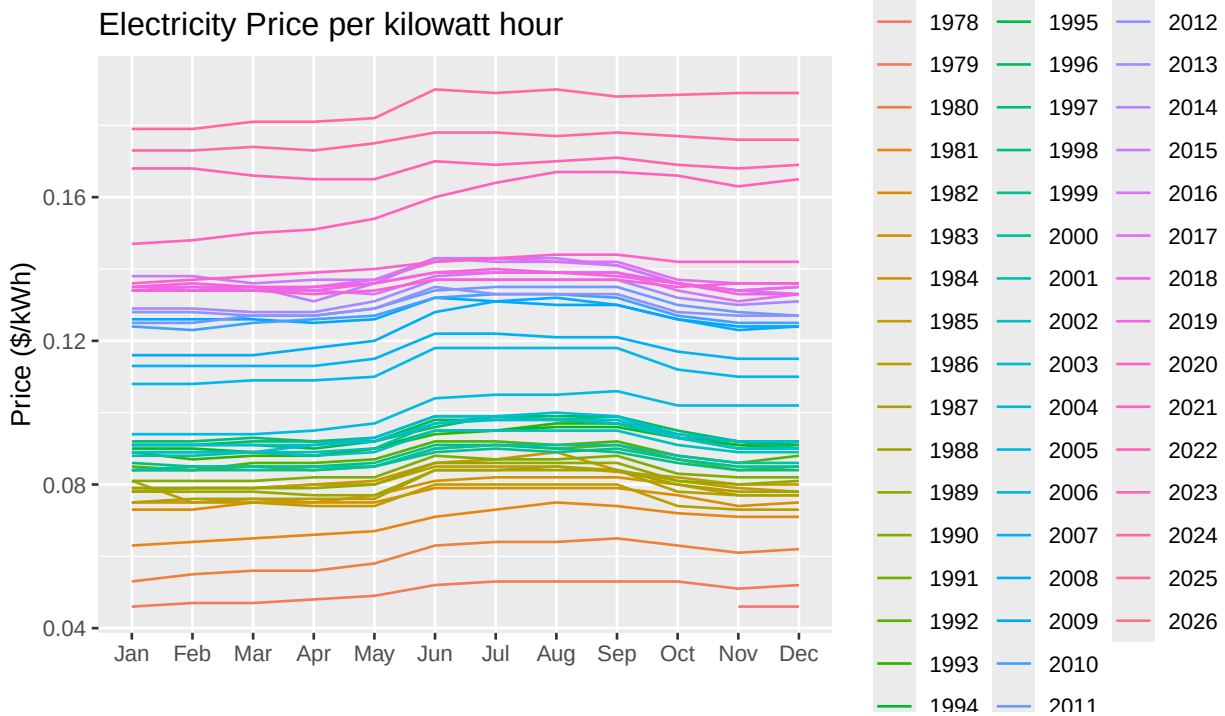
- There is a clear upward trend from 1978 to 2026.
- The increase appears mostly gradual and linear, with a sharper rise after 2020.
- The trend likely reflects rising production costs, infrastructure investment, and inflation over time.

There is a seasonal pattern.

- Monthly prices show regular fluctuations each year.
- Seasonal movements are relatively stable and do not increase significantly with the level of the series.
- There are no major structural breaks until the noticeable acceleration after 2020.
- Recent years show stronger price increases compared to earlier decades.







Summary Statistics

- Summary statistics don't have much meaning since data follows a linear trend.

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0460  0.0840  0.0950  0.1074  0.1340  0.1920
```

```
##     1     2     3     4     5     6     7     8     9    10    11    12
## 0.106 0.104 0.105 0.105 0.106 0.111 0.112 0.112 0.111 0.108 0.105 0.105
```

```
##     1     2     3     4     5     6     7     8     9    10    11    12
## 0.033 0.031 0.031 0.031 0.031 0.031 0.031 0.031 0.031 0.031 0.032 0.033
```

Data Transformation

- Because the trend is mostly linear rather than exponential, a log transformation is not necessary.
- Seasonal fluctuations do not grow proportionally with the level of the data.

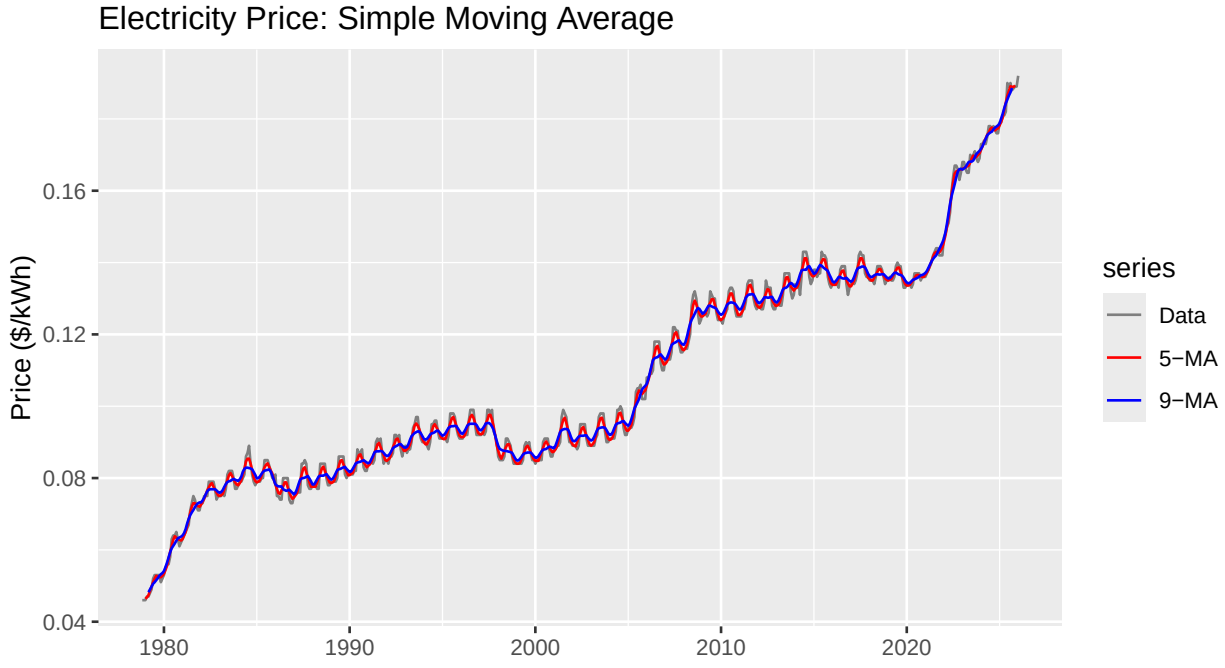
Therefore, we proceed using the original data with interpolated values when required for modeling.

Time Series Decomposition

Moving Averages

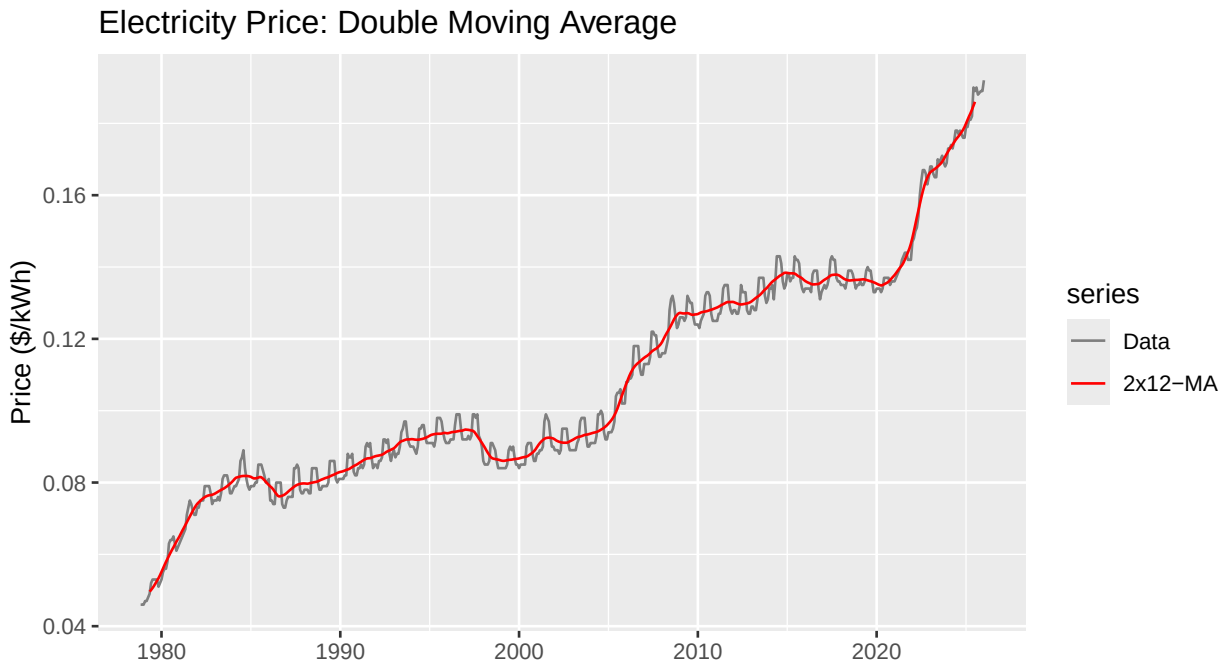
Simple Moving Average

We can highlight that in both the 5 month and the 9 month, seasonality is still prevalent, the 9 month is much smoother compared to the 5 month and raw data, which is to be expected.which means that it's not a good predictor for the trend cycle.

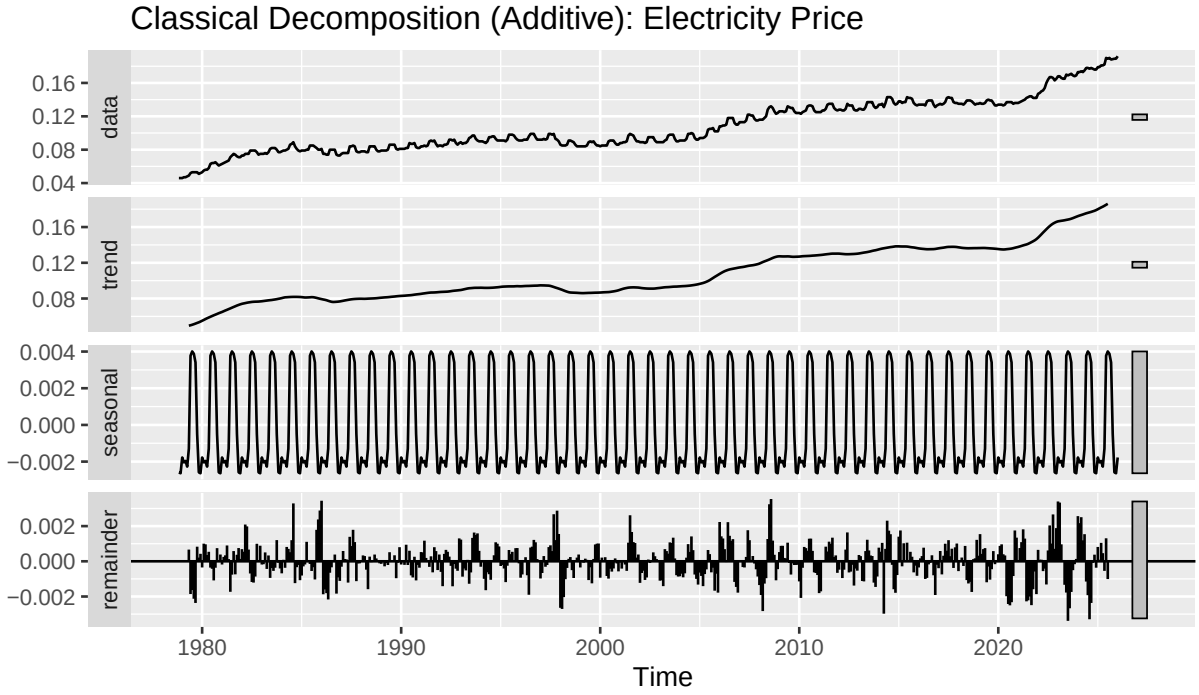


Double Moving Average

We can see that the double moving average removed the seasonality, and did a much better job of predicting the cycle trend component compared to the single moving average. Its also smoother than the single moving average as well. We can conclude that it's a good predictor of the trend cycle component.



Classical Decomposition



year	month	trend	seasonal	random
1978	11	NA	-0.0025859198	NA
1978	12	NA	-0.0026357387	NA
1979	1	NA	-0.0017752314	NA
1979	2	NA	-0.0020442532	NA
1979	3	NA	-0.0019645430	NA
1979	4	NA	-0.0022747785	NA
1979	5	0.04962500	-0.0012904758	6.654758e-04
1979	6	0.05008333	0.0037751270	-1.858460e-03
1979	7	0.05062500	0.0040082831	-1.633283e-03
1979	8	0.05125000	0.0038769425	-2.126942e-03
1979	9	0.05195833	0.0034077396	-2.366073e-03
1979	10	0.05266667	-0.0004971517	8.304851e-04
1979	11	0.05337500	-0.0025859198	2.109198e-04
1979	12	0.05420833	-0.0026357387	4.274053e-04
1980	1	0.05512500	-0.0017752314	-3.497686e-04
1980	2	0.05604167	-0.0020442532	1.002587e-03
1980	3	0.05700000	-0.0019645430	9.645430e-04
1980	4	0.05791667	-0.0022747785	3.581119e-04
1980	5	0.05875000	-0.0012904758	5.404758e-04
1980	6	0.05958333	0.0037751270	-3.584604e-04
1980	7	0.06041667	0.0040082831	-4.249497e-04
1980	8	0.06120833	0.0038769425	-1.085276e-03

1980	9	0.06195833	0.0034077396	-3.660729e-04
1980	10	0.06275000	-0.0004971517	7.471517e-04
1980	11	0.06354167	-0.0025859198	4.425317e-05
1980	12	0.06425000	-0.0026357387	3.857387e-04
1981	1	0.06495833	-0.0017752314	-1.831019e-04
1981	2	0.06579167	-0.0020442532	2.525865e-04
1981	3	0.06662500	-0.0019645430	3.395430e-04
1981	4	0.06737500	-0.0022747785	8.997785e-04
1981	5	0.06816667	-0.0012904758	1.238091e-04
1981	6	0.06895833	0.0037751270	-1.733460e-03
1981	7	0.06975000	0.0040082831	-7.582831e-04
1981	8	0.07054167	0.0038769425	5.813908e-04
1981	9	0.07133333	0.0034077396	-7.410729e-04
1981	10	0.07212500	-0.0004971517	3.721517e-04
1981	11	0.07283333	-0.0025859198	7.525865e-04
1981	12	0.07350000	-0.0026357387	1.357387e-04
1982	1	0.07408333	-0.0017752314	6.918981e-04
1982	2	0.07450000	-0.0020442532	5.442532e-04
1982	3	0.07487500	-0.0019645430	2.089543e-03
1982	4	0.07529167	-0.0022747785	1.983112e-03
1982	5	0.07562500	-0.0012904758	6.654758e-04
1982	6	0.07591667	0.0037751270	-6.917937e-04
1982	7	0.07616667	0.0040082831	-1.174950e-03
1982	8	0.07633333	0.0038769425	-1.210276e-03
1982	9	0.07645833	0.0034077396	-8.660729e-04
1982	10	0.07650000	-0.0004971517	9.971517e-04
1982	11	0.07658333	-0.0025859198	2.586500e-06
1982	12	0.07675000	-0.0026357387	8.857387e-04
1983	1	0.07695833	-0.0017752314	-1.831019e-04
1983	2	0.07720833	-0.0020442532	-1.640802e-04
1983	3	0.07745833	-0.0019645430	5.062097e-04
1983	4	0.07770833	-0.0022747785	-4.335548e-04
1983	5	0.07795833	-0.0012904758	3.321425e-04
1983	6	0.07816667	0.0037751270	-9.417937e-04
1983	7	0.07837500	0.0040082831	-3.832831e-04
1983	8	0.07866667	0.0038769425	-5.436092e-04
1983	9	0.07895833	0.0034077396	-3.660729e-04
1983	10	0.07929167	-0.0004971517	1.205485e-03
1983	11	0.07966667	-0.0025859198	-8.074683e-05
1983	12	0.08004167	-0.0026357387	-4.059280e-04
1984	1	0.08045833	-0.0017752314	-6.831019e-04
1984	2	0.08095833	-0.0020442532	8.591983e-05
1984	3	0.08133333	-0.0019645430	-3.687903e-04
1984	4	0.08145833	-0.0022747785	8.164452e-04

1984	5	0.08158333	-0.0012904758	7.071425e-04
1984	6	0.08170833	0.0037751270	5.165396e-04
1984	7	0.08179167	0.0040082831	1.200050e-03
1984	8	0.08183333	0.0038769425	3.289724e-03
1984	9	0.08183333	0.0034077396	-1.241073e-03
1984	10	0.08183333	-0.0004971517	-3.361816e-04
1984	11	0.08179167	-0.0025859198	-2.057468e-04
1984	12	0.08170833	-0.0026357387	-1.072595e-03
1985	1	0.08158333	-0.0017752314	-8.081019e-04
1985	2	0.08133333	-0.0020442532	-2.890802e-04
1985	3	0.08114583	-0.0019645430	-1.812903e-04
1985	4	0.08116667	-0.0022747785	1.108112e-03
1985	5	0.08125000	-0.0012904758	4.047580e-05
1985	6	0.08137500	0.0037751270	-1.501270e-04
1985	7	0.08154167	0.0040082831	-5.499497e-04
1985	8	0.08145833	0.0038769425	-3.352758e-04
1985	9	0.08112500	0.0034077396	-1.032740e-03
1985	10	0.08070833	-0.0004971517	1.788818e-03
1985	11	0.08020833	-0.0025859198	2.377587e-03
1985	12	0.07975000	-0.0026357387	2.885739e-03
1986	1	0.07933333	-0.0017752314	3.441898e-03
1986	2	0.07891667	-0.0020442532	-1.872413e-03
1986	3	0.07856250	-0.0019645430	-1.597957e-03
1986	4	0.07808333	-0.0022747785	-1.808555e-03
1986	5	0.07745833	-0.0012904758	-2.167858e-03
1986	6	0.07687500	0.0037751270	-6.501270e-04
1986	7	0.07633333	0.0040082831	-3.416164e-04
1986	8	0.07612500	0.0038769425	-1.942485e-06
1986	9	0.07620833	0.0034077396	3.839271e-04
1986	10	0.07633333	-0.0004971517	-1.836182e-03
1986	11	0.07650000	-0.0025859198	-9.140802e-04
1986	12	0.07675000	-0.0026357387	-1.114261e-03
1987	1	0.07708333	-0.0017752314	-3.081019e-04
1987	2	0.07745833	-0.0020442532	5.859198e-04
1987	3	0.07783333	-0.0019645430	1.312097e-04
1987	4	0.07816667	-0.0022747785	1.081119e-04
1987	5	0.07850000	-0.0012904758	-1.209524e-03
1987	6	0.07883333	0.0037751270	1.391540e-03
1987	7	0.07912500	0.0040082831	8.667169e-04
1987	8	0.07933333	0.0038769425	1.789724e-03
1987	9	0.07950000	0.0034077396	1.092260e-03
1987	10	0.07962500	-0.0004971517	-1.127848e-03
1987	11	0.07970833	-0.0025859198	-1.224135e-04
1987	12	0.07975000	-0.0026357387	-1.142613e-04

1988	1	0.07975000	-0.0017752314	2.523143e-05
1988	2	0.07970833	-0.0020442532	3.359198e-04
1988	3	0.07966667	-0.0019645430	2.978764e-04
1988	4	0.07975000	-0.0022747785	-4.752215e-04
1988	5	0.07987500	-0.0012904758	-1.584524e-03
1988	6	0.07995833	0.0037751270	2.665396e-04
1988	7	0.08004167	0.0040082831	-4.994973e-05
1988	8	0.08012500	0.0038769425	-1.942485e-06
1988	9	0.08020833	0.0034077396	3.839271e-04
1988	10	0.08033333	-0.0004971517	1.638184e-04
1988	11	0.08054167	-0.0025859198	4.425317e-05
1988	12	0.08075000	-0.0026357387	-1.142613e-04
1989	1	0.08091667	-0.0017752314	-1.414352e-04
1989	2	0.08108333	-0.0020442532	-3.908017e-05
1989	3	0.08125000	-0.0019645430	-2.854570e-04
1989	4	0.08137500	-0.0022747785	-1.002215e-04
1989	5	0.08150000	-0.0012904758	-2.095242e-04
1989	6	0.08170833	0.0037751270	5.165396e-04
1989	7	0.08191667	0.0040082831	7.505027e-05
1989	8	0.08208333	0.0038769425	3.972418e-05
1989	9	0.08225000	0.0034077396	3.422604e-04
1989	10	0.08245833	-0.0004971517	-9.611816e-04
1989	11	0.08266667	-0.0025859198	-8.074683e-05
1989	12	0.08283333	-0.0026357387	8.024053e-04
1990	1	0.08295833	-0.0017752314	-1.831019e-04
1990	2	0.08304167	-0.0020442532	2.586500e-06
1990	3	0.08316667	-0.0019645430	-2.021236e-04
1990	4	0.08333333	-0.0022747785	9.414452e-04
1990	5	0.08350000	-0.0012904758	-2.095242e-04
1990	6	0.08362500	0.0037751270	5.998730e-04
1990	7	0.08379167	0.0040082831	-7.999497e-04
1990	8	0.08404167	0.0038769425	-9.186092e-04
1990	9	0.08433333	0.0034077396	2.589271e-04
1990	10	0.08458333	-0.0004971517	-1.086182e-03
1990	11	0.08479167	-0.0025859198	-2.057468e-04
1990	12	0.08500000	-0.0026357387	-3.642613e-04
1991	1	0.08525000	-0.0017752314	5.252314e-04
1991	2	0.08554167	-0.0020442532	5.025865e-04
1991	3	0.08579167	-0.0019645430	1.172876e-03
1991	4	0.08608333	-0.0022747785	1.914452e-04
1991	5	0.08633333	-0.0012904758	-4.285753e-05
1991	6	0.08654167	0.0037751270	-3.167937e-04
1991	7	0.08670833	0.0040082831	2.833836e-04
1991	8	0.08675000	0.0038769425	-6.269425e-04

1991	9	0.08679167	0.0034077396	8.005937e-04
1991	10	0.08691667	-0.0004971517	5.804851e-04
1991	11	0.08708333	-0.0025859198	-4.974135e-04
1991	12	0.08725000	-0.0026357387	3.857387e-04
1992	1	0.08737500	-0.0017752314	-5.997686e-04
1992	2	0.08745833	-0.0020442532	-1.414080e-03
1992	3	0.08754167	-0.0019645430	4.228764e-04
1992	4	0.08762500	-0.0022747785	6.497785e-04
1992	5	0.08775000	-0.0012904758	5.404758e-04
1992	6	0.08795833	0.0037751270	2.665396e-04
1992	7	0.08825000	0.0040082831	-2.582831e-04
1992	8	0.08854167	0.0038769425	-1.418609e-03
1992	9	0.08875000	0.0034077396	-1.577396e-04
1992	10	0.08891667	-0.0004971517	-4.195149e-04
1992	11	0.08912500	-0.0025859198	-5.390802e-04
1992	12	0.08933333	-0.0026357387	1.302405e-03
1993	1	0.08954167	-0.0017752314	1.233565e-03
1993	2	0.08991667	-0.0020442532	-8.724135e-04
1993	3	0.09037500	-0.0019645430	-4.104570e-04
1993	4	0.09079167	-0.0022747785	-5.168881e-04
1993	5	0.09120833	-0.0012904758	8.214247e-05
1993	6	0.09150000	0.0037751270	-1.275127e-03
1993	7	0.09162500	0.0040082831	-6.332831e-04
1993	8	0.09179167	0.0038769425	1.331391e-03
1993	9	0.09195833	0.0034077396	1.633927e-03
1993	10	0.09200000	-0.0004971517	1.497152e-03
1993	11	0.09200000	-0.0025859198	1.585920e-03
1993	12	0.09204167	-0.0026357387	5.940720e-04
1994	1	0.09208333	-0.0017752314	-3.081019e-04
1994	2	0.09204167	-0.0020442532	2.586500e-06
1994	3	0.09195833	-0.0019645430	-9.937903e-04
1994	4	0.09191667	-0.0022747785	-1.641888e-03
1994	5	0.09191667	-0.0012904758	-6.261909e-04
1994	6	0.09195833	0.0037751270	-7.334604e-04
1994	7	0.09204167	0.0040082831	-1.049950e-03
1994	8	0.09212500	0.0038769425	-1.942485e-06
1994	9	0.09225000	0.0034077396	3.422604e-04
1994	10	0.09241667	-0.0004971517	1.080485e-03
1994	11	0.09258333	-0.0025859198	1.002587e-03
1994	12	0.09279167	-0.0026357387	8.440720e-04
1995	1	0.09304167	-0.0017752314	-2.664352e-04
1995	2	0.09325000	-0.0020442532	-2.057468e-04
1995	3	0.09337500	-0.0019645430	-4.104570e-04
1995	4	0.09345833	-0.0022747785	-1.183555e-03

1995	5	0.09354167	-0.0012904758	-2.511909e-04
1995	6	0.09358333	0.0037751270	6.415396e-04
1995	7	0.09358333	0.0040082831	4.083836e-04
1995	8	0.09358333	0.0038769425	5.397242e-04
1995	9	0.09362500	0.0034077396	-3.273959e-05
1995	10	0.09375000	-0.0004971517	7.471517e-04
1995	11	0.09383333	-0.0025859198	7.525865e-04
1995	12	0.09375000	-0.0026357387	-1.142613e-04
1996	1	0.09370833	-0.0017752314	-9.331019e-04
1996	2	0.09379167	-0.0020442532	-7.474135e-04
1996	3	0.09391667	-0.0019645430	4.787636e-05
1996	4	0.09404167	-0.0022747785	2.331119e-04
1996	5	0.09408333	-0.0012904758	-7.928575e-04
1996	6	0.09412500	0.0037751270	-1.900127e-03
1996	7	0.09420833	0.0040082831	7.833836e-04
1996	8	0.09429167	0.0038769425	8.313908e-04
1996	9	0.09437500	0.0034077396	1.217260e-03
1996	10	0.09441667	-0.0004971517	1.080485e-03
1996	11	0.09445833	-0.0025859198	1.275865e-04
1996	12	0.09462500	-0.0026357387	1.073867e-05
1997	1	0.09475000	-0.0017752314	-9.747686e-04
1997	2	0.09470833	-0.0020442532	-6.640802e-04
1997	3	0.09466667	-0.0019645430	2.978764e-04
1997	4	0.09458333	-0.0022747785	-3.085548e-04
1997	5	0.09450000	-0.0012904758	-2.095242e-04
1997	6	0.09441667	0.0037751270	8.082063e-04
1997	7	0.09408333	0.0040082831	9.083836e-04
1997	8	0.09354167	0.0038769425	5.813908e-04
1997	9	0.09291667	0.0034077396	2.675594e-03
1997	10	0.09229167	-0.0004971517	1.205485e-03
1997	11	0.09170833	-0.0025859198	2.877587e-03
1997	12	0.09108333	-0.0026357387	1.552405e-03
1998	1	0.09041667	-0.0017752314	-2.641435e-03
1998	2	0.08975000	-0.0020442532	-2.705747e-03
1998	3	0.08900000	-0.0019645430	-2.035457e-03
1998	4	0.08829167	-0.0022747785	-1.016888e-03
1998	5	0.08766667	-0.0012904758	-3.761909e-04
1998	6	0.08708333	0.0037751270	1.415396e-04
1998	7	0.08675000	0.0040082831	2.417169e-04
1998	8	0.08662500	0.0038769425	-5.019425e-04
1998	9	0.08654167	0.0034077396	-9.494063e-04
1998	10	0.08645833	-0.0004971517	3.881838e-05
1998	11	0.08637500	-0.0025859198	2.109198e-04
1998	12	0.08625000	-0.0026357387	3.857387e-04

1999	1	0.08612500	-0.0017752314	-3.497686e-04
1999	2	0.08604167	-0.0020442532	2.586500e-06
1999	3	0.08604167	-0.0019645430	-7.712364e-05
1999	4	0.08612500	-0.0022747785	1.497785e-04
1999	5	0.08620833	-0.0012904758	8.214247e-05
1999	6	0.08629167	0.0037751270	-1.066794e-03
1999	7	0.08633333	0.0040082831	-3.416164e-04
1999	8	0.08637500	0.0038769425	-1.251942e-03
1999	9	0.08645833	0.0034077396	1.339271e-04
1999	10	0.08654167	-0.0004971517	9.554851e-04
1999	11	0.08658333	-0.0025859198	1.002587e-03
1999	12	0.08662500	-0.0026357387	1.010739e-03
2000	1	0.08670833	-0.0017752314	-9.331019e-04
2000	2	0.08683333	-0.0020442532	2.109198e-04
2000	3	0.08695833	-0.0019645430	6.209689e-06
2000	4	0.08704167	-0.0022747785	2.331119e-04
2000	5	0.08712500	-0.0012904758	-8.345242e-04
2000	6	0.08720833	0.0037751270	-9.834604e-04
2000	7	0.08741667	0.0040082831	-4.249497e-04
2000	8	0.08770833	0.0038769425	-5.852758e-04
2000	9	0.08800000	0.0034077396	-4.077396e-04
2000	10	0.08833333	-0.0004971517	1.638184e-04
2000	11	0.08870833	-0.0025859198	-1.224135e-04
2000	12	0.08920833	-0.0026357387	-5.725947e-04
2001	1	0.08983333	-0.0017752314	-5.810191e-05
2001	2	0.09045833	-0.0020442532	-4.140802e-04
2001	3	0.09100000	-0.0019645430	-3.545698e-05
2001	4	0.09145833	-0.0022747785	-1.835548e-04
2001	5	0.09183333	-0.0012904758	-5.428575e-04
2001	6	0.09216667	0.0037751270	1.058206e-03
2001	7	0.09237500	0.0040082831	2.616717e-03
2001	8	0.09245833	0.0038769425	1.664724e-03
2001	9	0.09250000	0.0034077396	1.092260e-03
2001	10	0.09245833	-0.0004971517	1.038818e-03
2001	11	0.09237500	-0.0025859198	2.109198e-04
2001	12	0.09225000	-0.0026357387	3.857387e-04
2002	1	0.09200000	-0.0017752314	-1.224769e-03
2002	2	0.09170833	-0.0020442532	-6.640802e-04
2002	3	0.09150000	-0.0019645430	-5.354570e-04
2002	4	0.09133333	-0.0022747785	-1.058555e-03
2002	5	0.09120833	-0.0012904758	-9.178575e-04
2002	6	0.09112500	0.0037751270	9.987296e-05
2002	7	0.09108333	0.0040082831	-9.161640e-05
2002	8	0.09108333	0.0038769425	3.972418e-05

2002	9	0.09108333	0.0034077396	5.089271e-04
2002	10	0.09120833	-0.0004971517	2.888184e-04
2002	11	0.09145833	-0.0025859198	1.275865e-04
2002	12	0.09166667	-0.0026357387	-3.092799e-05
2003	1	0.09187500	-0.0017752314	-1.099769e-03
2003	2	0.09212500	-0.0020442532	-1.080747e-03
2003	3	0.09237500	-0.0019645430	-1.410457e-03
2003	4	0.09258333	-0.0022747785	6.914452e-04
2003	5	0.09270833	-0.0012904758	5.821425e-04
2003	6	0.09279167	0.0037751270	4.332063e-04
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2003	8	0.09308333	0.0038769425	1.039724e-03
2003	9	0.09325000	0.0034077396	1.342260e-03
2003	10	0.09333333	-0.0004971517	1.638184e-04
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2008	8	0.12458333	0.0038769425	3.539724e-03
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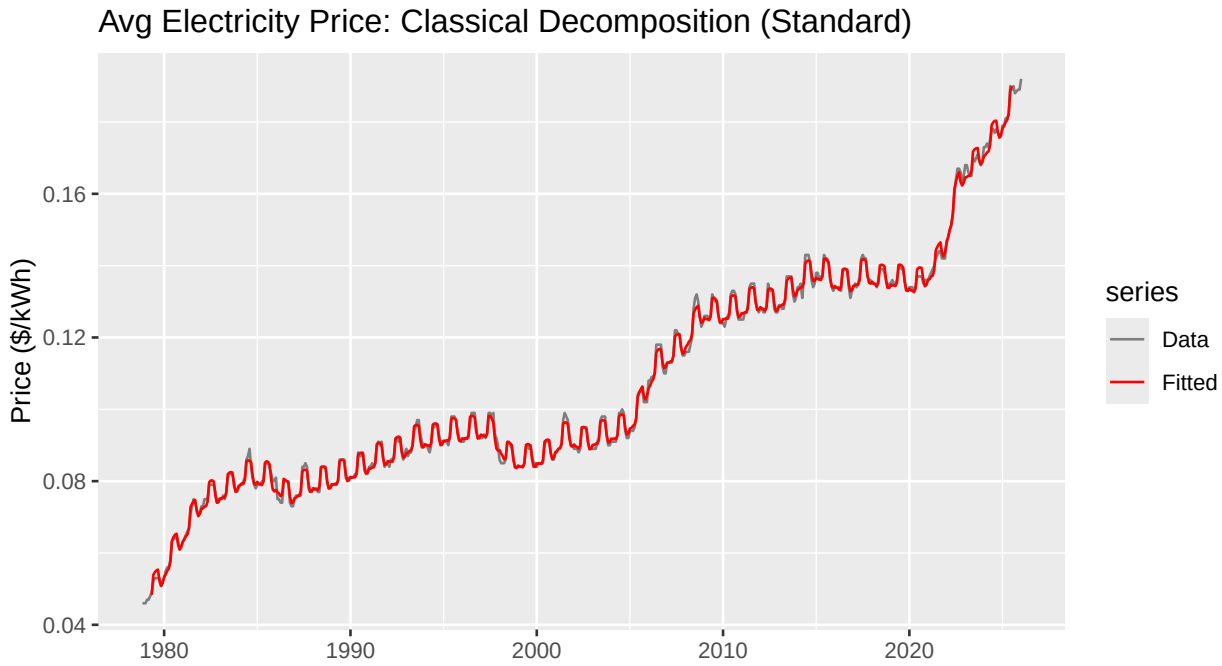
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2013	4	0.13141667	-0.0022747785	-1.141888e-03
2013	5	0.13170833	-0.0012904758	5.821425e-04
2013	6	0.13200000	0.0037751270	1.224873e-03
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2013	8	0.13279167	0.0038769425	3.313908e-04

2013	9	0.13329167	0.0034077396	3.005937e-04
2013	10	0.13370833	-0.0004971517	-1.211182e-03
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2014	12	0.13841667	-0.0026357387	-7.809280e-04
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2016	7	0.13516667	0.0040082831	-1.749497e-04
2016	8	0.13520833	0.0038769425	-8.527582e-05
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2017	4	0.13695833	-0.0022747785	3.164452e-04

2017	5	0.13729167	-0.0012904758	9.988091e-04
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2018	3	0.13675000	-0.0019645430	2.145430e-04
2018	4	0.13654167	-0.0022747785	-2.668881e-04
2018	5	0.13641667	-0.0012904758	8.738091e-04
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2020	4	0.13487500	-0.0022747785	3.997785e-04
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2021	4	0.13945833	-0.0022747785	1.816445e-03
2021	5	0.14000000	-0.0012904758	1.290476e-03
2021	6	0.14050000	0.0037751270	-2.275127e-03
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2021	10	0.14408333	-0.0004971517	-1.586182e-03
2021	11	0.14516667	-0.0025859198	-5.807468e-04
2021	12	0.14650000	-0.0026357387	-1.864261e-03
2022	1	0.14812500	-0.0017752314	6.502314e-04
2022	2	0.14995833	-0.0020442532	8.591983e-05
2022	3	0.15187500	-0.0019645430	8.954302e-05
2022	4	0.15383333	-0.0022747785	-5.585548e-04
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2022	10	0.16383333	-0.0004971517	2.663818e-03
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2022	12	0.16575000	-0.0026357387	1.885739e-03
2023	1	0.16637500	-0.0017752314	3.400231e-03
2023	2	0.16670833	-0.0020442532	3.335920e-03
2023	3	0.16700000	-0.0019645430	9.645430e-04
2023	4	0.16729167	-0.0022747785	-1.688814e-05
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2024	8	0.17641667	0.0038769425	-3.293609e-03

2024	9	0.17695833	0.0034077396	-2.366073e-03
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2025	3	0.18191667	-0.0019645430	1.047876e-03
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2025	5	0.18383333	-0.0012904758	-5.428575e-04
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2025	7	0.18600000	0.0040082831	-1.008283e-03
2025	8	NA	0.0038769425	NA
2025	9	NA	0.0034077396	NA
2025	10	NA	-0.0004971517	NA
2025	11	NA	-0.0025859198	NA
2025	12	NA	-0.0026357387	NA
2026	1	NA	-0.0017752314	NA

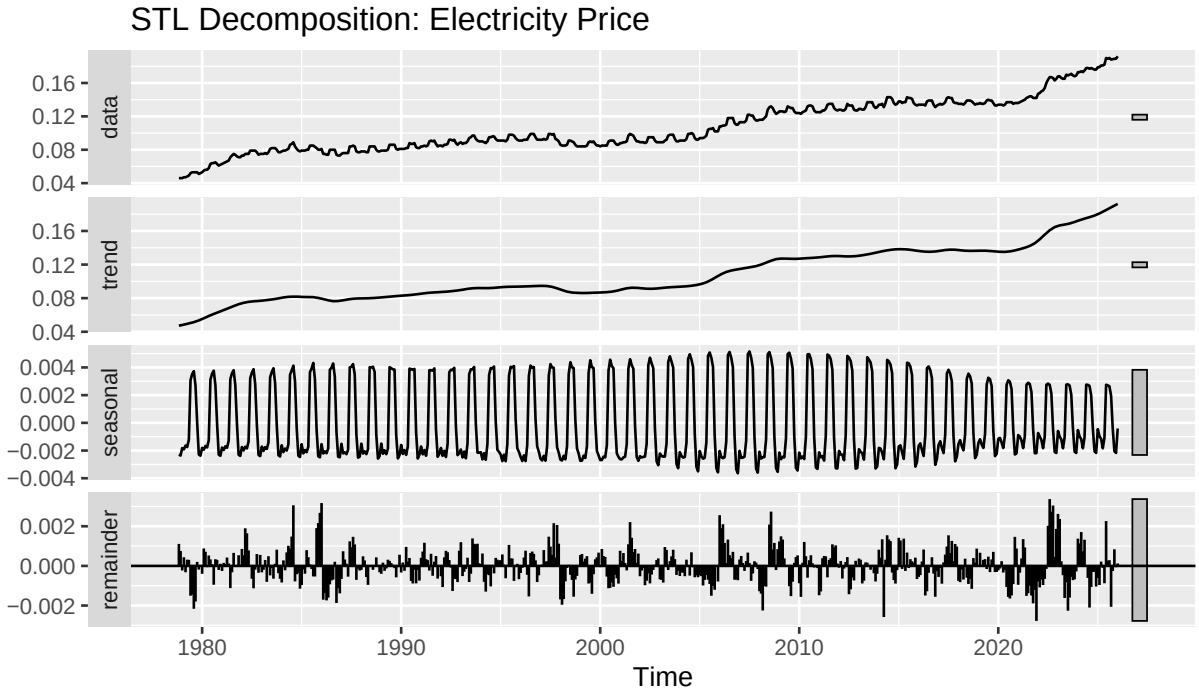


Examine extreme outliers in the remainder component:

```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.       NA's
## -0.0033833 -0.0006641  0.0000026 -0.0000083  0.0006579  0.0035397         12
##
## No extreme outliers detected (3 × IQR rule).
```

STL Decomposition

For both the STL, additive, and multiplicative decomposition there was barely any change, which could be because the seasonality of the electricity price is not due to price levels, but to the season itself, which means, if it's summer, people use AC, thus consuming more electricity, and in winter, We found out that a majority of the houses use gas for heating, which can explain this issue. This all means that the seasonal component is constant over time.



year	month	trend	seasonal	remainder
1978	11	0.04717768	-2.287454e-03	1.109776e-03
1978	12	0.04761293	-2.364152e-03	7.512224e-04
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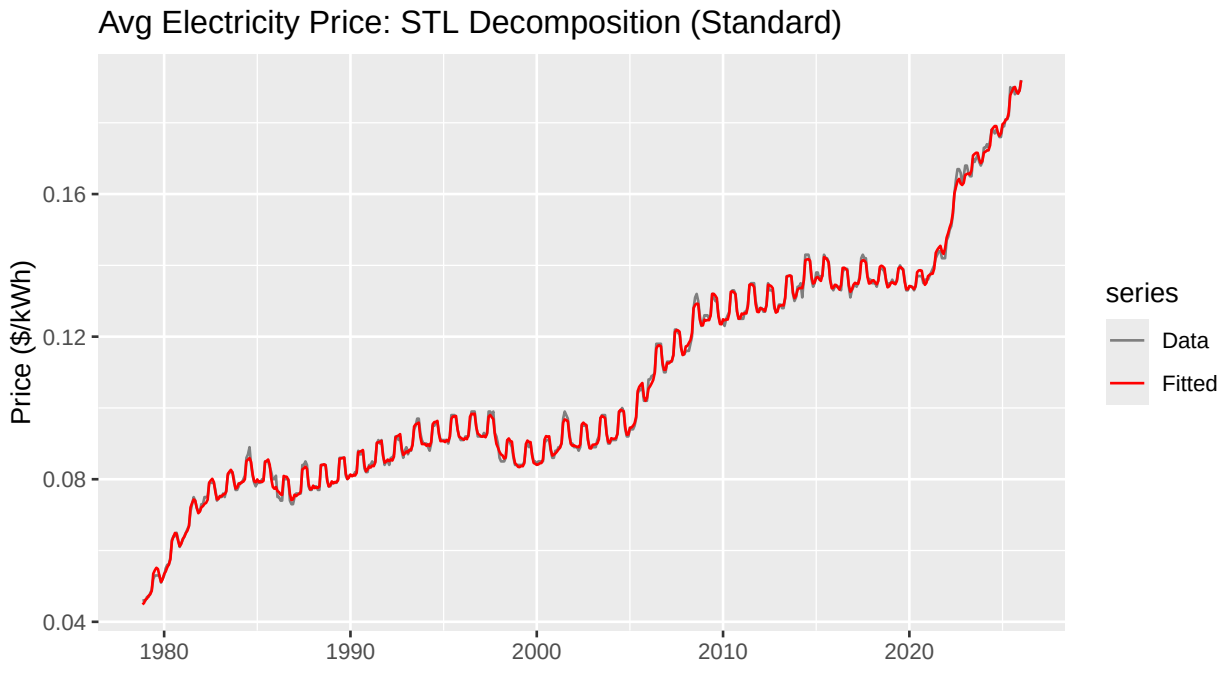
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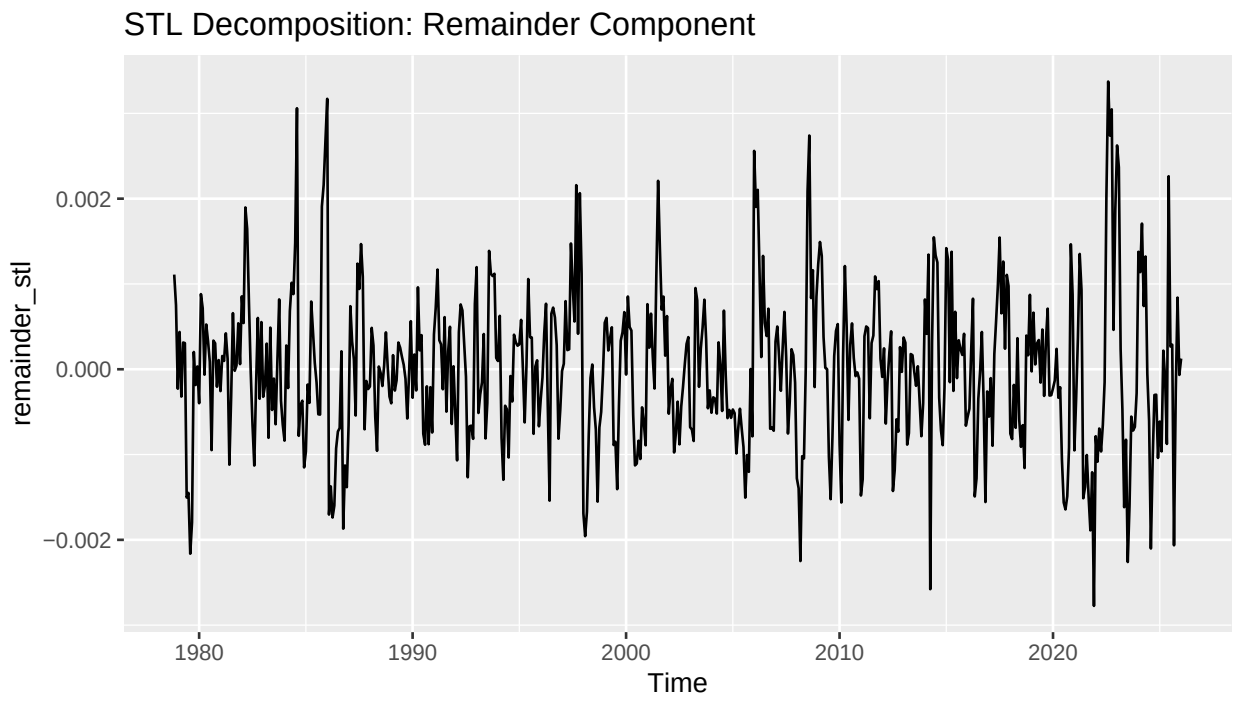
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2023	4	0.16730605	-1.835779e-03	-4.702682e-04
2023	5	0.16764772	-1.030314e-03	-1.617402e-03
2023	6	0.16806496	2.761204e-03	-8.261613e-04
2023	7	0.16848220	2.778100e-03	-2.260298e-03
2023	8	0.16889944	2.696542e-03	-1.595983e-03
2023	9	0.16951862	2.036289e-03	-5.549139e-04
2023	10	0.17013781	-4.193046e-04	-7.185039e-04
2023	11	0.17075699	-2.081022e-03	-6.759701e-04
2023	12	0.17145115	-2.171717e-03	-2.794291e-04
2024	1	0.17214530	-5.225251e-04	1.377225e-03
2024	2	0.17283945	-9.781062e-04	1.138653e-03
2024	3	0.17346861	-1.176490e-03	1.707879e-03

2024	4	0.17409777	-1.838558e-03	7.407895e-04
2024	5	0.17472693	-1.049011e-03	1.322084e-03
2024	6	0.17529531	2.756443e-03	-5.175153e-05
2024	7	0.17586369	2.742087e-03	-6.057772e-04
2024	8	0.17643207	2.669467e-03	-2.101539e-03
2024	9	0.17707183	1.993759e-03	-1.065594e-03
2024	10	0.17771160	-4.098524e-04	-3.017449e-04
2024	11	0.17835136	-2.053316e-03	-2.980439e-04
2024	12	0.17921075	-2.171130e-03	-1.039623e-03
2025	1	0.18007014	-4.573605e-04	-6.127844e-04
2025	2	0.18092954	-9.665718e-04	-9.629655e-04
2025	3	0.18193404	-1.152281e-03	2.182368e-04
2025	4	0.18293855	-1.841353e-03	-9.719893e-05
2025	5	0.18394306	-1.067366e-03	-8.756935e-04
2025	6	0.18498461	2.752486e-03	2.262899e-03
2025	7	0.18602617	2.707339e-03	2.664902e-04
2025	8	0.18706773	2.644020e-03	2.882542e-04
2025	9	0.18811094	1.953219e-03	-2.064158e-03
2025	10	0.18915415	-3.982223e-04	-2.559295e-04
2025	11	0.19019736	-2.037536e-03	8.401709e-04
2025	12	0.19123982	-2.173413e-03	-6.640969e-05
2026	1	0.19228228	-4.069496e-04	1.246680e-04



Examine extreme outliers in the STL remainder component:




```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -2.775e-03 -6.336e-04 -2.856e-05 -8.840e-06  4.841e-04  3.373e-03
##
## No extreme outliers detected (3 × IQR rule).
```

Project II

Time Series Cross Validation (TSCV)

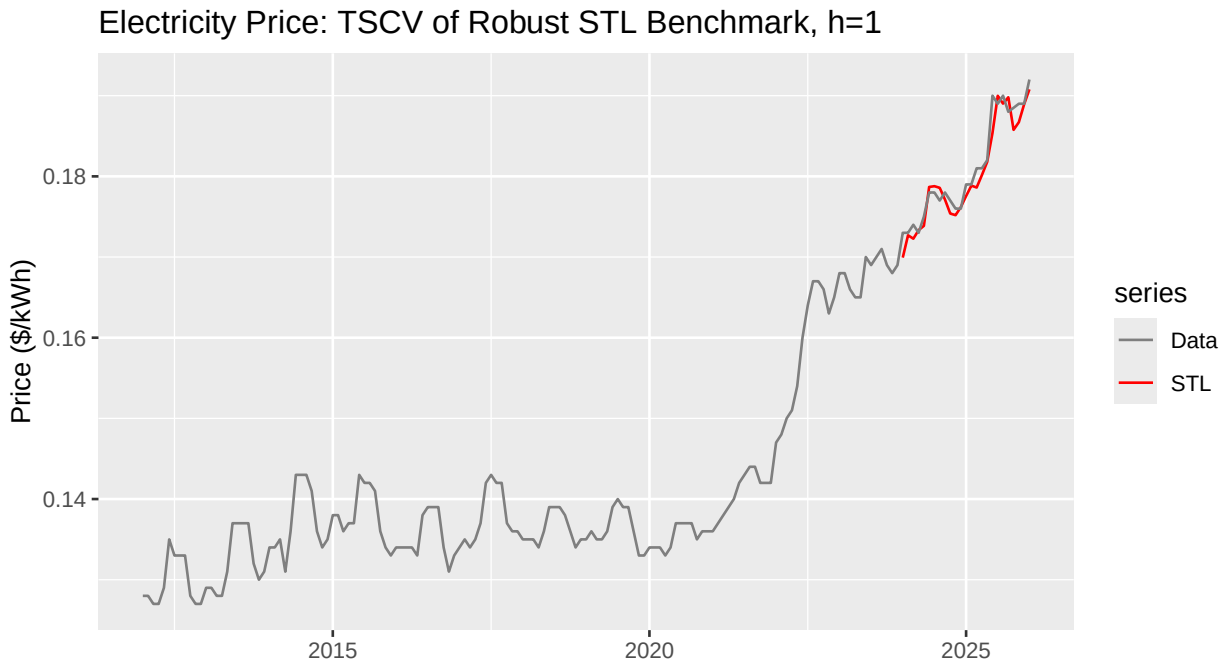
a) We chose a moving window for a few reasons. The main one being that demand of data centers are massive and growing. Data centers use massive amounts of electricity, so we use a moving window which makes the recent observations more informative than the older ones.

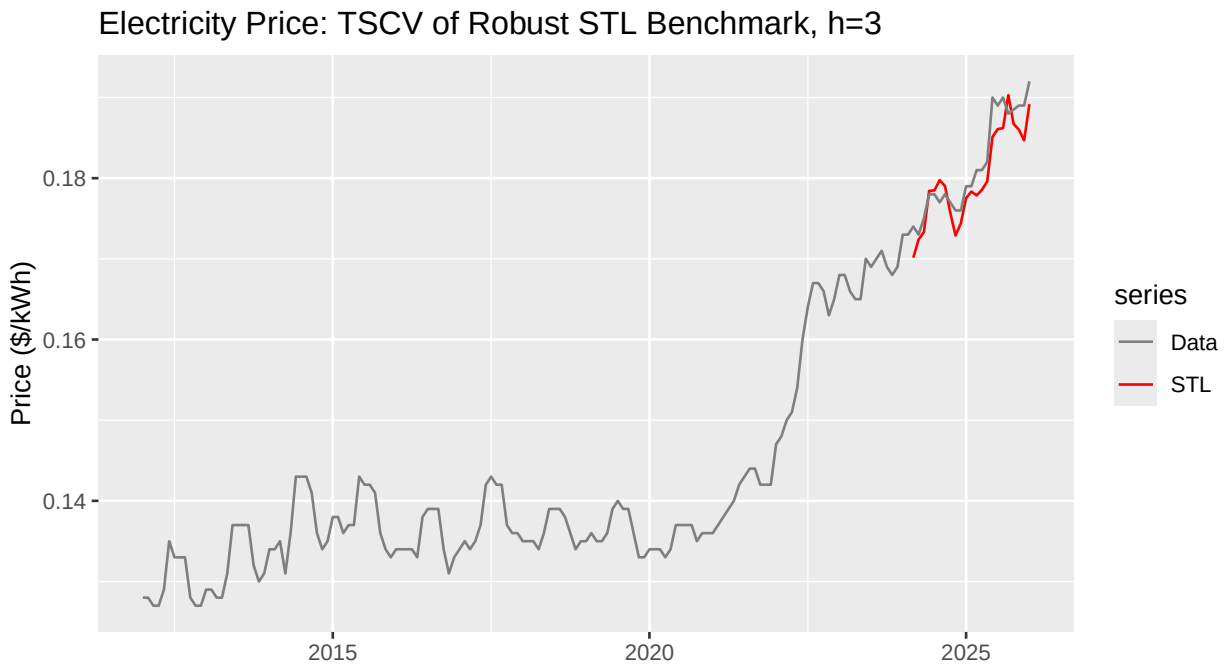
B) We chose our start year to be 2010 because it includes a more relevant electricity market: shale gas revolution, rise in renewable energy, and a rise in AI demand. Thus, having it start in 2010 rather than 2005, our estimates drift towards a more relevant market.

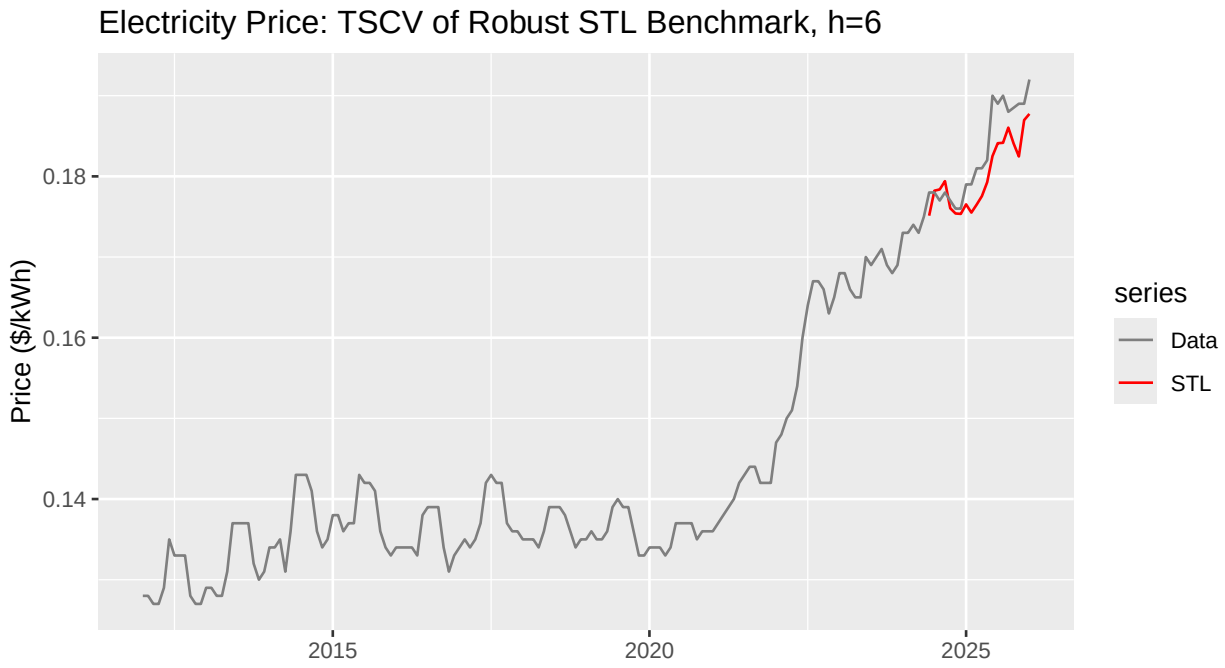
C) 144 months = 12 years = ~ 2 business cycles. 2 cycles ensure the models sees at least one full expansion and one full contraction.

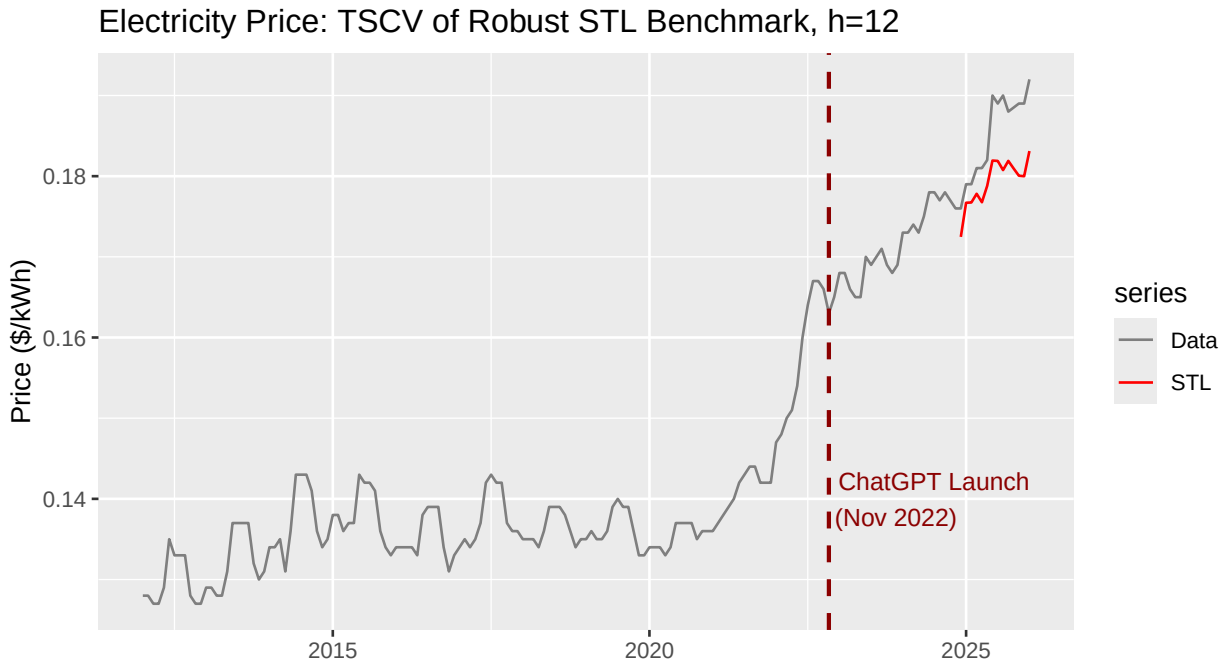
D) Forecast Horizons: h = 1 –> Next month h = 3 –> Quarterly h = 6 –> Semi-annual h = 12 –> Annual

E) Since the data has a trend, we use percentage metrics (MPE, MAPE). They tell more of the story than raw errors (ME, MAE, RMSE). Ignore Theil's U, since it is unreliable when the data has a trend.


Electricity Price: TSCV of Robust STL Benchmark, h=1
Price ($/kWh)
0.18
0.16
0.14
2015
2020
2025
series
Data
STL


Electricity Price: TSCV of Robust STL Benchmark, h=3
Price ($/kWh)
0.18
0.16
0.14
2015
2020
2025
series
Data
STL






```
##          ME  RMSE   MAE   MPE  MAPE   ACF1 Theil's U
## h=1   0.001 0.002 0.001 0.441 0.721 -0.239     0.227
## h=3   0.002 0.003 0.002 0.916 1.251  0.346     0.332
## h=6   0.003 0.004 0.003 1.507 1.677  0.541     0.450
## h=12  0.006 0.007 0.006 3.190 3.190  0.680     0.670
```

MPE increases from 0.417 –> 3.570 across time horizons, indicating a consistent positive bias -

meaning the model underestimates actual prices as time horizon increases. We believe this is in part because of massive and growing demand for data centers.

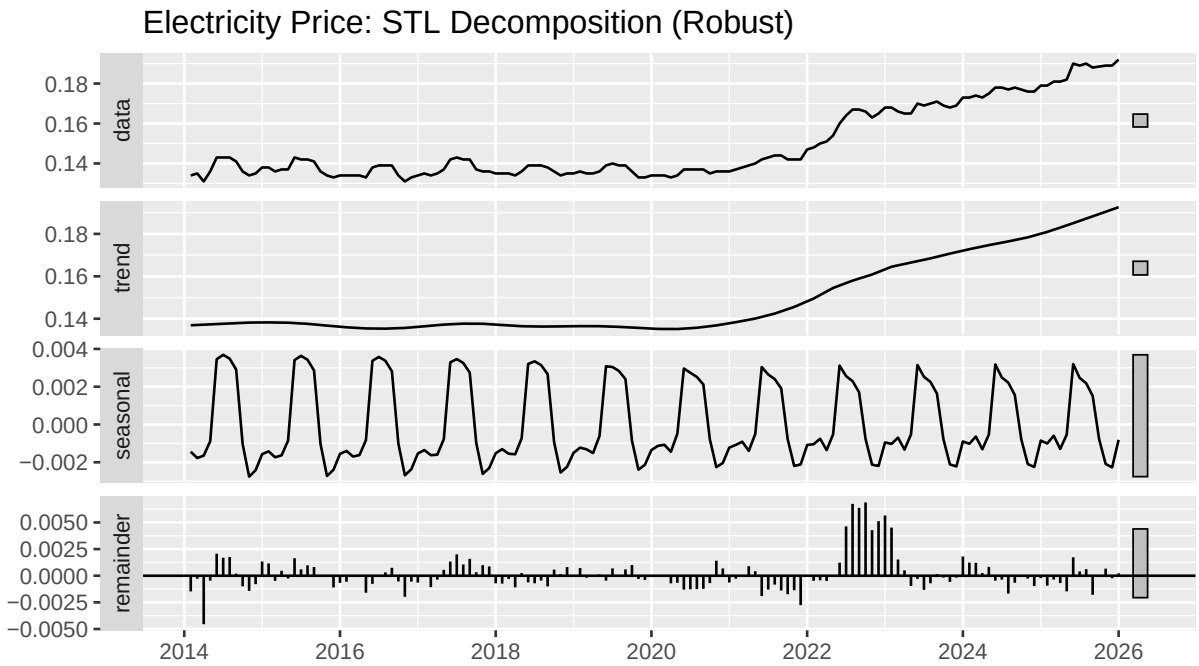
MAPE increases from 0.841 –> 0.3577 across time horizons, indicating that errors increases as time horizon increases.

Forecast the Future

Question 3:

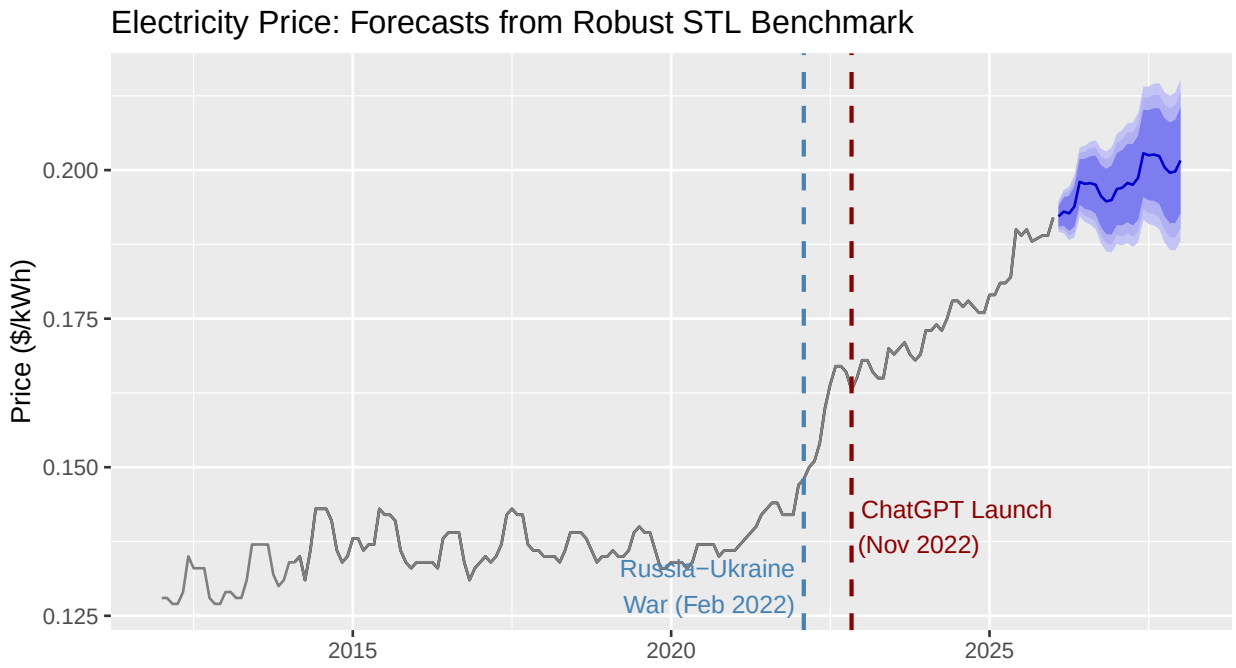
Consistent with the moving window used in TSCV, the robust STL decomposition is applied to the last 144 months of historical data (2014M2–2026M1) to generate forecasts for the next 24 months.

In the figure below, The trend component shows a clear acceleration in early 2022 as a result from Russia-Ukraine war and an increase in trend towards the end of 2022 (could be from LLM's being released), the seasonal component remains stable, and the remainder component captures the 2022 energy price shock without distorting the trend estimate.



In the figure below, we forecast 24 months ahead (24 months is a reasonable horizon for electricity prices) from the robust STL benchmark with 80%, 90%, 95%, and 99% prediction intervals.

The blue shaded show the forecast confidence intervals at respective levels through the period 2028. The dashed blue line highlights the start of the Russia-Ukraine war. The dashed red line highlights the launch of ChatGPT in November 2022, which helps see how AI and the demand for data centers are a tailwind for these increasing electricity prices.



	Point Forecast	Lo 80	Hi 80	Lo 90	Hi 90	Lo 95	Hi 95
Feb 2026	0.1922079	0.1905213	0.1938945	0.1900431	0.1943726	0.1896284	0.1947873
Mar 2026	0.1930208	0.1906273	0.1954143	0.1899488	0.1960928	0.1893603	0.1966813
Apr 2026	0.1927216	0.1897801	0.1956631	0.1889462	0.1964969	0.1882230	0.1972202
May 2026	0.1938803	0.1904721	0.1972884	0.1895059	0.1982546	0.1886679	0.1990926
Jun 2026	0.1980184	0.1941951	0.2018418	0.1931112	0.2029257	0.1921711	0.2038658
Jul 2026	0.1976821	0.1934797	0.2018846	0.1922884	0.2030759	0.1912551	0.2041092
Aug 2026	0.1978041	0.1932498	0.2023584	0.1919587	0.2036495	0.1908388	0.2047694
Sep 2026	0.1975375	0.1926525	0.2024225	0.1912677	0.2038074	0.1900666	0.2050085
Oct 2026	0.1956373	0.1904388	0.2008357	0.1889651	0.2023094	0.1876869	0.2035876
Nov 2026	0.1947355	0.1892379	0.2002332	0.1876794	0.2017917	0.1863276	0.2031434
Dec 2026	0.1949553	0.1891705	0.2007401	0.1875306	0.2023800	0.1861082	0.2038023
Jan 2027	0.1968144	0.1907528	0.2028760	0.1890344	0.2045943	0.1875440	0.2060848
Feb 2027	0.1970222	0.1906928	0.2033516	0.1888985	0.2051460	0.1873422	0.2067022
Mar 2027	0.1978352	0.1912458	0.2044245	0.1893778	0.2062925	0.1877576	0.2079127
Apr 2027	0.1975359	0.1906936	0.2043783	0.1887539	0.2063180	0.1870715	0.2080004
May 2027	0.1986946	0.1916056	0.2057837	0.1895959	0.2077933	0.1878528	0.2095364
Jun 2027	0.2028328	0.1955026	0.2101630	0.1934246	0.2122410	0.1916223	0.2140433
Jul 2027	0.2024965	0.1949303	0.2100627	0.1927854	0.2122077	0.1909250	0.2140681
Aug 2027	0.2026185	0.1948208	0.2104161	0.1926103	0.2126267	0.1906930	0.2145440
Sep 2027	0.2023519	0.1943270	0.2103768	0.1920521	0.2126517	0.1900789	0.2146249
Oct 2027	0.2004516	0.1922034	0.2086999	0.1898651	0.2110382	0.1878370	0.2130662
Nov 2027	0.1995499	0.1910818	0.2080179	0.1886812	0.2104185	0.1865991	0.2125007
Dec 2027	0.1997696	0.1910851	0.2084542	0.1886231	0.2109162	0.1864877	0.2130516

Jan 2028		0.2016287	0.1927307	0.2105268	0.1902082	0.2130492	0.1880204	0.2152371

year	month	seasonal	trend	remainder
2014	2	-0.001441	0.136914	-0.001473
2014	3	-0.001775	0.137056	-0.000281
2014	4	-0.001647	0.137199	-0.004552
2014	5	-0.000885	0.137341	-0.000456
2014	6	0.003454	0.137479	0.002067
2014	7	0.003691	0.137616	0.001693
2014	8	0.003485	0.137754	0.001761
2014	9	0.002908	0.137896	0.000196
2014	10	-0.001039	0.138038	-0.000999
2014	11	-0.002760	0.138181	-0.001421
2014	12	-0.002421	0.138209	-0.000788
2015	1	-0.001571	0.138237	0.001334
2015	2	-0.001420	0.138265	0.001154
2015	3	-0.001736	0.138216	-0.000480
2015	4	-0.001634	0.138167	0.000467
2015	5	-0.000857	0.138117	-0.000260
2015	6	0.003414	0.137943	0.001643
2015	7	0.003635	0.137769	0.000596
2015	8	0.003428	0.137595	0.000976
2015	9	0.002864	0.137317	0.000819
2015	10	-0.001025	0.137039	-0.000014
2015	11	-0.002726	0.136760	-0.000034
2015	12	-0.002394	0.136493	-0.001099
2016	1	-0.001555	0.136227	-0.000672
2016	2	-0.001397	0.135960	-0.000563
2016	3	-0.001696	0.135781	-0.000085
2016	4	-0.001621	0.135602	0.000019
2016	5	-0.000830	0.135423	-0.001593
2016	6	0.003373	0.135388	-0.000761
2016	7	0.003579	0.135353	0.000068
2016	8	0.003371	0.135318	0.000311
2016	9	0.002821	0.135433	0.000746
2016	10	-0.001009	0.135548	-0.000538
2016	11	-0.002690	0.135662	-0.001972
2016	12	-0.002366	0.135919	-0.000553
2017	1	-0.001535	0.136175	-0.000640
2017	2	-0.001352	0.136432	-0.000079
2017	3	-0.001623	0.136697	-0.001073
2017	4	-0.001598	0.136962	-0.000363
2017	5	-0.000778	0.137227	0.000551

2017	6	0.003290	0.137380	0.001330
2017	7	0.003464	0.137533	0.002003
2017	8	0.003257	0.137686	0.001057
2017	9	0.002745	0.137666	0.001589
2017	10	-0.000972	0.137646	0.000326
2017	11	-0.002617	0.137626	0.000991
2017	12	-0.002305	0.137432	0.000873
2018	1	-0.001522	0.137238	-0.000717
2018	2	-0.001305	0.137045	-0.000740
2018	3	-0.001549	0.136854	-0.000305
2018	4	-0.001576	0.136664	-0.001088
2018	5	-0.000728	0.136474	0.000254
2018	6	0.003206	0.136419	-0.000625
2018	7	0.003346	0.136364	-0.000710
2018	8	0.003141	0.136309	-0.000449
2018	9	0.002668	0.136334	-0.001002
2018	10	-0.000935	0.136359	0.000577
2018	11	-0.002542	0.136384	0.000158
2018	12	-0.002240	0.136424	0.000815
2019	1	-0.001501	0.136465	0.000036
2019	2	-0.001223	0.136505	0.000718
2019	3	-0.001315	0.136497	-0.000183
2019	4	-0.001514	0.136489	0.000025
2019	5	-0.000607	0.136482	0.000125
2019	6	0.003084	0.136373	-0.000457
2019	7	0.003043	0.136265	0.000692
2019	8	0.002831	0.136156	0.000013
2019	9	0.002399	0.136003	0.000599
2019	10	-0.000854	0.135850	0.001005
2019	11	-0.002392	0.135697	-0.000305
2019	12	-0.002130	0.135532	-0.000402
2020	1	-0.001352	0.135368	-0.000017
2020	2	-0.001131	0.135204	-0.000073
2020	3	-0.001072	0.135184	-0.000112
2020	4	-0.001446	0.135163	-0.000717
2020	5	-0.000482	0.135142	-0.000660
2020	6	0.002964	0.135341	-0.001305
2020	7	0.002738	0.135540	-0.001278
2020	8	0.002517	0.135738	-0.001255
2020	9	0.002122	0.136107	-0.001229
2020	10	-0.000783	0.136477	-0.000693
2020	11	-0.002255	0.136846	0.001409
2020	12	-0.002036	0.137352	0.000685
2021	1	-0.001220	0.137858	-0.000638

2021	2	-0.001082	0.138363	-0.000281
2021	3	-0.000909	0.138935	-0.000026
2021	4	-0.001399	0.139507	0.000891
2021	5	-0.000503	0.140079	0.000423
2021	6	0.003040	0.140864	-0.001904
2021	7	0.002648	0.141648	-0.001295
2021	8	0.002403	0.142432	-0.000835
2021	9	0.001912	0.143475	-0.001387
2021	10	-0.000780	0.144518	-0.001738
2021	11	-0.002195	0.145561	-0.001366
2021	12	-0.002117	0.146881	-0.002764
2022	1	-0.001086	0.148201	-0.000115
2022	2	-0.001043	0.149521	-0.000478
2022	3	-0.000752	0.151183	-0.000430
2022	4	-0.001355	0.152844	-0.000490
2022	5	-0.000523	0.154506	0.000017
2022	6	0.003119	0.155657	0.001224
2022	7	0.002561	0.156808	0.004631
2022	8	0.002295	0.157959	0.006747
2022	9	0.001708	0.158923	0.006369
2022	10	-0.000772	0.159887	0.006885
2022	11	-0.002130	0.160851	0.004279
2022	12	-0.002194	0.162071	0.005123
2023	1	-0.000949	0.163291	0.005658
2023	2	-0.001028	0.164511	0.004517
2023	3	-0.000694	0.165170	0.001523
2023	4	-0.001332	0.165829	0.000502
2023	5	-0.000527	0.166488	-0.000961
2023	6	0.003149	0.167144	-0.000294
2023	7	0.002526	0.167800	-0.001326
2023	8	0.002255	0.168456	-0.000711
2023	9	0.001635	0.169203	0.000161
2023	10	-0.000777	0.169950	-0.000173
2023	11	-0.002113	0.170697	-0.000584
2023	12	-0.002220	0.171393	-0.000173
2024	1	-0.000899	0.172089	0.001810
2024	2	-0.001017	0.172785	0.001232
2024	3	-0.000639	0.173422	0.001217
2024	4	-0.001311	0.174059	0.000252
2024	5	-0.000533	0.174696	0.000837
2024	6	0.003179	0.175283	-0.000462
2024	7	0.002490	0.175871	-0.000361
2024	8	0.002215	0.176458	-0.001674
2024	9	0.001563	0.177095	-0.000658

2024	10	-0.000782	0.177731	0.000051
2024	11	-0.002096	0.178367	-0.000272
2024	12	-0.002247	0.179223	-0.000975
2025	1	-0.000850	0.180078	-0.000228
2025	2	-0.001005	0.180933	-0.000927
2025	3	-0.000594	0.181957	-0.000364
2025	4	-0.001294	0.182982	-0.000688
2025	5	-0.000537	0.184006	-0.001470
2025	6	0.003200	0.185070	0.001730
2025	7	0.002463	0.186133	0.000404
2025	8	0.002184	0.187197	0.000619
2025	9	0.001516	0.188272	-0.001788
2025	10	-0.000786	0.189348	-0.000062
2025	11	-0.002088	0.190424	0.000665
2025	12	-0.002270	0.191506	-0.000236
2026	1	-0.000812	0.192588	0.000224

Manual Verification

- Y_1^{sa} (2014M2) = 0.136914 + (-0.001473) = 0.135441
- Y_T^{sa} (2026M1) = 0.192588 + 0.000224 = 0.192812
- T = 144, drift per month = (0.192812 - 0.135441) / 143 = 0.000401

Forecast 1: February 2026 (h = 1)

Seasonally adjusted forecast:

$$\hat{Y}_{T+1}^{sa} = 0.192812 + 0.000401 \times 1 = 0.193213$$

Re-seasonalize using Yfeb2026 = -0.001005:

$$\hat{Y}_{Feb2026} = 0.193213 + (-0.001005) = 0.192208$$

R table output: 0.1922079

Forecast 2: March 2026 (h = 2)

Seasonally adjusted forecast:

$$\hat{Y}_{T+2}^{sa} = 0.192812 + 0.000401 \times 2 = 0.193614$$

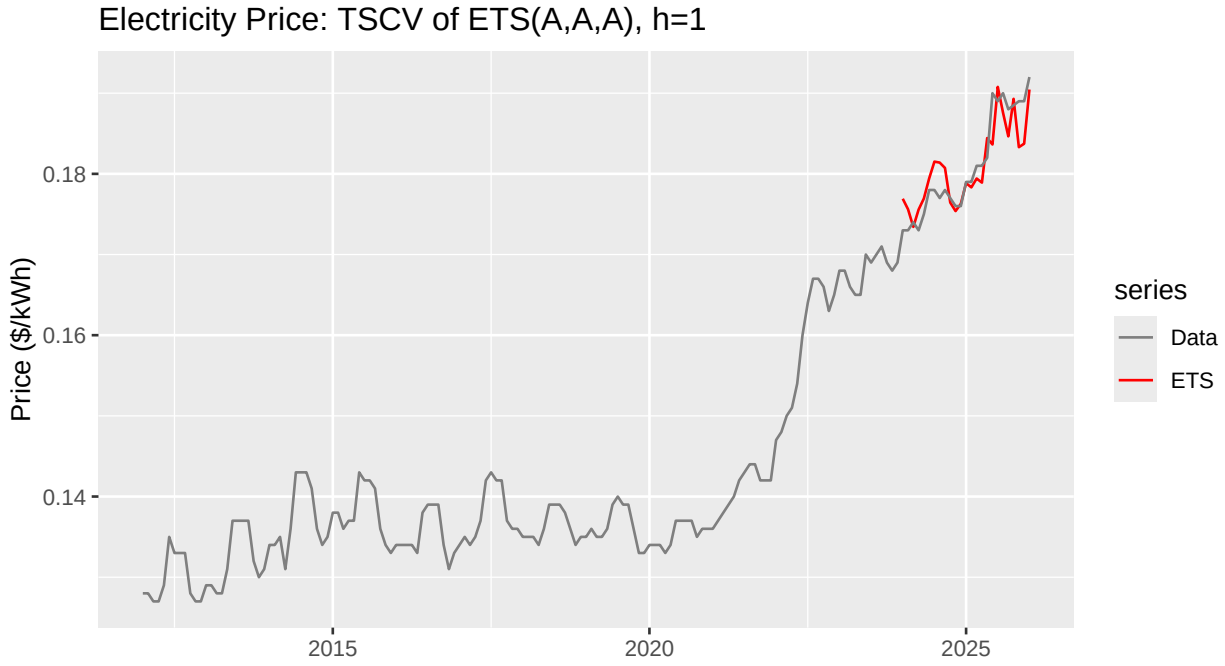
Re-seasonalize using S_{2025M3} = -0.000594:

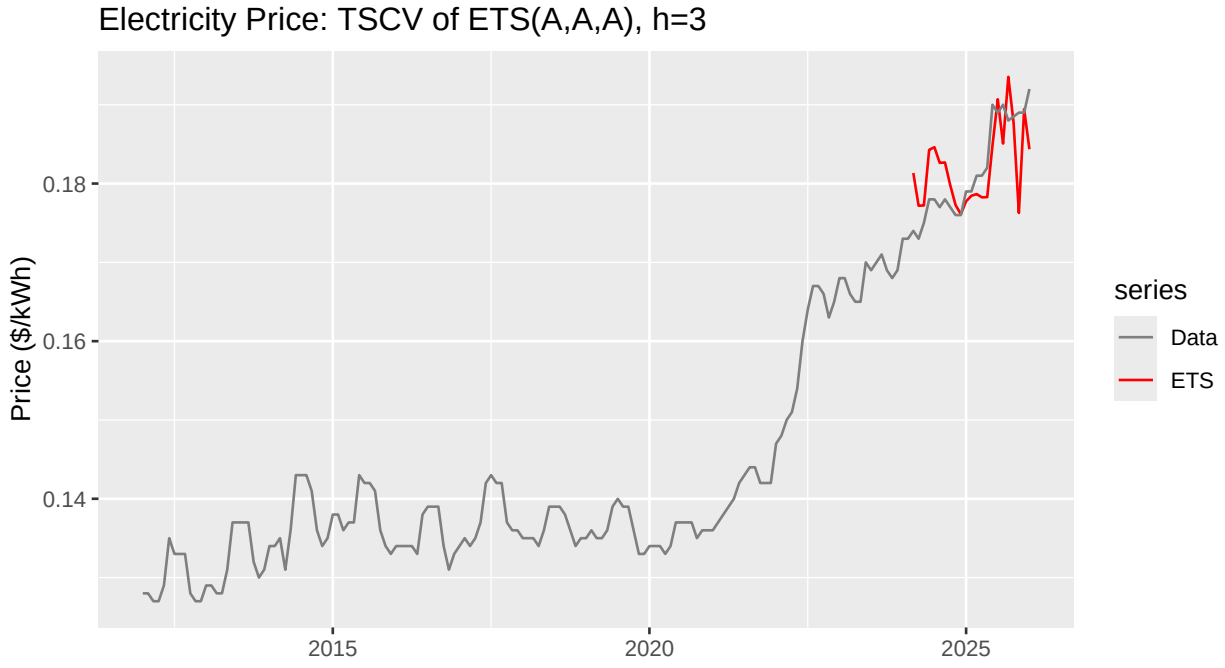
$$\hat{Y}_{Mar2026} = 0.193614 + (-0.000594) = 0.193020$$

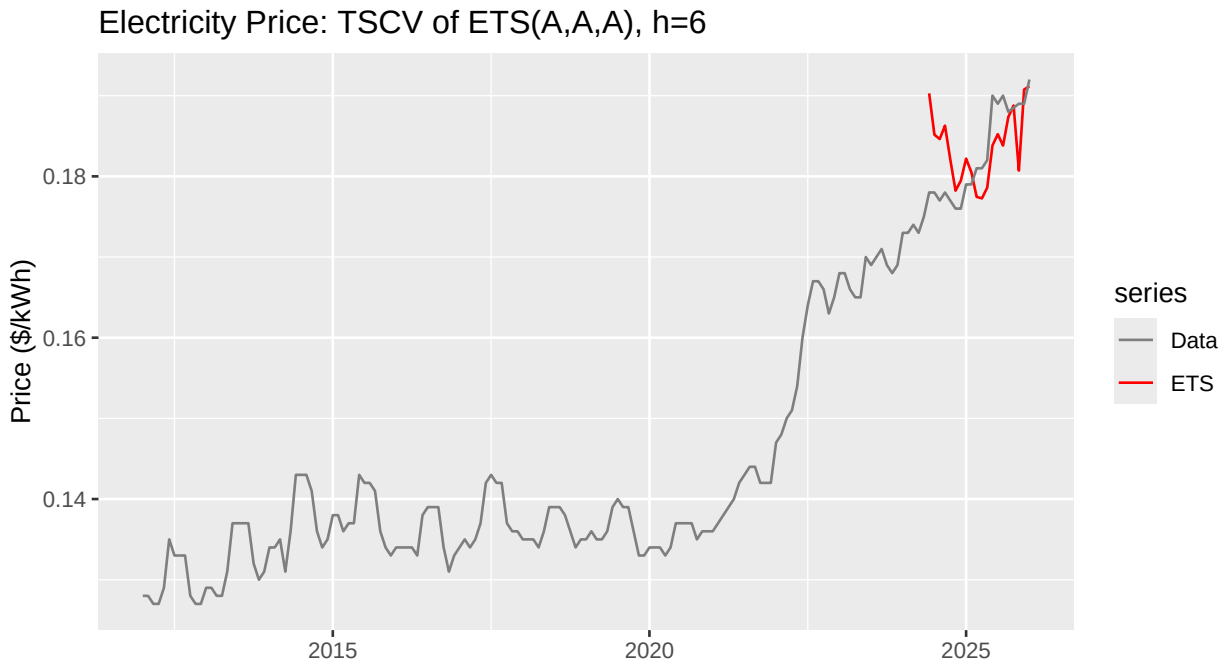
R table output: 0.1930208

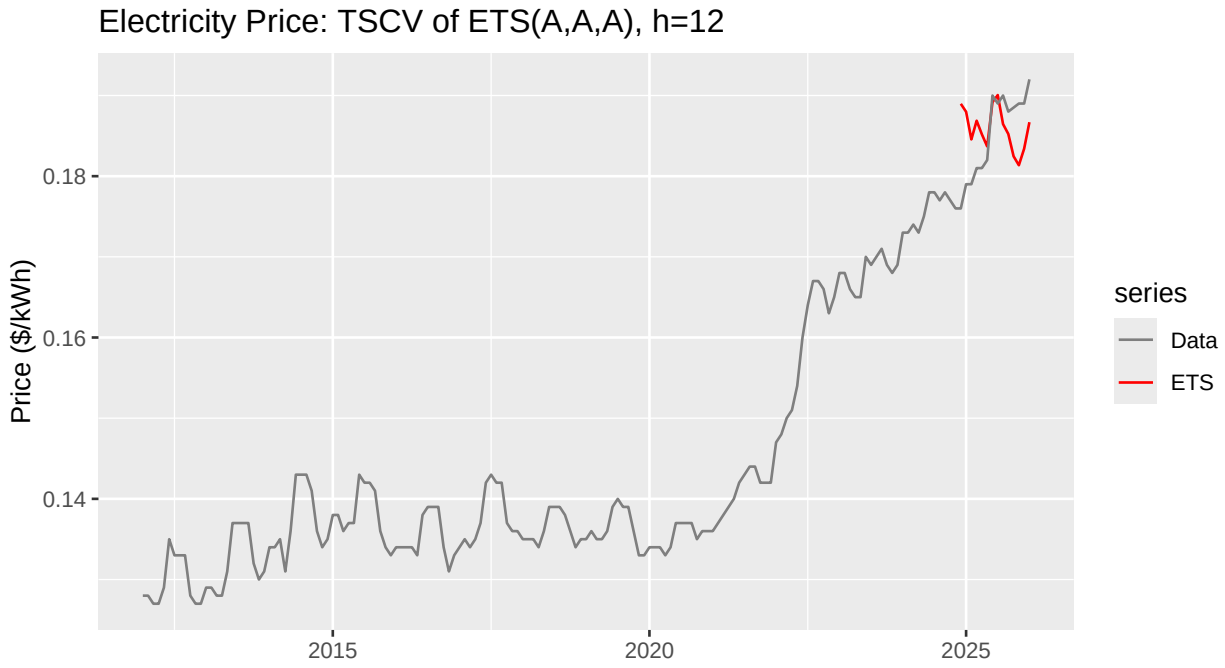
Project III

For our first model we chose ETS(A,A,A) as there is a linear upward trend in electricty prices, as a result from the Russia/Ukraine and rising demand to expand our electricty grid for the use of AI. We initally chose ETS (A,Ad,A) for our seconed model, however the parameter phi converged to 0.9799 at the 0.98 ceiling, meaning the trend was barely being dampened at all. For this reason, for our seconed model we chose ETS(M,A,M).

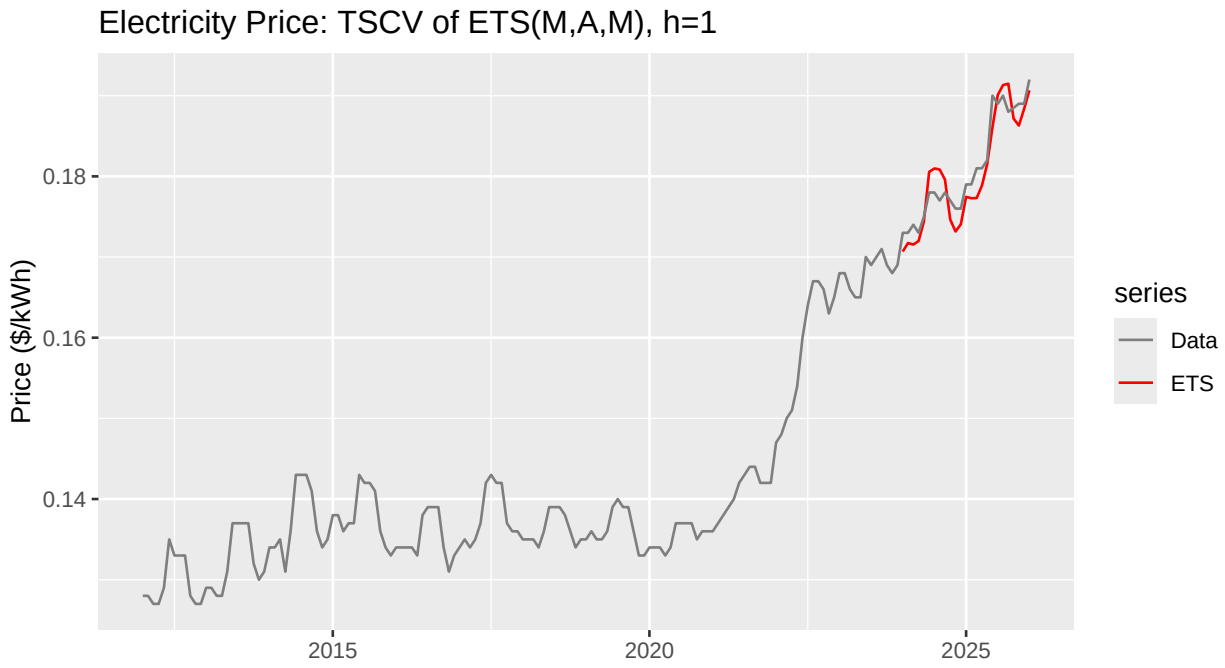


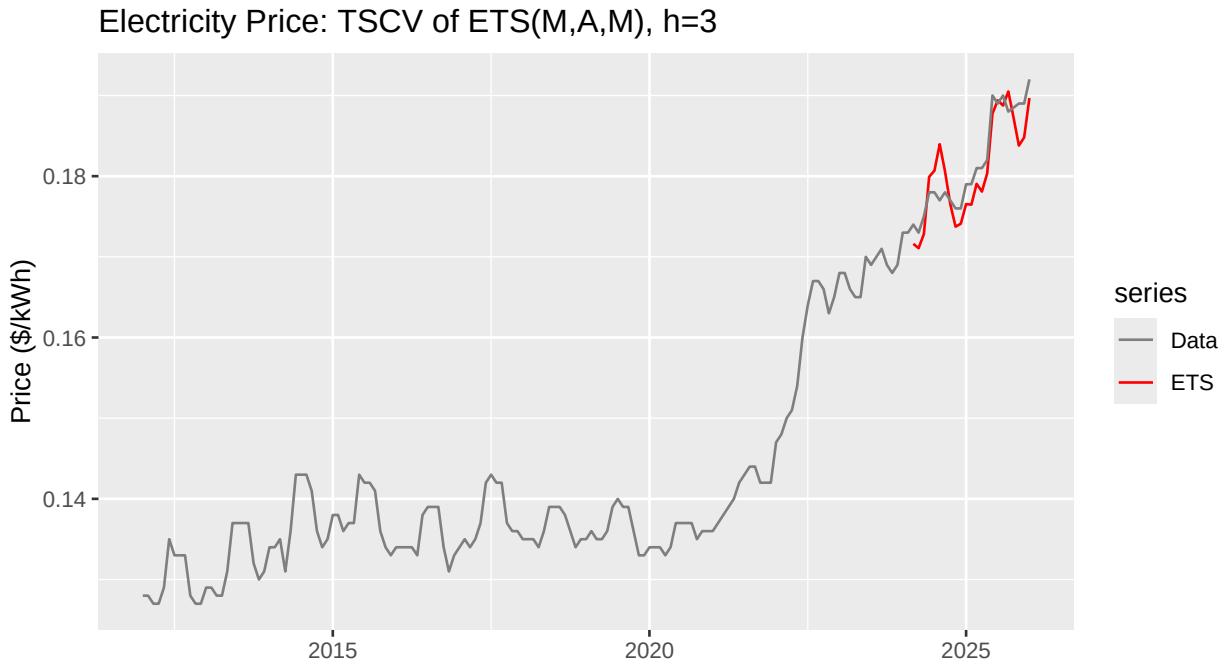


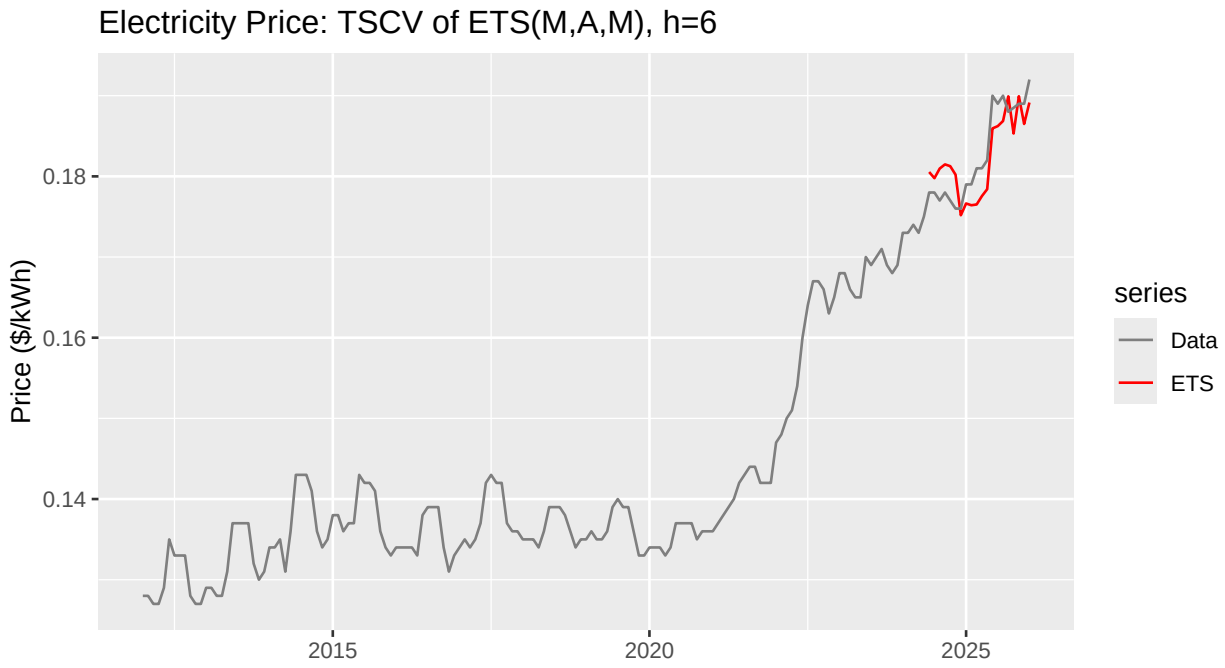


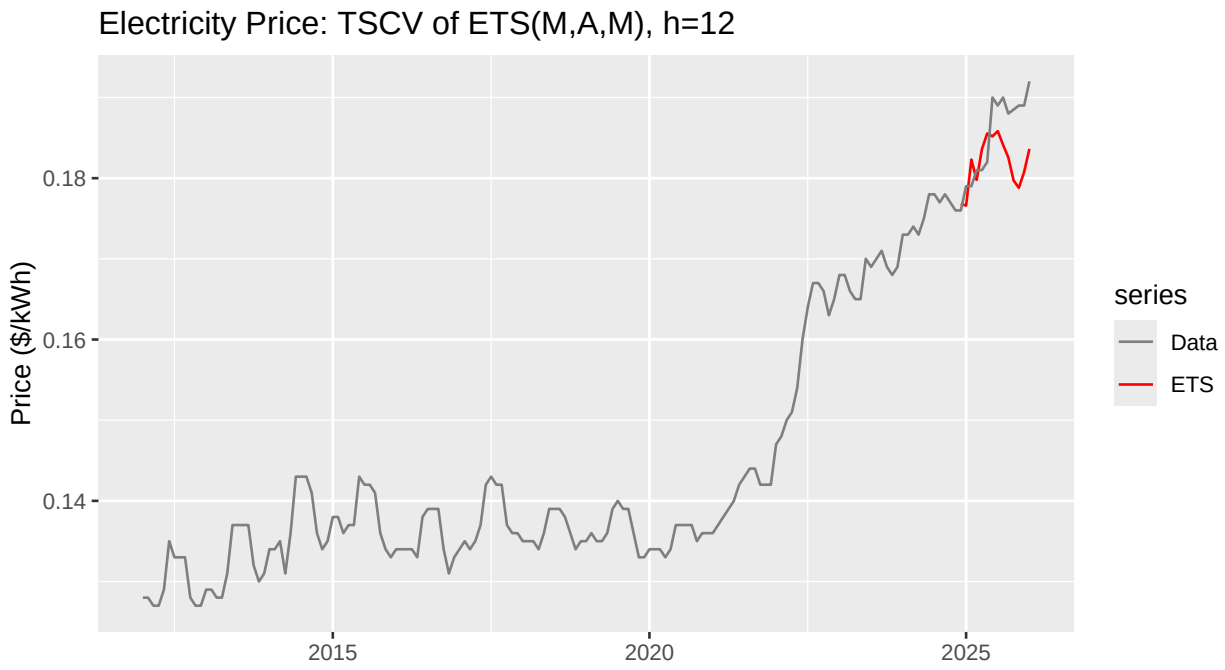



```
##           ME   RMSE    MAE      MPE   MAPE   ACF1 Theil's U
## h=1    1e-04 0.0029 0.0024  0.0185 1.3012 0.2496    0.3888
## h=3   -3e-04 0.0049 0.0039 -0.2213 2.1613 0.2948    0.6282
## h=6   -8e-04 0.0054 0.0045 -0.5078 2.4639 0.6143    0.6765
## h=12  -6e-04 0.0060 0.0051 -0.4123 2.8054 0.7294    0.6415
```


Electricity Price: TSCV of ETS(M,A,M), h=1
Price ($/kWh)
0.18
0.16
0.14
2015
2020
2025
series
Data
ETS


Electricity Price: TSCV of ETS(M,A,M), h=3
Price ($/kWh)
0.18
0.16
0.14
2015
2020
2025
series
Data
ETS






```
##           ME   RMSE    MAE    MPE   MAPE   ACF1 Theil's U
## h=1   0.0007 0.0023 0.0020 0.3876 1.1319 0.4859    0.3080
## h=3   0.0010 0.0028 0.0024 0.5168 1.3455 0.5851    0.3614
## h=6   0.0006 0.0031 0.0029 0.3207 1.6135 0.6342    0.3862
## h=12  0.0034 0.0057 0.0049 1.8057 2.6191 0.6205    0.5827

##       ETS(A,A,A) ETS(M,A,M)
```

```
## h=1      0.0185     0.3876
## h=3     -0.2213     0.5168
## h=6     -0.5078     0.3207
## h=12    -0.4123     1.8057

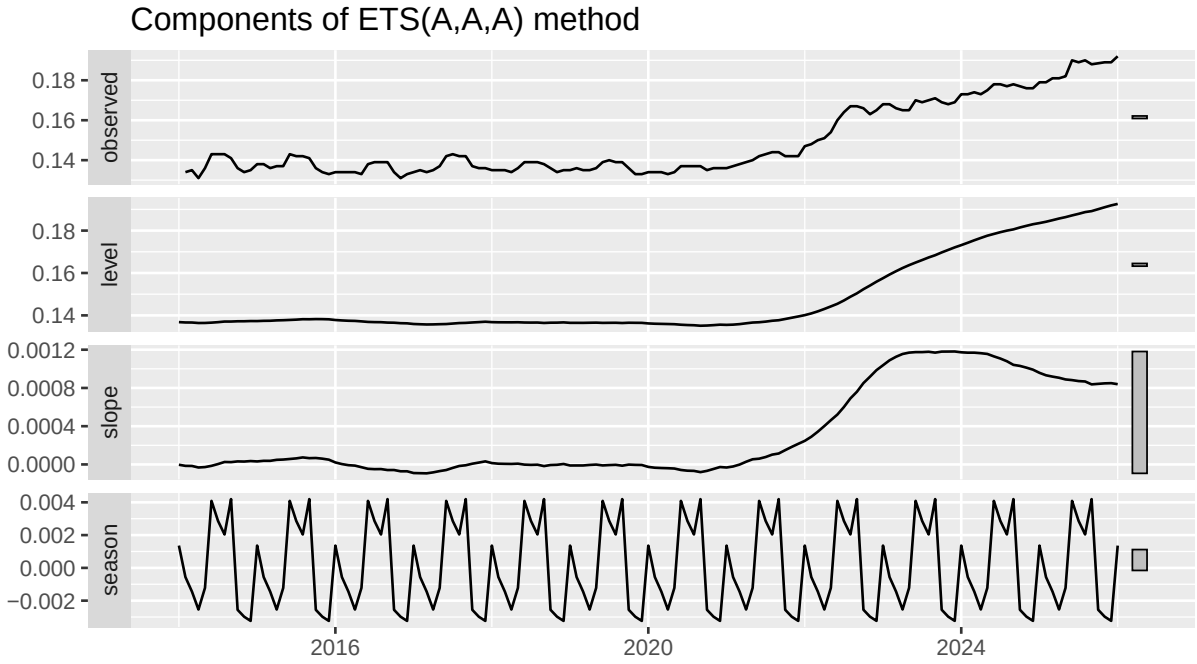
##      ETS(A,A,A) ETS(M,A,M)
## h=1      1.3012     1.1319
## h=3      2.1613     1.3455
## h=6      2.4639     1.6135
## h=12     2.8054     2.6191
```

Our ETS(M,A,M) model outperforms our ETS (A,A,A) model in the MAPE which indicates higher accuracy.The MAPE for the (M,A,M) are 1.1319 is lower than the (A,A,A) model's of 1.3012 which indicates greater accuracy in the short term. This pattern continues for h=3, h=6 and h=12 which indicates greater accuracy in the mid term also.

However the (M,A,M) model is outperformed by the (A,A,A) model in the MPE statistic. For h=1 the (M,A,M) model under estimates by more than the (A,A,A) model with a value of .3876 compared to .0185 for the (A,A,A) model. For h=3 the (M,A,M) model underestimates by more than the (A,A,A) model over estimates with values of 0.5168 and -0.2213. At the horizon of h=6 the (M,A,M) has a better performing MPE but at h=12 the (A,A,A) is far better than the (M,A,M). From these statistics we can conclude that the (M,A,M) model is more biased but has a higher accuracy compared to the (A,A,A) model.

```
## ETS(A,A,A)
##
## Call:
## ets(y = tail(price_clean, window.size), model = "AAA", damped = FALSE)
##
##   Smoothing parameters:
##     alpha = 0.0683
##     beta  = 0.0052
##     gamma = 1e-04
##
##   Initial states:
##     l = 0.1368
##     b = 0
##     s = 0.0014 -0.0032 -0.003 -0.0026 0.0042 0.002
##            0.0029 0.0041 -0.0012 -0.0025 -0.0014 -6e-04
##
##   sigma:  0.005
##
##       AIC      AICc       BIC
## -795.6405 -790.7833 -745.1536
##
## Training set error measures:
##                       ME        RMSE         MAE       MPE     MAPE      MASE
## Training set 0.001122518 0.004674642 0.003094078 0.7406679 2.021731 0.5449209
```

```
##                      ACF1
## Training set 0.8639903
```



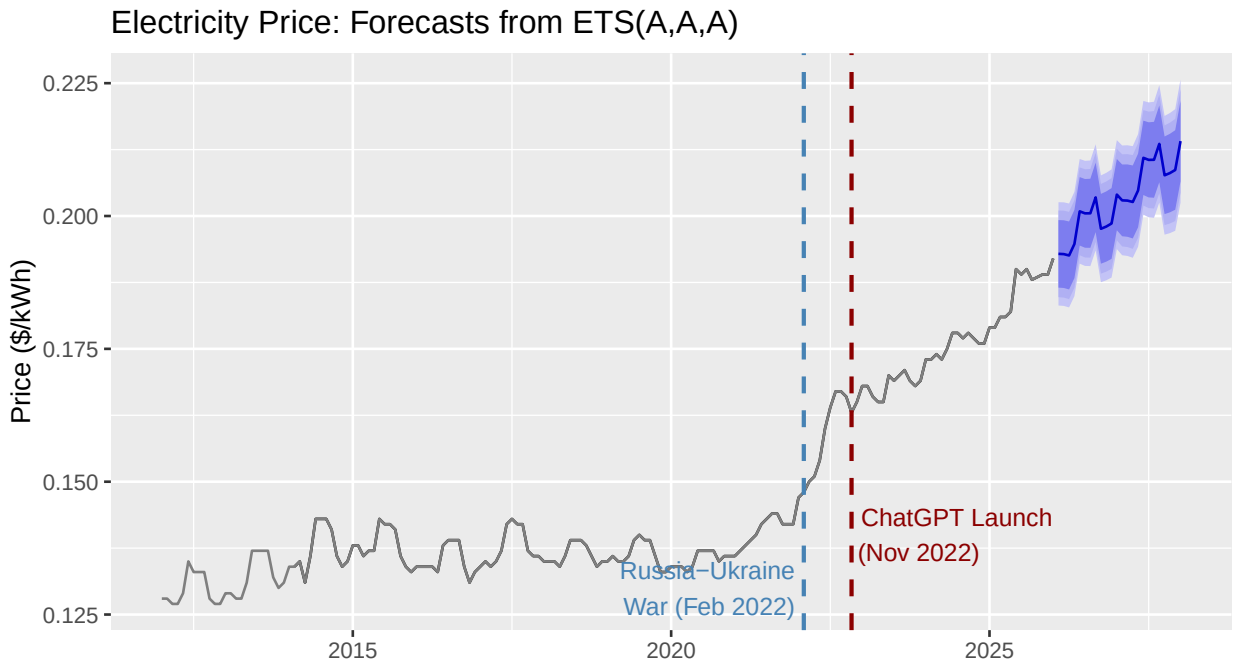
The fitted model for the (A,A,A) model shows a alpha level of .0683 which indicates almost no change in the level of data overtime. However this contradicts our data's long term trend. Due to the fact that our model is only fitted to the last 12 years it's probable that the value is skewed from more recent data.

The beta value is at .0052 which is a very small value indicating that the slope changes slowly overtime

The gamma value being at 1e-04 indicates that the seasonal variations change very slowly overtime.

The Russia-Ukraine energy shock caused extreme residuals, consistent with what the data shows. When Russia invaded Ukraine in February 2022, Russia restricted natural gas supply to Europe. Since natural gas is one of the primary fuels used to generate electricity, as the price of natural gas spiked, price of electricity followed. This remainder component is consistent with what we observed in Project 1. Another thing to note on extreme outliers is that it takes time for electricity prices to absorb these shocks, which is why we see outliers in July, August, and October, instead of February.

```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -7.266e-03 -1.629e-03  9.235e-05  1.123e-03  2.329e-03  1.739e-02
##
## Extreme Outliers (3 × IQR rule):
##       date      value
## 1 Jul 2022 0.01515411
## 2 Aug 2022 0.01734321
## 3 Oct 2022 0.01739481
```



	Point Forecast	Lo 80	Hi 80	Lo 90	Hi 90	Lo 95	Hi 95
Feb 2026	0.1928798	0.1865256	0.1992340	0.1847242	0.2010353	0.1831618	0.2025977
Mar 2026	0.1928309	0.1864595	0.1992023	0.1846533	0.2010084	0.1830867	0.2025751
Apr 2026	0.1925767	0.1861857	0.1989676	0.1843739	0.2007794	0.1828025	0.2023508
May 2026	0.1947355	0.1883223	0.2011487	0.1865042	0.2029668	0.1849273	0.2045437
Jun 2026	0.2008837	0.1944454	0.2073219	0.1926203	0.2091470	0.1910373	0.2107301
Jul 2026	0.2004995	0.1940333	0.2069656	0.1922003	0.2087986	0.1906104	0.2103885
Aug 2026	0.2005137	0.1940167	0.2070108	0.1921749	0.2088526	0.1905774	0.2104501
Sep 2026	0.2035087	0.1969776	0.2100398	0.1951262	0.2118912	0.1935203	0.2134971
Oct 2026	0.1975952	0.1910269	0.2041636	0.1891648	0.2060257	0.1875498	0.2076407
Nov 2026	0.1980130	0.1914039	0.2046221	0.1895303	0.2064957	0.1879053	0.2081207
Dec 2026	0.1985950	0.1919417	0.2052482	0.1900556	0.2071344	0.1884197	0.2087703
Jan 2027	0.2040278	0.1973268	0.2107289	0.1954272	0.2126285	0.1937795	0.2142762
Feb 2027	0.2029493	0.1961968	0.2097018	0.1942825	0.2116161	0.1926222	0.2132764
Mar 2027	0.2029004	0.1960928	0.2097081	0.1941629	0.2116380	0.1924890	0.2133119
Apr 2027	0.2026462	0.1957796	0.2095129	0.1938330	0.2114595	0.1921446	0.2131479
May 2027	0.2048051	0.1978756	0.2117345	0.1959112	0.2136990	0.1942073	0.2154028
Jun 2027	0.2109532	0.2039570	0.2179495	0.2019736	0.2199328	0.2002534	0.2216531
Jul 2027	0.2105690	0.2035020	0.2176360	0.2014987	0.2196394	0.1997610	0.2213770
Aug 2027	0.2105833	0.2034416	0.2177250	0.2014170	0.2197496	0.1996610	0.2215056
Sep 2027	0.2135783	0.2063578	0.2207987	0.2043109	0.2228456	0.2025355	0.2246210
Oct 2027	0.2076648	0.2003615	0.2149681	0.1982912	0.2170384	0.1964954	0.2188342
Nov 2027	0.2080826	0.2006924	0.2154727	0.1985974	0.2175677	0.1967803	0.2193848
Dec 2027	0.2086645	0.2011835	0.2161456	0.1990627	0.2182664	0.1972232	0.2201058

Jan 2028		0.2140974	0.2065213	0.2216735	0.2043736	0.2238212	0.2025108	0.2256840	

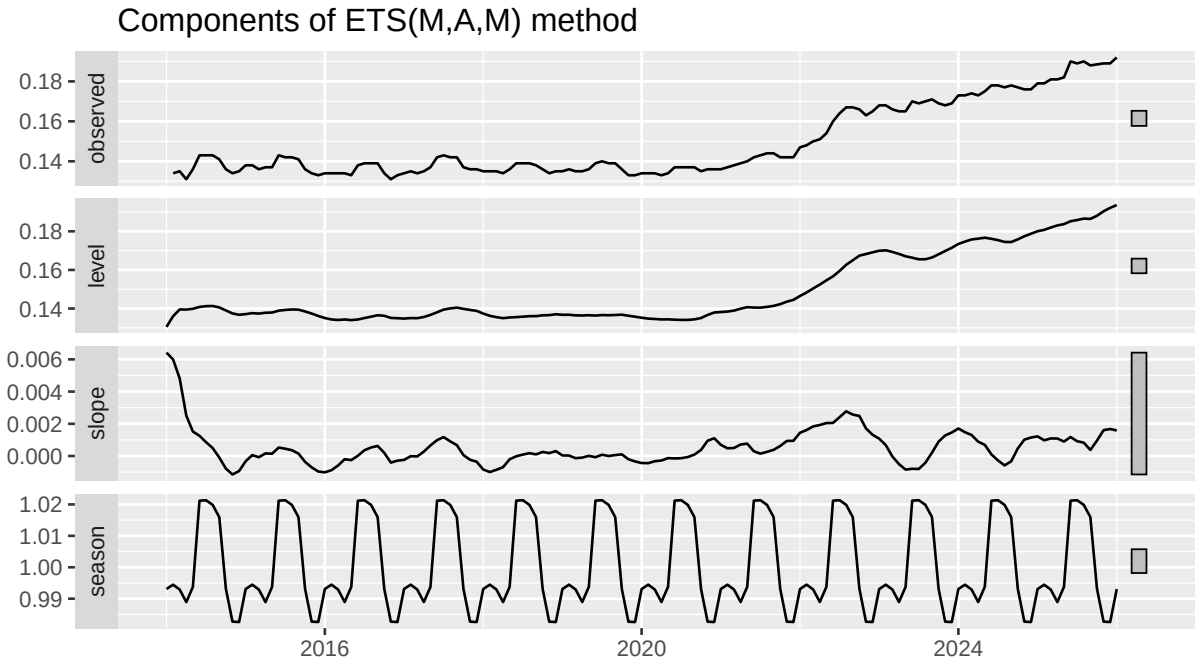
year	month	l	b	s1	s2	s3	s4	s5	s6	
2014	1	0.136818	-0.000003	0.001361	-0.003237	-0.002980	-0.002560	0.004198	0.002038	0
2014	2	0.136660	-0.000015	-0.000560	0.001361	-0.003237	-0.002980	-0.002560	0.004198	0
2014	3	0.136632	-0.000016	-0.001448	-0.000560	0.001361	-0.003237	-0.002980	-0.002560	0
2014	4	0.136405	-0.000032	-0.002542	-0.001448	-0.000560	0.001361	-0.003237	-0.002980	-0
2014	5	0.136431	-0.000028	-0.001222	-0.002542	-0.001448	-0.000560	0.001361	-0.003237	-0
2014	6	0.136575	-0.000015	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0.001361	-0
2014	7	0.136804	0.000004	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0
2014	8	0.137092	0.000026	0.002038	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0
2014	9	0.137096	0.000024	0.004198	0.002038	0.002864	0.004089	-0.001222	-0.002542	-0
2014	10	0.137219	0.000031	-0.002560	0.004198	0.002038	0.002864	0.004089	-0.001222	-0
2014	11	0.137232	0.000030	-0.002980	-0.002560	0.004198	0.002038	0.002864	0.004089	-0
2014	12	0.137328	0.000035	-0.003237	-0.002980	-0.002560	0.004198	0.002038	0.002864	0
2015	1	0.137314	0.000031	0.001361	-0.003237	-0.002980	-0.002560	0.004198	0.002038	0
2015	2	0.137428	0.000038	-0.000559	0.001361	-0.003237	-0.002980	-0.002560	0.004198	0
2015	3	0.137465	0.000038	-0.001448	-0.000559	0.001361	-0.003237	-0.002980	-0.002560	0
2015	4	0.137642	0.000048	-0.002542	-0.001448	-0.000559	0.001361	-0.003237	-0.002980	-0
2015	5	0.137726	0.000051	-0.001222	-0.002542	-0.001448	-0.000559	0.001361	-0.003237	-0
2015	6	0.137855	0.000057	0.004089	-0.001222	-0.002542	-0.001448	-0.000559	0.001361	-0
2015	7	0.137995	0.000063	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0.000559	0
2015	8	0.138189	0.000073	0.002038	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0
2015	9	0.138162	0.000066	0.004197	0.002038	0.002864	0.004089	-0.001222	-0.002542	-0
2015	10	0.138250	0.000067	-0.002560	0.004197	0.002038	0.002864	0.004089	-0.001222	-0
2015	11	0.138226	0.000060	-0.002980	-0.002560	0.004197	0.002038	0.002864	0.004089	-0
2015	12	0.138147	0.000050	-0.003237	-0.002980	-0.002560	0.004197	0.002038	0.002864	0
2016	1	0.137816	0.000021	0.001360	-0.003237	-0.002980	-0.002560	0.004197	0.002038	0
2016	2	0.137613	0.000004	-0.000560	0.001360	-0.003237	-0.002980	-0.002560	0.004197	0
2016	3	0.137469	-0.000008	-0.001448	-0.000560	0.001360	-0.003237	-0.002980	-0.002560	0
2016	4	0.137398	-0.000012	-0.002542	-0.001448	-0.000560	0.001360	-0.003237	-0.002980	-0
2016	5	0.137169	-0.000029	-0.001222	-0.002542	-0.001448	-0.000560	0.001360	-0.003237	-0
2016	6	0.136920	-0.000046	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0.001360	-0
2016	7	0.136823	-0.000050	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0
2016	8	0.136787	-0.000049	0.002038	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0
2016	9	0.136606	-0.000059	0.004197	0.002038	0.002864	0.004089	-0.001222	-0.002542	-0
2016	10	0.136548	-0.000059	-0.002560	0.004197	0.002038	0.002864	0.004089	-0.001222	-0
2016	11	0.136318	-0.000072	-0.002980	-0.002560	0.004197	0.002038	0.002864	0.004089	-0
2016	12	0.136245	-0.000072	-0.003237	-0.002980	-0.002560	0.004197	0.002038	0.002864	0
2017	1	0.135932	-0.000090	0.001360	-0.003237	-0.002980	-0.002560	0.004197	0.002038	0
2017	2	0.135822	-0.000092	-0.000560	0.001360	-0.003237	-0.002980	-0.002560	0.004197	0
2017	3	0.135711	-0.000093	-0.001448	-0.000560	0.001360	-0.003237	-0.002980	-0.002560	0
2017	4	0.135750	-0.000083	-0.002542	-0.001448	-0.000560	0.001360	-0.003237	-0.002980	-0

2017	5	0.135841	-0.000070	-0.001222	-0.002542	-0.001448	-0.000560	0.001360	-0.003237	-0
2017	6	0.135918	-0.000059	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0.001360	-0
2017	7	0.136151	-0.000036	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0
2017	8	0.136378	-0.000016	0.002039	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0
2017	9	0.136460	-0.000009	0.004197	0.002039	0.002864	0.004089	-0.001222	-0.002542	-0
2017	10	0.136664	0.000007	-0.002560	0.004197	0.002039	0.002864	0.004089	-0.001222	-0
2017	11	0.136829	0.000019	-0.002980	-0.002560	0.004197	0.002039	0.002864	0.004089	-0
2017	12	0.137012	0.000032	-0.003237	-0.002980	-0.002560	0.004197	0.002039	0.002864	0
2018	1	0.136811	0.000014	0.001359	-0.003237	-0.002980	-0.002560	0.004197	0.002039	0
2018	2	0.136739	0.000008	-0.000560	0.001359	-0.003237	-0.002980	-0.002560	0.004197	0
2018	3	0.136726	0.000006	-0.001448	-0.000560	0.001359	-0.003237	-0.002980	-0.002560	0
2018	4	0.136719	0.000005	-0.002542	-0.001448	-0.000560	0.001359	-0.003237	-0.002980	-0
2018	5	0.136758	0.000008	-0.001222	-0.002542	-0.001448	-0.000560	0.001359	-0.003237	-0
2018	6	0.136639	-0.000002	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0.001359	-0
2018	7	0.136602	-0.000005	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0.000560	0
2018	8	0.136623	-0.000003	0.002039	0.002864	0.004089	-0.001222	-0.002542	-0.001448	-0
2018	9	0.136427	-0.000017	0.004197	0.002039	0.002864	0.004089	-0.001222	-0.002542	-0
2018	10	0.136557	-0.000006	-0.002559	0.004197	0.002039	0.002864	0.004089	-0.001222	-0
2018	11	0.136580	-0.000004	-0.002980	-0.002559	0.004197	0.002039	0.002864	0.004089	-0
2018	12	0.136689	0.000005	-0.003237	-0.002980	-0.002559	0.004197	0.002039	0.002864	0
2019	1	0.136485	-0.000011	0.001359	-0.003237	-0.002980	-0.002559	0.004197	0.002039	0
2019	2	0.136480	-0.000011	-0.000560	0.001359	-0.003237	-0.002980	-0.002559	0.004197	0
2019	3	0.136468	-0.000011	-0.001448	-0.000560	0.001359	-0.003237	-0.002980	-0.002559	0
2019	4	0.136531	-0.000005	-0.002542	-0.001448	-0.000560	0.001359	-0.003237	-0.002980	-0
2019	5	0.136573	-0.000002	-0.001222	-0.002542	-0.001448	-0.000560	0.001359	-0.003237	-0
2019	6	0.136458	-0.000010	0.004088	-0.001222	-0.002542	-0.001448	-0.000560	0.001359	-0
2019	7	0.136495	-0.000007	0.002864	0.004088	-0.001222	-0.002542	-0.001448	-0.000560	0
2019	8	0.136520	-0.000004	0.002039	0.002864	0.004088	-0.001222	-0.002542	-0.001448	-0
2019	9	0.136399	-0.000013	0.004197	0.002039	0.002864	0.004088	-0.001222	-0.002542	-0
2019	10	0.136534	-0.000002	-0.002559	0.004197	0.002039	0.002864	0.004088	-0.001222	-0
2019	11	0.136495	-0.000005	-0.002980	-0.002559	0.004197	0.002039	0.002864	0.004088	-0
2019	12	0.136473	-0.000006	-0.003237	-0.002980	-0.002559	0.004197	0.002039	0.002864	0
2020	1	0.136205	-0.000026	0.001359	-0.003237	-0.002980	-0.002559	0.004197	0.002039	0
2020	2	0.136069	-0.000034	-0.000560	0.001359	-0.003237	-0.002980	-0.002559	0.004197	0
2020	3	0.135994	-0.000037	-0.001449	-0.000560	0.001359	-0.003237	-0.002980	-0.002559	0
2020	4	0.135928	-0.000040	-0.002542	-0.001449	-0.000560	0.001359	-0.003237	-0.002980	-0
2020	5	0.135843	-0.000043	-0.001222	-0.002542	-0.001449	-0.000560	0.001359	-0.003237	-0
2020	6	0.135603	-0.000058	0.004088	-0.001222	-0.002542	-0.001449	-0.000560	0.001359	-0
2020	7	0.135448	-0.000066	0.002864	0.004088	-0.001222	-0.002542	-0.001449	-0.000560	0
2020	8	0.135354	-0.000068	0.002039	0.002864	0.004088	-0.001222	-0.002542	-0.001449	-0
2020	9	0.135116	-0.000081	0.004197	0.002039	0.002864	0.004088	-0.001222	-0.002542	-0
2020	10	0.135208	-0.000068	-0.002559	0.004197	0.002039	0.002864	0.004088	-0.001222	-0
2020	11	0.135403	-0.000047	-0.002979	-0.002559	0.004197	0.002039	0.002864	0.004088	-0
2020	12	0.135621	-0.000027	-0.003236	-0.002979	-0.002559	0.004197	0.002039	0.002864	0

2021	1	0.135528	-0.000032	0.001359	-0.003236	-0.002979	-0.002559	0.004197	0.002039
2021	2	0.135637	-0.000021	-0.000560	0.001359	-0.003236	-0.002979	-0.002559	0.004197
2021	3	0.135878	-0.000001	-0.001448	-0.000560	0.001359	-0.003236	-0.002979	-0.002559
2021	4	0.136263	0.000028	-0.002541	-0.001448	-0.000560	0.001359	-0.003236	-0.002979
2021	5	0.136628	0.000054	-0.001221	-0.002541	-0.001448	-0.000560	0.001359	-0.003236
2021	6	0.136766	0.000060	0.004088	-0.001221	-0.002541	-0.001448	-0.000560	0.001359
2021	7	0.137052	0.000077	0.002865	0.004088	-0.001221	-0.002541	-0.001448	-0.000560
2021	8	0.137460	0.000103	0.002039	0.002865	0.004088	-0.001221	-0.002541	-0.001448
2021	9	0.137716	0.000114	0.004197	0.002039	0.002865	0.004088	-0.001221	-0.002541
2021	10	0.138290	0.000149	-0.002558	0.004197	0.002039	0.002865	0.004088	-0.001221
2021	11	0.138886	0.000183	-0.002979	-0.002558	0.004197	0.002039	0.002865	0.004088
2021	12	0.139491	0.000216	-0.003236	-0.002979	-0.002558	0.004197	0.002039	0.002865
2022	1	0.140112	0.000246	0.001359	-0.003236	-0.002979	-0.002558	0.004197	0.002039
2022	2	0.140919	0.000289	-0.000559	0.001359	-0.003236	-0.002979	-0.002558	0.004197
2022	3	0.141908	0.000343	-0.001447	-0.000559	0.001359	-0.003236	-0.002979	-0.002558
2022	4	0.143022	0.000401	-0.002540	-0.001447	-0.000559	0.001359	-0.003236	-0.002979
2022	5	0.144230	0.000463	-0.001220	-0.002540	-0.001447	-0.000559	0.001359	-0.003236
2022	6	0.145460	0.000521	0.004090	-0.001220	-0.002540	-0.001447	-0.000559	0.001359
2022	7	0.147017	0.000600	0.002866	0.004090	-0.001220	-0.002540	-0.001447	-0.000559
2022	8	0.148803	0.000691	0.002041	0.002866	0.004090	-0.001220	-0.002540	-0.001447
2022	9	0.150403	0.000760	0.004198	0.002041	0.002866	0.004090	-0.001220	-0.002540
2022	10	0.152352	0.000851	-0.002556	0.004198	0.002041	0.002866	0.004090	-0.001220
2022	11	0.154076	0.000917	-0.002977	-0.002556	0.004198	0.002041	0.002866	0.004090
2022	12	0.155898	0.000986	-0.003234	-0.002977	-0.002556	0.004198	0.002041	0.002866
2023	1	0.157552	0.001037	0.001360	-0.003234	-0.002977	-0.002556	0.004198	0.002041
2023	2	0.159270	0.001089	-0.000558	0.001360	-0.003234	-0.002977	-0.002556	0.004198
2023	3	0.160844	0.001126	-0.001446	-0.000558	0.001360	-0.003234	-0.002977	-0.002556
2023	4	0.162351	0.001155	-0.002539	-0.001446	-0.000558	0.001360	-0.003234	-0.002977
2023	5	0.163692	0.001169	-0.001220	-0.002539	-0.001446	-0.000558	0.001360	-0.003234
2023	6	0.164933	0.001175	0.004090	-0.001220	-0.002539	-0.001446	-0.000558	0.001360
2023	7	0.166109	0.001175	0.002866	0.004090	-0.001220	-0.002539	-0.001446	-0.000558
2023	8	0.167331	0.001179	0.002041	0.002866	0.004090	-0.001220	-0.002539	-0.001446
2023	9	0.168392	0.001170	0.004198	0.002041	0.002866	0.004090	-0.001220	-0.002539
2023	10	0.169698	0.001180	-0.002556	0.004198	0.002041	0.002866	0.004090	-0.001220
2023	11	0.170885	0.001181	-0.002977	-0.002556	0.004198	0.002041	0.002866	0.004090
2023	12	0.172077	0.001181	-0.003234	-0.002977	-0.002556	0.004198	0.002041	0.002866
2024	1	0.173148	0.001173	0.001360	-0.003234	-0.002977	-0.002556	0.004198	0.002041
2024	2	0.174269	0.001169	-0.000558	0.001360	-0.003234	-0.002977	-0.002556	0.004198
2024	3	0.175438	0.001169	-0.001446	-0.000558	0.001360	-0.003234	-0.002977	-0.002556
2024	4	0.176534	0.001163	-0.002540	-0.001446	-0.000558	0.001360	-0.003234	-0.002977
2024	5	0.177597	0.001156	-0.001220	-0.002540	-0.001446	-0.000558	0.001360	-0.003234
2024	6	0.178422	0.001131	0.004089	-0.001220	-0.002540	-0.001446	-0.000558	0.001360
2024	7	0.179250	0.001108	0.002866	0.004089	-0.001220	-0.002540	-0.001446	-0.000558
2024	8	0.179989	0.001079	0.002041	0.002866	0.004089	-0.001220	-0.002540	-0.001446

2024	9	0.180572	0.001041	0.004197	0.002041	0.002866	0.004089	-0.001220	-0.002540	-0
2024	10	0.181472	0.001031	-0.002556	0.004197	0.002041	0.002866	0.004089	-0.001220	-0
2024	11	0.182262	0.001012	-0.002977	-0.002556	0.004197	0.002041	0.002866	0.004089	-0
2024	12	0.182998	0.000991	-0.003235	-0.002977	-0.002556	0.004197	0.002041	0.002866	0
2025	1	0.183556	0.000958	0.001359	-0.003235	-0.002977	-0.002556	0.004197	0.002041	0
2025	2	0.184175	0.000932	-0.000558	0.001359	-0.003235	-0.002977	-0.002556	0.004197	0
2025	3	0.184926	0.000919	-0.001446	-0.000558	0.001359	-0.003235	-0.002977	-0.002556	0
2025	4	0.185687	0.000907	-0.002540	-0.001446	-0.000558	0.001359	-0.003235	-0.002977	-0
2025	5	0.186363	0.000889	-0.001220	-0.002540	-0.001446	-0.000558	0.001359	-0.003235	-0
2025	6	0.187160	0.000882	0.004089	-0.001220	-0.002540	-0.001446	-0.000558	0.001359	-0
2025	7	0.187912	0.000872	0.002866	0.004089	-0.001220	-0.002540	-0.001446	-0.000558	0
2025	8	0.188727	0.000868	0.002041	0.002866	0.004089	-0.001220	-0.002540	-0.001446	-0
2025	9	0.189199	0.000838	0.004197	0.002041	0.002866	0.004089	-0.001220	-0.002540	-0
2025	10	0.190106	0.000843	-0.002556	0.004197	0.002041	0.002866	0.004089	-0.001220	-0
2025	11	0.191019	0.000848	-0.002977	-0.002556	0.004197	0.002041	0.002866	0.004089	-0
2025	12	0.191893	0.000850	-0.003235	-0.002977	-0.002556	0.004197	0.002041	0.002866	0
2026	1	0.192599	0.000839	0.001359	-0.003235	-0.002977	-0.002556	0.004197	0.002041	0

```
## ETS(M,A,M)
##
## Call:
## ets(y = tail(price_clean, window.size), model = "MAM", damped = FALSE)
##
##   Smoothing parameters:
##     alpha = 0.4118
##     beta  = 0.1942
##     gamma = 5e-04
##
##   Initial states:
##     l = 0.1305
##     b = 0.0064
##     s = 0.9931 0.9826 0.9827 0.9932 1.016 1.0199
##            1.0213 1.0212 0.9938 0.989 0.9929 0.9945
##
##   sigma:  0.0142
##
##        AIC       AICc        BIC
## -1042.1151 -1037.2580  -991.6283
##
## Training set error measures:
##                          ME        RMSE         MAE       MPE      MAPE
## Training set -0.0001717469 0.001983707 0.001422429 -0.132189 0.9610181
##                    MASE      ACF1
## Training set 0.2505144 0.6094306
```



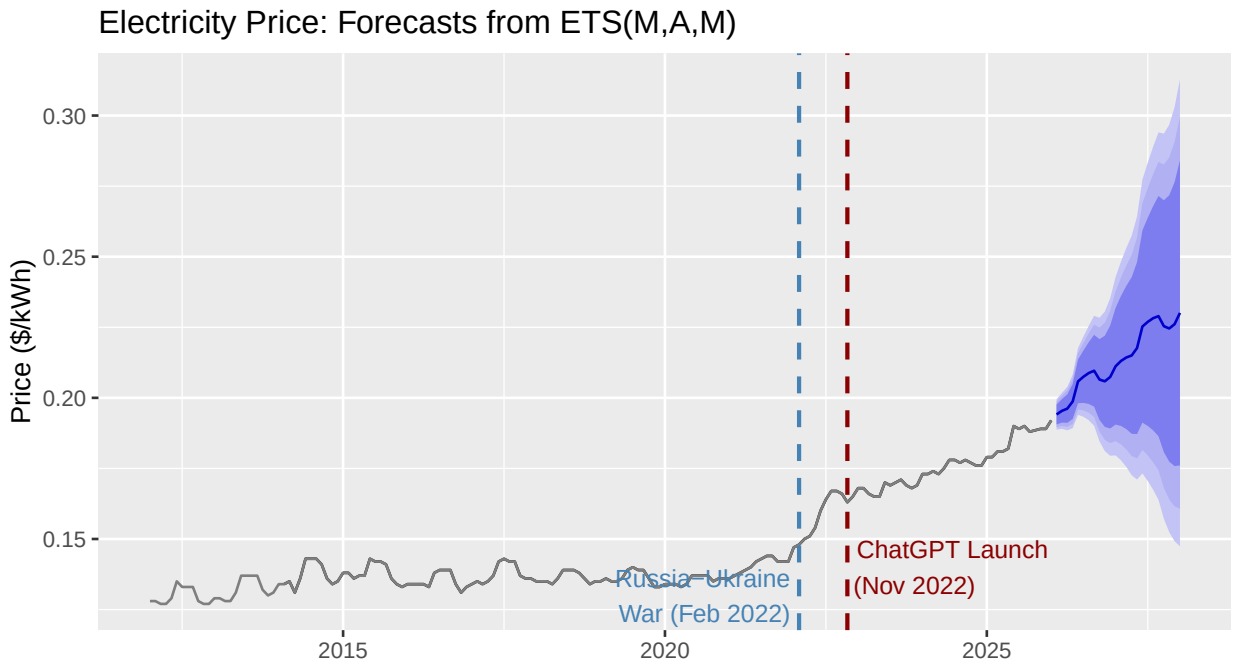
alpha = 0.4118: the level updates at a moderate pace slower than ETS(A,A,A), which indicates.

beta = 0.1942: the trend updates over time, which is the key difference from ETS(A,A,A). The model allows the slope to evolve, which captures the steeper post-2022 trend that our fixed trend in Project 1 would miss.

gamma = 5e-0 signals the seasonal component is nearly fixed, which is consistent with stable seasonality.

sigma = 0.0142 vs 0.005 for AAA; the larger sigma reflects wider multiplicative uncertainty, explaining the wider forecast bands you see in the graph.

```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0823275 -0.0078370  0.0003851 -0.0011371  0.0066013  0.0236242
##
## Extreme Outliers (3 × IQR rule):
##       date       value
## 1 Apr 2014 -0.08232752
```



	Point Forecast	Lo 80	Hi 80	Lo 90	Hi 90	Lo 95	Hi 95
Feb 2026	0.1941166	0.1905815	0.1976516	0.1895794	0.1986538	0.1887102	0.1995230
Mar 2026	0.1953907	0.1912390	0.1995424	0.1900620	0.2007194	0.1890412	0.2017402
Apr 2026	0.1961772	0.1911482	0.2012062	0.1897225	0.2026319	0.1884860	0.2038684
May 2026	0.1987112	0.1925230	0.2048994	0.1907687	0.2066537	0.1892472	0.2081753
Jun 2026	0.2058177	0.1980903	0.2135451	0.1958997	0.2157357	0.1939997	0.2176357
Jul 2026	0.2074666	0.1981981	0.2167351	0.1955706	0.2193626	0.1932916	0.2216415
Aug 2026	0.2087857	0.1978433	0.2197282	0.1947412	0.2228302	0.1920507	0.2255207
Sep 2026	0.2095998	0.1968847	0.2223149	0.1932802	0.2259194	0.1901537	0.2290458
Oct 2026	0.2064726	0.1921493	0.2207960	0.1880888	0.2248564	0.1845670	0.2283783
Nov 2026	0.2058578	0.1897011	0.2220145	0.1851209	0.2265947	0.1811483	0.2305673
Dec 2026	0.2073916	0.1891486	0.2256346	0.1839769	0.2308062	0.1794913	0.2352918
Jan 2027	0.2111833	0.1905342	0.2318325	0.1846804	0.2376862	0.1796032	0.2427635
Feb 2027	0.2130544	0.1900646	0.2360442	0.1835474	0.2425615	0.1778946	0.2482143
Mar 2027	0.2142991	0.1889444	0.2396539	0.1817566	0.2468417	0.1755223	0.2530759
Apr 2027	0.2150099	0.1872754	0.2427443	0.1794130	0.2506067	0.1725936	0.2574261
May 2027	0.2176358	0.1871844	0.2480872	0.1785518	0.2567197	0.1710644	0.2642072
Jun 2027	0.2252647	0.1912311	0.2592983	0.1815831	0.2689463	0.1732149	0.2773146
Jul 2027	0.2269162	0.1900494	0.2637831	0.1795982	0.2742343	0.1705333	0.2832992
Aug 2027	0.2282073	0.1884841	0.2679305	0.1772231	0.2791915	0.1674559	0.2889587
Sep 2027	0.2289471	0.1863941	0.2715002	0.1743309	0.2835634	0.1638679	0.2940264
Oct 2027	0.2253858	0.1807926	0.2699790	0.1681511	0.2826206	0.1571864	0.2935852
Nov 2027	0.2245718	0.1774065	0.2717372	0.1640358	0.2851079	0.1524387	0.2967050
Dec 2027	0.2261033	0.1758251	0.2763814	0.1615720	0.2906346	0.1492095	0.3029970

Jan 2028		0.2300950	0.1760504	0.2841395	0.1607295	0.2994604	0.1474410	0.3127490		

year	month	l	b	s1	s2	s3	s4	s5	s6	
2014	1	0.130508	0.006420	0.993081	0.982578	0.982712	0.993159	1.015967	1.019857	1.0213
2014	2	0.136029	0.005996	0.994459	0.993081	0.982578	0.982712	0.993159	1.015967	1.0198
2014	3	0.139528	0.004818	0.992913	0.994459	0.993081	0.982578	0.982712	0.993159	1.0159
2014	4	0.139452	0.002510	0.988916	0.992913	0.994459	0.993081	0.982578	0.982712	0.9931
2014	5	0.139859	0.001518	0.993743	0.988916	0.992913	0.994459	0.993081	0.982578	0.9827
2014	6	0.140824	0.001257	1.021183	0.993743	0.988916	0.992913	0.994459	0.993081	0.9825
2014	7	0.141230	0.000856	1.021322	1.021183	0.993743	0.988916	0.992913	0.994459	0.9930
2014	8	0.141315	0.000493	1.019850	1.021322	1.021183	0.993743	0.988916	0.992913	0.9944
2014	9	0.140563	-0.000095	1.015955	1.019850	1.021322	1.021183	0.993743	0.988916	0.9929
2014	10	0.139014	-0.000780	0.993146	1.015955	1.019850	1.021322	1.021183	0.993743	0.9889
2014	11	0.137461	-0.001145	0.982705	0.993146	1.015955	1.019850	1.021322	1.021183	0.9937
2014	12	0.136760	-0.000936	0.982583	0.982705	0.993146	1.015955	1.019850	1.021322	1.0211
2015	1	0.137116	-0.000326	0.993094	0.982583	0.982705	0.993146	1.015955	1.019850	1.0213
2015	2	0.137605	0.000058	0.994466	0.993094	0.982583	0.982705	0.993146	1.015955	1.0198
2015	3	0.137378	-0.000076	0.992911	0.994466	0.993094	0.982583	0.982705	0.993146	1.0159
2015	4	0.137810	0.000163	0.988921	0.992911	0.994466	0.993094	0.982583	0.982705	0.9931
2015	5	0.137927	0.000142	0.993742	0.988921	0.992911	0.994466	0.993094	0.982583	0.9827
2015	6	0.138878	0.000523	1.021191	0.993742	0.988921	0.992911	0.994466	0.993094	0.9825
2015	7	0.139251	0.000452	1.021320	1.021191	0.993742	0.988921	0.992911	0.994466	0.9930
2015	8	0.139511	0.000362	1.019848	1.021320	1.021191	0.993742	0.988921	0.992911	0.9944
2015	9	0.139425	0.000151	1.015951	1.019848	1.021320	1.021191	0.993742	0.988921	0.9929
2015	10	0.138490	-0.000362	0.993135	1.015951	1.019848	1.021320	1.021191	0.993742	0.9889
2015	11	0.137399	-0.000705	0.982698	0.993135	1.015951	1.019848	1.021320	1.021191	0.9937
2015	12	0.136144	-0.000965	0.982577	0.982698	0.993135	1.015951	1.019848	1.021320	1.0211
2016	1	0.135077	-0.001013	0.993093	0.982577	0.982698	0.993135	1.015951	1.019848	1.0213
2016	2	0.134345	-0.000880	0.994469	0.993093	0.982577	0.982698	0.993135	1.015951	1.0198
2016	3	0.134079	-0.000591	0.992917	0.994469	0.993093	0.982577	0.982698	0.993135	1.0159
2016	4	0.134317	-0.000200	0.988929	0.992917	0.994469	0.993093	0.982577	0.982698	0.9931
2016	5	0.134002	-0.000254	0.993741	0.988929	0.992917	0.994469	0.993093	0.982577	0.9826
2016	6	0.134320	0.000015	1.021197	0.993741	0.988929	0.992917	0.994469	0.993093	0.9825
2016	7	0.135061	0.000358	1.021328	1.021197	0.993741	0.988929	0.992917	0.994469	0.9930
2016	8	0.135780	0.000528	1.019852	1.021328	1.021197	0.993741	0.988929	0.992917	0.9944
2016	9	0.136518	0.000627	1.015953	1.019852	1.021328	1.021197	0.993741	0.988929	0.9929
2016	10	0.136231	0.000196	0.993127	1.015953	1.019852	1.021328	1.021197	0.993741	0.9889
2016	11	0.135142	-0.000410	0.982686	0.993127	1.015953	1.019852	1.021328	1.021197	0.9937
2016	12	0.134990	-0.000288	0.982580	0.982686	0.993127	1.015953	1.019852	1.021328	1.0211
2017	1	0.134797	-0.000244	0.993094	0.982580	0.982686	0.993127	1.015953	1.019852	1.0213
2017	2	0.135046	-0.000011	0.994474	0.993094	0.982580	0.982686	0.993127	1.015953	1.0198
2017	3	0.135003	-0.000026	0.992916	0.994474	0.993094	0.982580	0.982686	0.993127	1.0159
2017	4	0.135608	0.000272	0.988935	0.992916	0.994474	0.993094	0.982580	0.982686	0.9931

2017	5	0.136697	0.000657	0.993749	0.988935	0.992916	0.994474	0.993094	0.982580	0.9826
2017	6	0.138053	0.000987	1.021203	0.993749	0.988935	0.992916	0.994474	0.993094	0.9825
2017	7	0.139441	0.001176	1.021331	1.021203	0.993749	0.988935	0.992916	0.994474	0.9930
2017	8	0.140048	0.000908	1.019846	1.021331	1.021203	0.993749	0.988935	0.992916	0.9944
2017	9	0.140468	0.000678	1.015948	1.019846	1.021331	1.021203	0.993749	0.988935	0.9929
2017	10	0.139829	0.000057	0.993114	1.015948	1.019846	1.021331	1.021203	0.993749	0.9889
2017	11	0.139272	-0.000233	0.982680	0.993114	1.015948	1.019846	1.021331	1.021203	0.9937
2017	12	0.138781	-0.000355	0.982577	0.982680	0.993114	1.015948	1.019846	1.021331	1.0212
2018	1	0.137402	-0.000838	0.993084	0.982577	0.982680	0.993114	1.015948	1.019846	1.0213
2018	2	0.136229	-0.000996	0.994471	0.993084	0.982577	0.982680	0.993114	1.015948	1.0198
2018	3	0.135534	-0.000854	0.992919	0.994471	0.993084	0.982577	0.982680	0.993114	1.0159
2018	4	0.135017	-0.000695	0.988939	0.992919	0.994471	0.993084	0.982577	0.982680	0.9931
2018	5	0.135365	-0.000203	0.993759	0.988939	0.992919	0.994471	0.993084	0.982577	0.9826
2018	6	0.135554	-0.000018	1.021207	0.993759	0.988939	0.992919	0.994471	0.993084	0.9825
2018	7	0.135767	0.000091	1.021334	1.021207	0.993759	0.988939	0.992919	0.994471	0.9930
2018	8	0.136038	0.000176	1.019848	1.021334	1.021207	0.993759	0.988939	0.992919	0.9944
2018	9	0.136057	0.000102	1.015947	1.019848	1.021334	1.021207	0.993759	0.988939	0.9929
2018	10	0.136482	0.000254	0.993117	1.015947	1.019848	1.021334	1.021207	0.993759	0.9889
2018	11	0.136582	0.000181	0.982679	0.993117	1.015947	1.019848	1.021334	1.021207	0.9937
2018	12	0.137023	0.000304	0.982580	0.982679	0.993117	1.015947	1.019848	1.021334	1.0212
2019	1	0.136756	0.000035	0.993079	0.982580	0.982679	0.993117	1.015947	1.019848	1.0213
2019	2	0.136776	0.000028	0.994471	0.993079	0.982580	0.982679	0.993117	1.015947	1.0198
2019	3	0.136458	-0.000135	0.992916	0.994471	0.993079	0.982580	0.982679	0.993117	1.0159
2019	4	0.136400	-0.000099	0.988939	0.992916	0.994471	0.993079	0.982580	0.982679	0.9931
2019	5	0.136528	0.000008	0.993761	0.988939	0.992916	0.994471	0.993079	0.982580	0.9826
2019	6	0.136363	-0.000074	1.021206	0.993761	0.988939	0.992916	0.994471	0.993079	0.9825
2019	7	0.136613	0.000079	1.021337	1.021206	0.993761	0.988939	0.992916	0.994471	0.9930
2019	8	0.136528	0.000002	1.019846	1.021337	1.021206	0.993761	0.988939	0.992916	0.9944
2019	9	0.136649	0.000058	1.015948	1.019846	1.021337	1.021206	0.993761	0.988939	0.9929
2019	10	0.136804	0.000104	0.993118	1.015948	1.019846	1.021337	1.021206	0.993761	0.9889
2019	11	0.136264	-0.000200	0.982672	0.993118	1.015948	1.019846	1.021337	1.021206	0.9937
2019	12	0.135773	-0.000337	0.982577	0.982672	0.993118	1.015948	1.019846	1.021337	1.0212
2020	1	0.135229	-0.000435	0.993077	0.982577	0.982672	0.993118	1.015948	1.019846	1.0213
2020	2	0.134774	-0.000444	0.994470	0.993077	0.982577	0.982672	0.993118	1.015948	1.0198
2020	3	0.134588	-0.000323	0.992918	0.994470	0.993077	0.982577	0.982672	0.993118	1.0159
2020	4	0.134357	-0.000279	0.988940	0.992918	0.994470	0.993077	0.982577	0.982672	0.9931
2020	5	0.134392	-0.000131	0.993764	0.988940	0.992918	0.994470	0.993077	0.982577	0.9826
2020	6	0.134217	-0.000152	1.021205	0.993764	0.988940	0.992918	0.994470	0.993077	0.9825
2020	7	0.134095	-0.000138	1.021337	1.021205	0.993764	0.988940	0.992918	0.994470	0.9930
2020	8	0.134113	-0.000065	1.019848	1.021337	1.021205	0.993764	0.988940	0.992918	0.9944
2020	9	0.134378	0.000091	1.015951	1.019848	1.021337	1.021205	0.993764	0.988940	0.9929
2020	10	0.135073	0.000376	0.993124	1.015951	1.019848	1.021337	1.021205	0.993764	0.9889
2020	11	0.136663	0.000949	0.982684	0.993124	1.015951	1.019848	1.021337	1.021205	0.9937
2020	12	0.137941	0.001104	0.982580	0.982684	0.993124	1.015951	1.019848	1.021337	1.0212

2021	1	0.138182	0.000697	0.993068	0.982580	0.982684	0.993124	1.015951	1.019848	1.0213
2021	2	0.138418	0.000480	0.994466	0.993068	0.982580	0.982684	0.993124	1.015951	1.0198
2021	3	0.138934	0.000497	0.992919	0.994466	0.993068	0.982580	0.982684	0.993124	1.0159
2021	4	0.139893	0.000715	0.988945	0.992919	0.994466	0.993068	0.982580	0.982684	0.9931
2021	5	0.140719	0.000767	0.993765	0.988945	0.992919	0.994466	0.993068	0.982580	0.9826
2021	6	0.140484	0.000294	1.021196	0.993765	0.988945	0.992919	0.994466	0.993068	0.9825
2021	7	0.140463	0.000146	1.021334	1.021196	0.993765	0.988945	0.992919	0.994466	0.9930
2021	8	0.140851	0.000260	1.019850	1.021334	1.021196	0.993765	0.988945	0.992919	0.9944
2021	9	0.141370	0.000382	1.015954	1.019850	1.021334	1.021196	0.993765	0.988945	0.9929
2021	10	0.142259	0.000621	0.993129	1.015954	1.019850	1.021334	1.021196	0.993765	0.9889
2021	11	0.143548	0.000936	0.982690	0.993129	1.015954	1.019850	1.021334	1.021196	0.9937
2021	12	0.144498	0.000943	0.982580	0.982690	0.993129	1.015954	1.019850	1.021334	1.0211
2022	1	0.146505	0.001445	0.993078	0.982580	0.982690	0.993129	1.015954	1.019850	1.0213
2022	2	0.148310	0.001614	0.994469	0.993078	0.982580	0.982690	0.993129	1.015954	1.0198
2022	3	0.150396	0.001837	0.992923	0.994469	0.993078	0.982580	0.982690	0.993129	1.0159
2022	4	0.152420	0.001925	0.988946	0.992923	0.994469	0.993078	0.982580	0.982690	0.9931
2022	5	0.154601	0.002046	0.993768	0.988946	0.992923	0.994469	0.993078	0.982580	0.9826
2022	6	0.156660	0.002052	1.021196	0.993768	0.988946	0.992923	0.994469	0.993078	0.9825
2022	7	0.159479	0.002414	1.021341	1.021196	0.993768	0.988946	0.992923	0.994469	0.9930
2022	8	0.162657	0.002774	1.019857	1.021341	1.021196	0.993768	0.988946	0.992923	0.9944
2022	9	0.164998	0.002570	1.015950	1.019857	1.021341	1.021196	0.993768	0.988946	0.9929
2022	10	0.167395	0.002488	0.993128	1.015950	1.019857	1.021341	1.021196	0.993768	0.9889
2022	11	0.168231	0.001709	0.982678	0.993128	1.015950	1.019857	1.021341	1.021196	0.9937
2022	12	0.169111	0.001318	0.982574	0.982678	0.993128	1.015950	1.019857	1.021341	1.0211
2023	1	0.169911	0.001074	0.993074	0.982574	0.982678	0.993128	1.015950	1.019857	1.0213
2023	2	0.170140	0.000676	0.994463	0.993074	0.982574	0.982678	0.993128	1.015950	1.0198
2023	3	0.169320	-0.000030	0.992911	0.994463	0.993074	0.982574	0.982678	0.993128	1.0159
2023	4	0.168283	-0.000505	0.988939	0.992911	0.994463	0.993074	0.982574	0.982678	0.9931
2023	5	0.167060	-0.000843	0.993762	0.988939	0.992911	0.994463	0.993074	0.982574	0.9826
2023	6	0.166322	-0.000794	1.021197	0.993762	0.988939	0.992911	0.994463	0.993074	0.9825
2023	7	0.165503	-0.000805	1.021341	1.021197	0.993762	0.988939	0.992911	0.994463	0.9930
2023	8	0.165518	-0.000419	1.019863	1.021341	1.021197	0.993762	0.988939	0.992911	0.9944
2023	9	0.166424	0.000206	1.015961	1.019863	1.021341	1.021197	0.993762	0.988939	0.9929
2023	10	0.168087	0.000893	0.993139	1.015961	1.019863	1.021341	1.021197	0.993762	0.9889
2023	11	0.169796	0.001278	0.982684	0.993139	1.015961	1.019863	1.021341	1.021197	0.9937
2023	12	0.171454	0.001457	0.982577	0.982684	0.993139	1.015961	1.019863	1.021341	1.0211
2024	1	0.173445	0.001709	0.993078	0.982577	0.982684	0.993139	1.015961	1.019863	1.0213
2024	2	0.174663	0.001478	0.994459	0.993078	0.982577	0.982684	0.993139	1.015961	1.0198
2024	3	0.175771	0.001303	0.992909	0.994459	0.993078	0.982577	0.982684	0.993139	1.0159
2024	4	0.176193	0.000888	0.988932	0.992909	0.994459	0.993078	0.982577	0.982684	0.9931
2024	5	0.176676	0.000697	0.993759	0.988932	0.992909	0.994459	0.993078	0.982577	0.9826
2024	6	0.176110	0.000101	1.021187	0.993759	0.988932	0.992909	0.994459	0.993078	0.9825
2024	7	0.175416	-0.000274	1.021335	1.021187	0.993759	0.988932	0.992909	0.994459	0.9930
2024	8	0.174488	-0.000582	1.019858	1.021335	1.021187	0.993759	0.988932	0.992909	0.9944

2024	9	0.174440	-0.000330	1.015965	1.019858	1.021335	1.021187	0.993759	0.988932	0.9929
2024	10	0.175803	0.000468	0.993152	1.015965	1.019858	1.021335	1.021187	0.993759	0.9889
2024	11	0.177437	0.001018	0.982692	0.993152	1.015965	1.019858	1.021335	1.021187	0.9937
2024	12	0.178729	0.001147	0.982579	0.982692	0.993152	1.015965	1.019858	1.021335	1.0211
2025	1	0.180029	0.001219	0.993079	0.982579	0.982692	0.993152	1.015965	1.019858	1.0213
2025	2	0.180733	0.000976	0.994455	0.993079	0.982579	0.982692	0.993152	1.015965	1.0198
2025	3	0.181950	0.001090	0.992910	0.994455	0.993079	0.982579	0.982692	0.993152	1.0159
2025	4	0.183034	0.001087	0.988932	0.992910	0.994455	0.993079	0.982579	0.982692	0.9931
2025	5	0.183718	0.000897	0.993756	0.988932	0.992910	0.994455	0.993079	0.982579	0.9826
2025	6	0.185209	0.001177	1.021191	0.993756	0.988932	0.992910	0.994455	0.993079	0.9825
2025	7	0.185837	0.000918	1.021331	1.021191	0.993756	0.988932	0.992910	0.994455	0.9930
2025	8	0.186568	0.000830	1.019857	1.021331	1.021191	0.993756	0.988932	0.992910	0.9944
2025	9	0.186429	0.000373	1.015958	1.019857	1.021331	1.021191	0.993756	0.988932	0.9929
2025	10	0.188037	0.000955	0.993161	1.015958	1.019857	1.021331	1.021191	0.993756	0.9889
2025	11	0.190366	0.001603	0.982702	0.993161	1.015958	1.019857	1.021331	1.021191	0.9937
2025	12	0.192126	0.001677	0.982580	0.982702	0.993161	1.015958	1.019857	1.021331	1.0211
2026	1	0.193612	0.001587	0.993078	0.982580	0.982702	0.993161	1.015958	1.019857	1.0213

Manual Verification

- Y_1^{sa} (2014M2) = 0.136914 + (-0.001473) = 0.135441
- Y_T^{sa} (2026M1) = 0.192588 + 0.000224 = 0.192812
- T = 144, drift per month = (0.192812 - 0.135441) / 143 = 0.000401

Forecast 1: February 2026 (h = 1) Seasonally adjusted forecast:

$$\hat{Y}_{T+1}^{sa} = 0.192812 + 0.000401 \times 1 = 0.193213$$

Re-seasonalize using $S_{Feb2026}$ = -0.001005:

$$\hat{Y}_{Feb2026} = 0.193213 + (-0.001005) = 0.192208$$

R table output: 0.1922079

Forecast 2: March 2026 (h = 2)** Seasonally adjusted forecast:

$$\hat{Y}_{T+2}^{sa} = 0.192812 + 0.000401 \times 2 = 0.193614$$

Re-seasonalize using $S_{Mar2026}$ = -0.000594:

$$\hat{Y}_{Mar2026} = 0.193614 + (-0.000594) = 0.193020$$

R table output: 0.1930208

Project IV

ARIMA Model 1

ARIMA Model 1

order of non-seasonal differencing:1

order of seasonal differencing: 1

non-seasonal autoregressive order: 2

non-seasonal moving average order: 2

seasonal autoregressive order: 1

seasonal moving average order: 1

The model includes a drift.

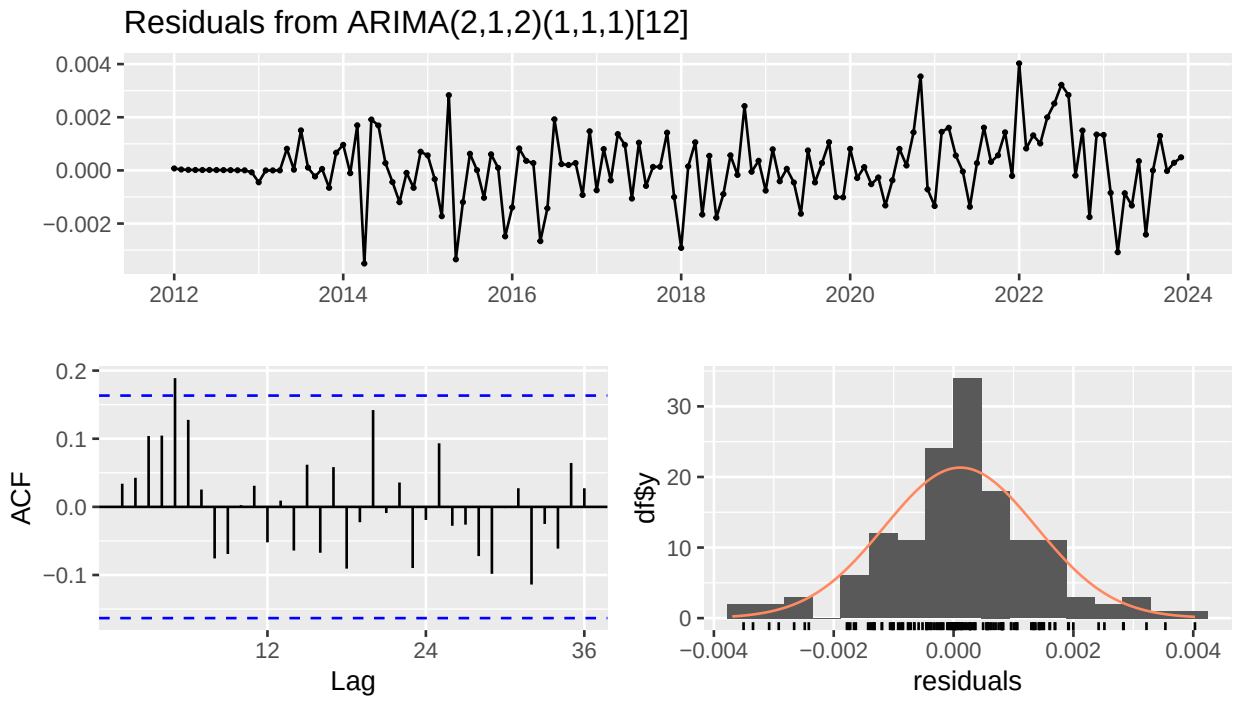
This model has a significant AR1, AR2, MA1,MA2,and SMA1 at the 5% level but the SAR1 is insignificant which is a poor indication for the model.This indicates the model is not useable.

One of the first 24 autocorrelation are significant, and the p-value is above .05 so the model is not serial correlated.

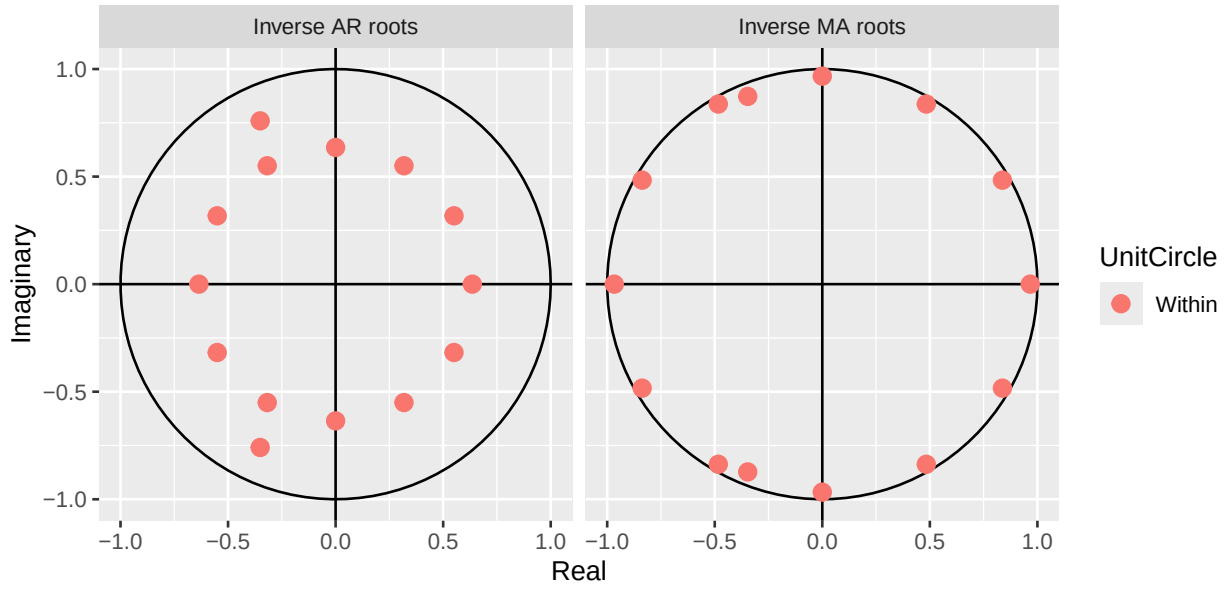
However, the AR and the MA components contain common roots, and some of the inverted roots are very close to the unit circle.

Due to the common roots and insignificant parameter this model is not useable and will not be considered for further analysis.

```
## Series: train
## ARIMA(2,1,2)(1,1,1)[12]
##
## Coefficients:
##           ar1      ar2     ma1     ma2    sar1     sma1
##       -0.7012  -0.6992  0.6930  0.8819  0.0043  -0.6675
## s.e.   0.1222   0.1444  0.0817  0.1006  0.1696   0.1517
##
## sigma^2 = 1.855e-06:  log likelihood = 678.15
## AIC=-1342.29   AICc=-1341.38   BIC=-1322.17
##
## Training set error measures:
##                         ME        RMSE          MAE        MPE      MAPE
## Training set 0.0001118811 0.001268903 0.0009149016 0.07186709 0.6433874
##                    MASE       ACF1
## Training set 0.1938475 0.03396824
```



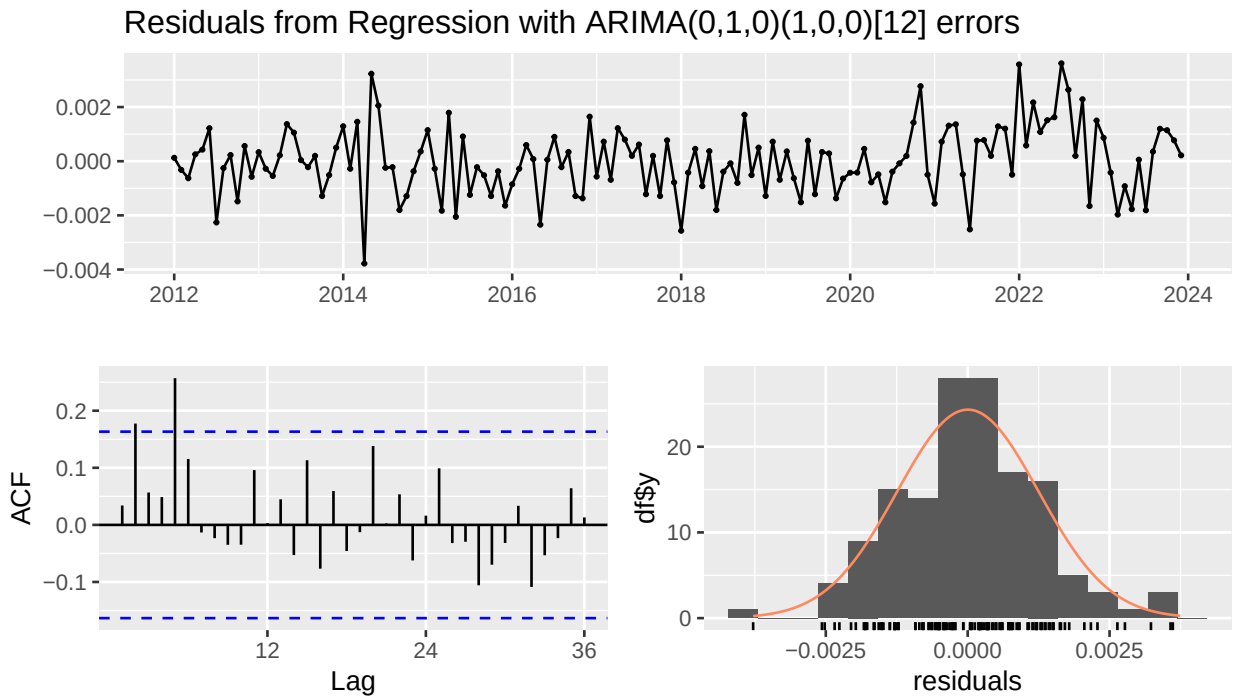

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,1,2)(1,1,1)[12]
## Q* = 23.076, df = 18, p-value = 0.1877
##
## Model df: 6.   Total lags used: 24
```



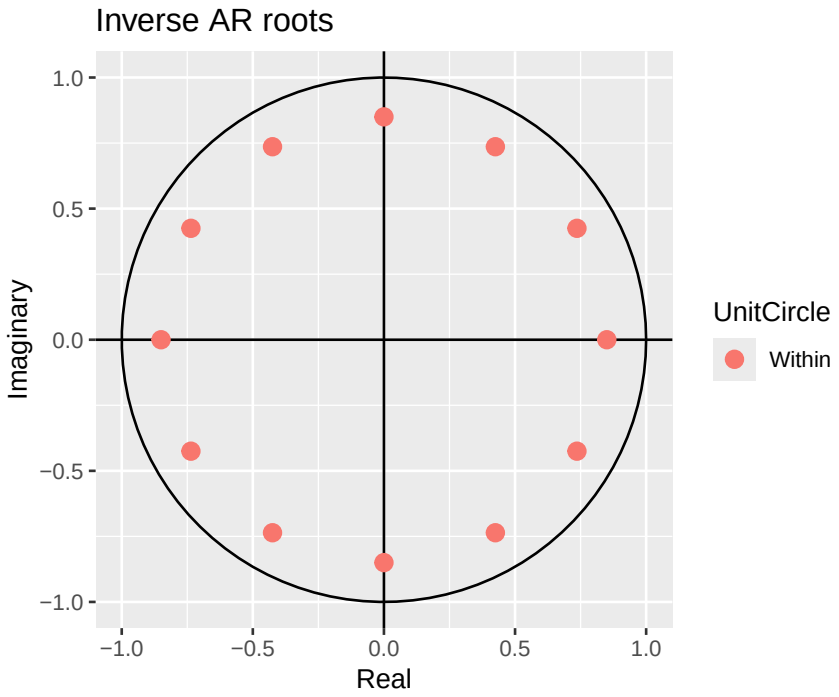

```
##  [1] 0.967 0.967 0.967 0.967 0.967 0.967 0.967 0.967 0.967 0.967 0.967 0.967
## [13] 0.939 0.939 0.836 0.836 0.636 0.636 0.636 0.636 0.636 0.636 0.636 0.636
## [25] 0.636 0.636 0.636 0.636
```

Model 3

```
## Series: train
## Regression with ARIMA(0,1,0)(1,0,0)[12] errors
##
## Coefficients:
##         sar1  drift     Jan     Feb    Mar    Apr     May     Jun     Jul
##       0.1421  3e-04  0.0014  0.0014  7e-04  2e-04  0.0015  0.0059  0.0059
## s.e.  0.0842  1e-04  0.0004  0.0006  6e-04  7e-04  0.0007  0.0007  0.0007
##          Aug     Sep     Oct     Nov
##       0.0059  0.0054  0.0016  -3e-04
## s.e.  0.0007  0.0006  0.0005   4e-04
##
## sigma^2 = 1.712e-06:  log likelihood = 753.17
## AIC=-1478.34   AICc=-1475.06   BIC=-1436.86
##
## Training set error measures:
##                        ME        RMSE         MAE        MPE      MAPE
## Training set 3.575422e-06 0.001243085 0.000972347 -0.0115907 0.6884675
##                    MASE       ACF1
## Training set 0.2060189 0.03397643
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(0,1,0)(1,0,0)[12] errors
## Q* = 28.884, df = 23, p-value = 0.1842
##
## Model df: 1.   Total lags used: 24
```




```
##  [1] 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85

## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -3.780e-03 -6.870e-04  4.817e-05  3.575e-06  7.600e-04  3.614e-03
##
## No extreme outliers detected (3 × IQR rule).
```

ARIMA Model 3 : ARIMAX with Seasonal Dummies

Model 3 extends ARIMA by including seasonal dummy variables as exogenous regressors:

ARIMA(0,1,0)(1,0,0)[12] with seasonal dummies

First differencing ($d = 1$) removes the trend, while a seasonal AR(1) term captures yearly dependence. Monthly dummy variables account for stable seasonal patterns, which is appropriate when seasonality does not vary significantly over time.

The seasonal SAR(1) coefficient is not statistically significant, but most monthly dummies are significant, with higher values in summer months (June–August), consistent with higher electricity demand. Our only not significant months are March, April and November.

The Ljung-Box test yields a p-value of **0.1841 ($>$ 0.05)**, indicating no residual autocorrelation. Residual plots confirm no remaining structure.

All inverted AR roots lie well within the unit circle (≈ 0.881), indicating a stable and stationary model. No extreme outliers are detected.

Model 3 satisfies most diagnostic criteria:

- Significant and meaningful drift but insignificant SAR1
- No residual autocorrelation
- Stable and invertible
- No extreme outliers

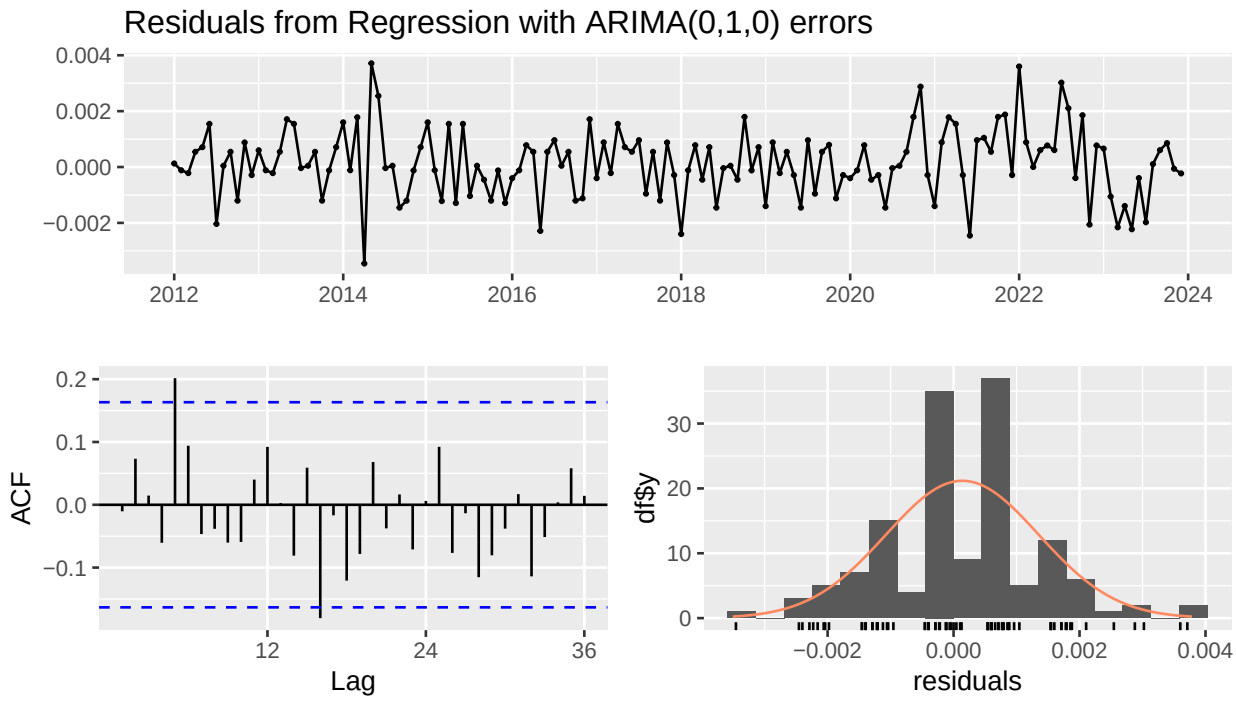
Therefore, Model M4 is selected as the preferred model. While Model M3 is already a strong model which achieves a strong performance. Model 4 is an ARIMA (0,1,0) that includes a trend and level variable which controls for the jump in level and change in trend after the Russia/Ukraine Shock. M4's trend control was found to be significant but the level control was not. This may be due to the late spike in the data which the model cannot accurately pick up. It also includes seasonal dummies to control for the seasonality of the data. The M4 model has greater accuracy and lower bias compared to the M3 which proves it's greater performance over the M3.

Since the model includes seasonal dummies for January through November, December serves as the baseline. The dummies are positive and significant for Jan., Feb., May, June, July, Aug., Sep., and Oct., and insignificant for the other months. That is, using December as a baseline, avg. electricity price are higher in Jan., Feb., May, June, July, Aug., Sep., and Oct. and not significantly different in the other months.

As shown below, none of the first 24 autocorrelation is significant (outside the 95% confidence band marked by the dashed lines). In addition, the Ljung-Box test has a p-value above 0.05, indicating that the residuals are not serially correlated.

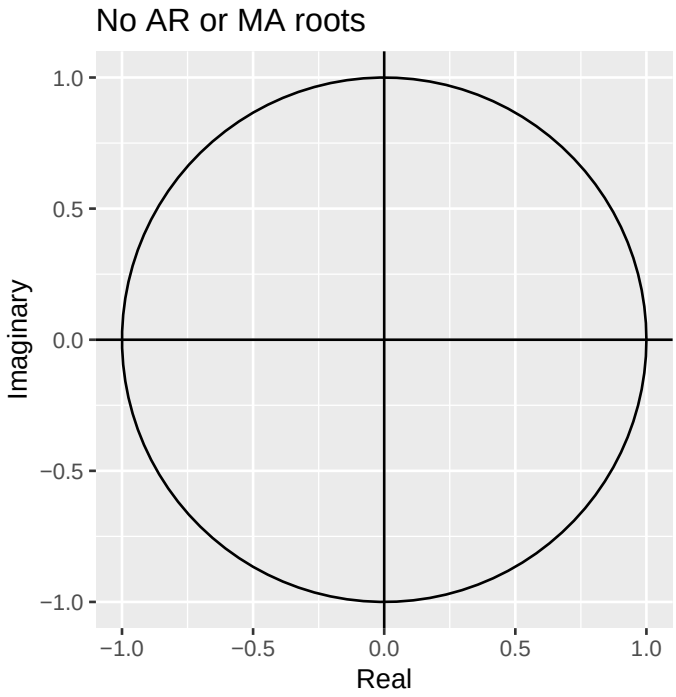
There are no MA or AR roots to test so this model passes all criteria and will is our best ARIMA model.

```
## Series: train
## Regression with ARIMA(0,1,0) errors
##
## Coefficients:
##          Jan     Feb    Mar    Apr     May     Jun    Jul     Aug     Sep
##       0.0014  0.0015  7e-04  2e-04  0.0015  0.0059  6e-03  0.0059  0.0054
## s.e.  0.0004  0.0005  6e-04  6e-04  0.0006  0.0006  6e-04  0.0006  0.0005
##          Oct     Nov  D_level  D_trend
##       0.0016  -3e-04   0.0018    9e-04
## s.e.  0.0005   3e-04   0.0013    3e-04
##
## sigma^2 = 1.646e-06:  log likelihood = 756.08
## AIC=-1484.17   AICc=-1480.89   BIC=-1442.69
##
## Training set error measures:
##                        ME        RMSE          MAE        MPE      MAPE
## Training set 0.000138956 0.001219041 0.0009320864 0.09833079 0.6616112
##                    MASE        ACF1
## Training set 0.1974886 -0.01039139
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(0,1,0) errors
## Q* = 24.624, df = 24, p-value = 0.4264
##
## Model df: 0.   Total lags used: 24

## numeric(0)
```




```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -3.455e-03 -4.003e-04  4.471e-05  1.390e-04  7.848e-04  3.711e-03
##
## No extreme outliers detected (3 × IQR rule).
```

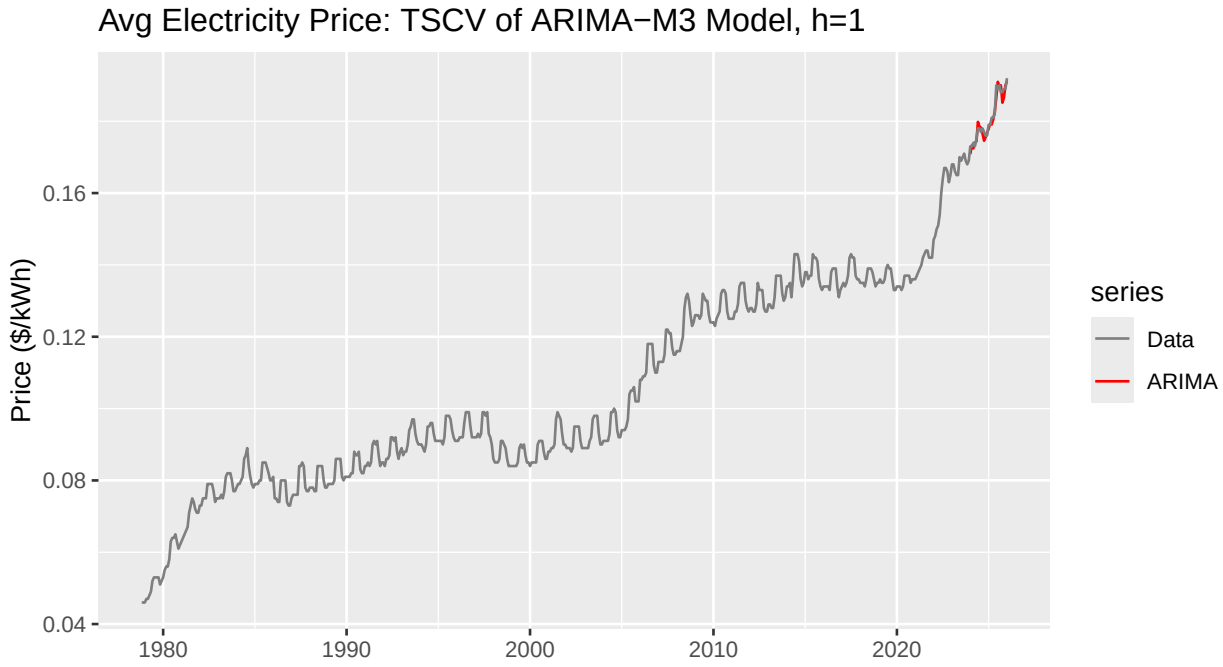
Project V

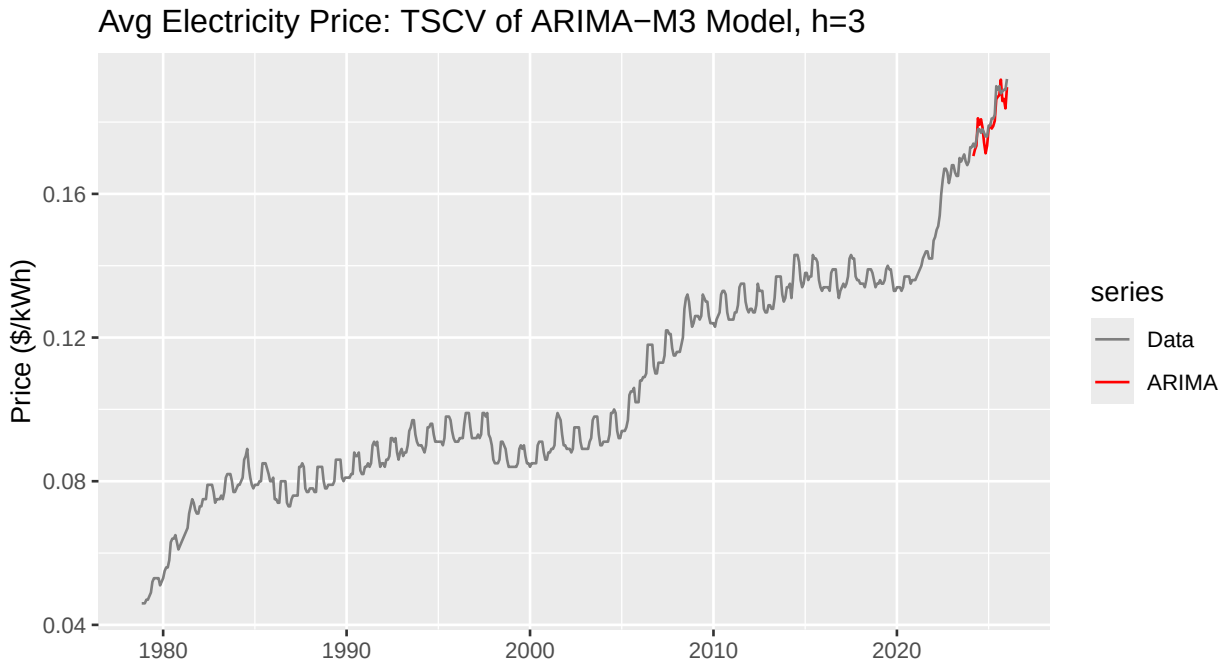
1. Model Selection

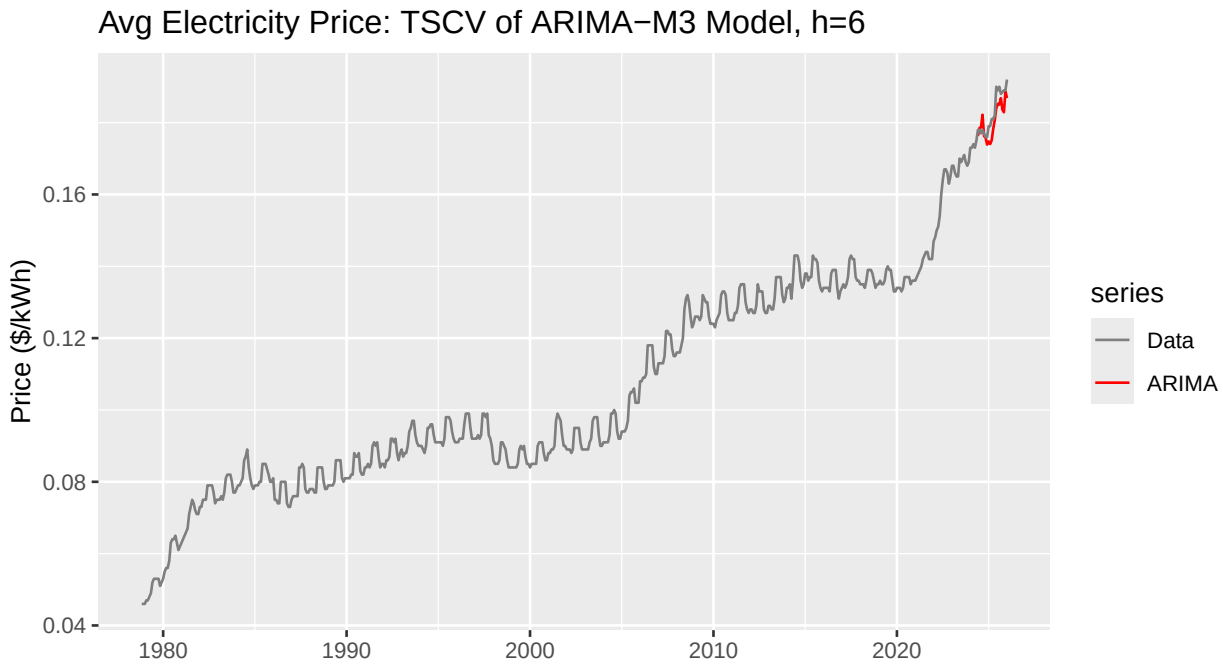
For Project 5, we selected ARIMA-M3 and ARIMA-M4 from Project 4. ARIMA-M3 is an ARIMAX model that includes seasonal dummy variables as exogenous regressors. ARIMA-M4 is an ARIMAX model that includes seasonal dummy variables, a level adjustment, and a trend adjustment.

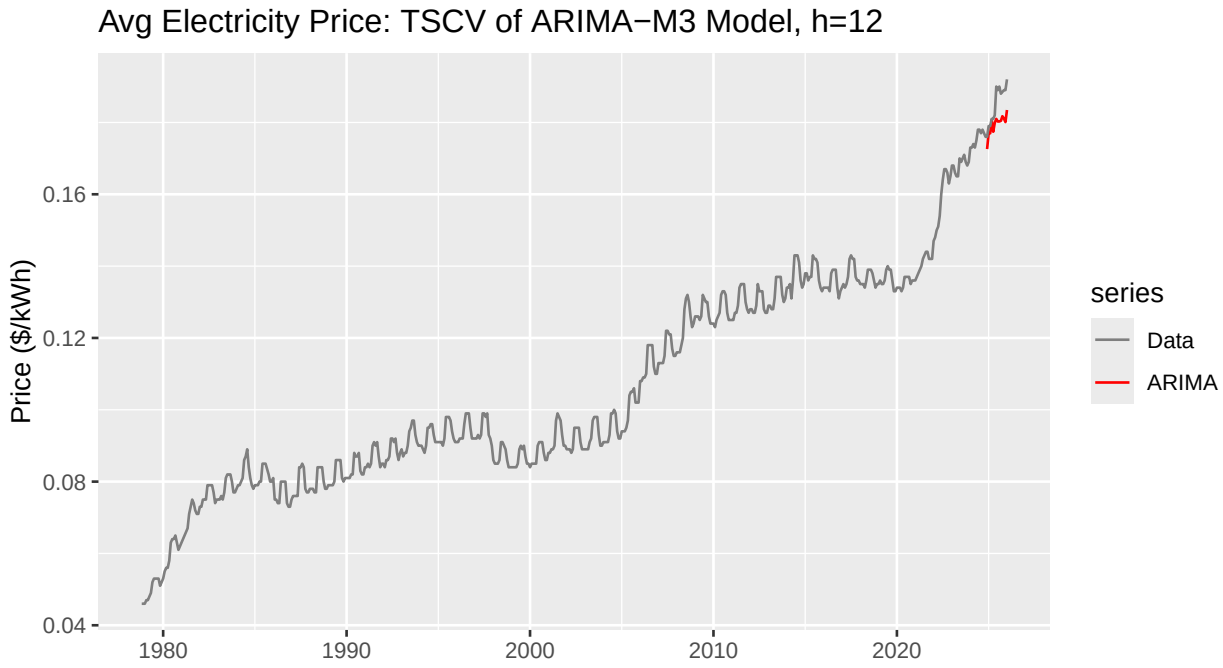
These two models were selected because they represent two different ways of modeling the average electricity price series. ARIMA-M3 captures the monthly seasonal pattern in electricity prices, while ARIMA-M4 adds level and trend adjustments to account for the stronger change in the series after 2022

ARIMA Model 3 TSCV










```
##          ME  RMSE   MAE   MPE  MAPE   ACF1 Theil's U
## h=1   0.000 0.002 0.001 0.238 0.754 -0.274     0.219
## h=3   0.001 0.003 0.002 0.740 1.352  0.276     0.355
## h=6   0.002 0.004 0.003 1.339 1.697  0.410     0.457
## h=12  0.006 0.007 0.006 3.068 3.068  0.695     0.674
```

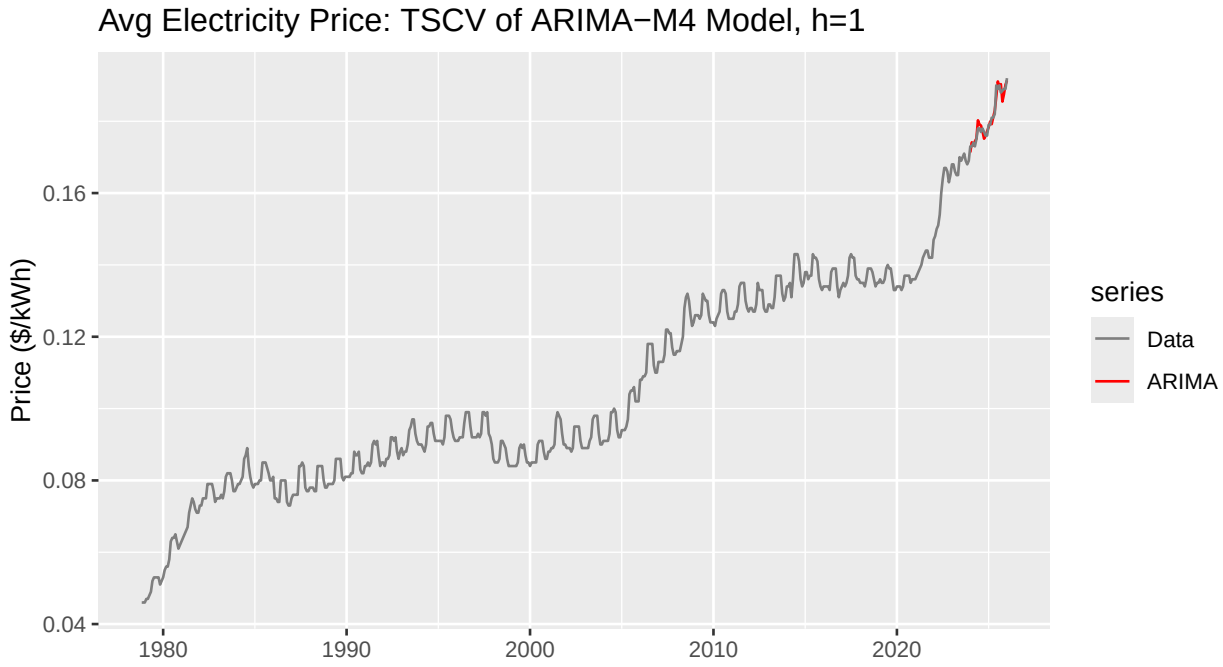
For ARIMA-M3, the time series cross validation results show that forecast accuracy becomes worse

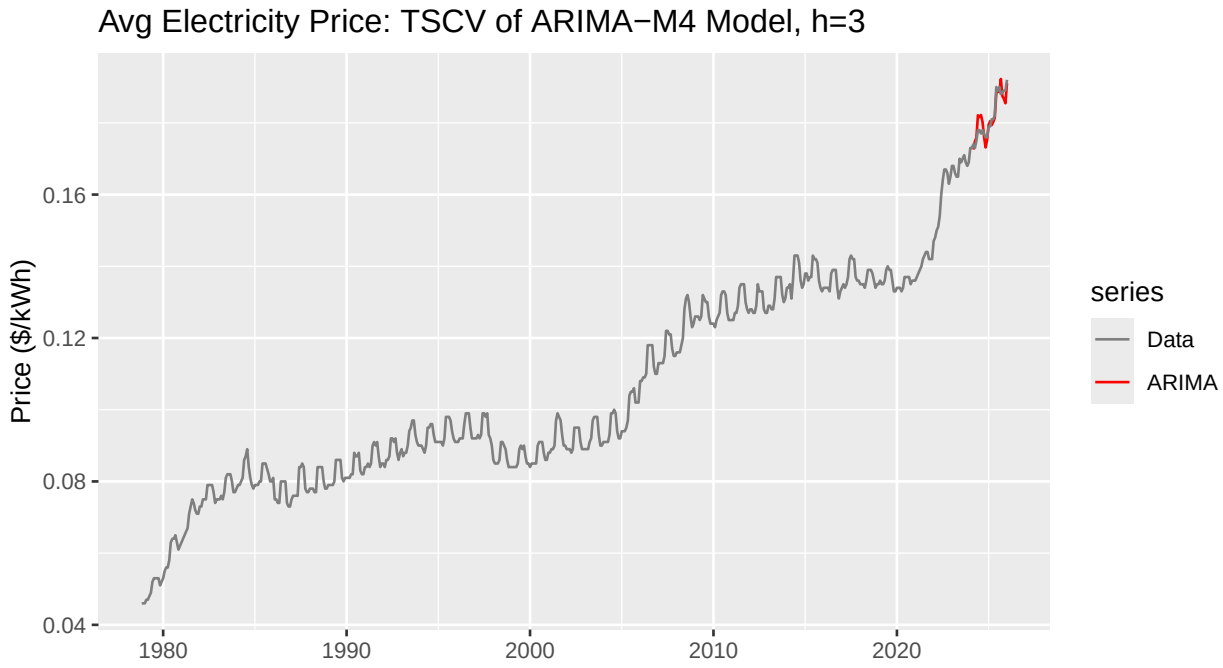
as the forecast horizon increases. At h = 1, the model has a MAPE = 0.754. At h = 12, these values increase to a MAPE = 3.068. This means ARIMA-M3 is much more accurate for short-term forecasts than for long-term forecasts.

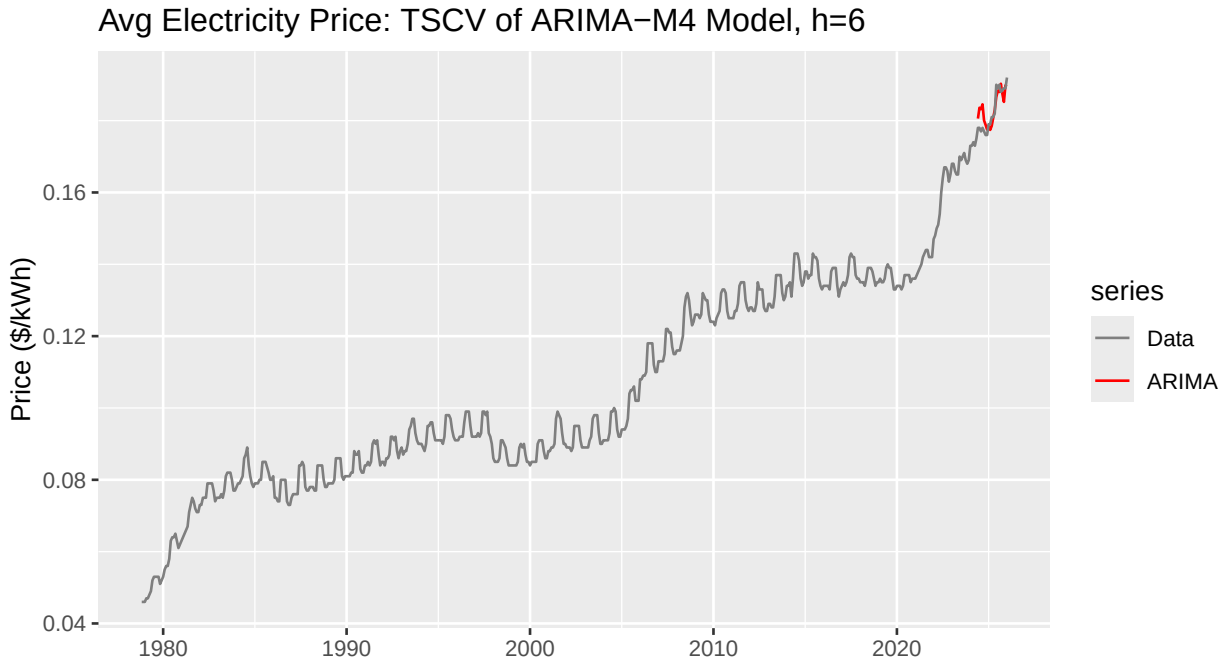
The MPE values are positive for all four forecast horizons: 0.238 at h = 1, 0.740 at h = 3, 1.339 at h = 6, and 2.246 at h = 12. Since MPE is positive, ARIMA-M3 tends to underforecast the actual electricity price.

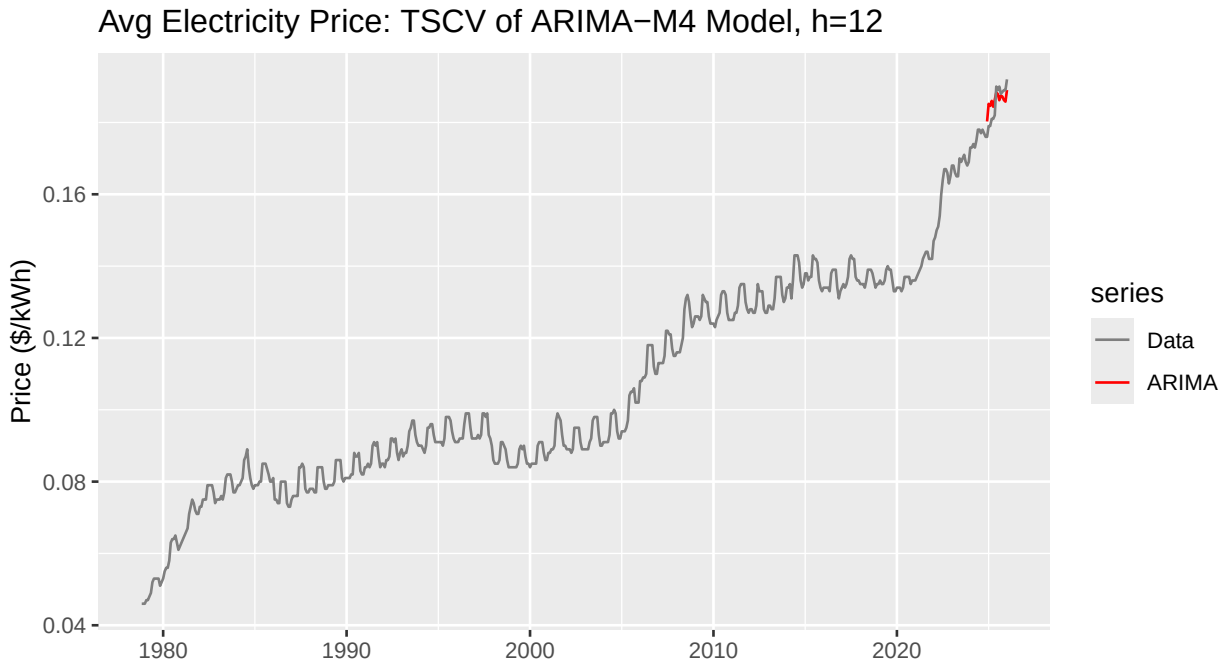
The overlay graph for ARIMA-M3 is consistent with the accuracy table. The model tracks the actual electricity price series better at shorter horizons, but the forecast becomes less accurate as the horizon increases. This is shown by the increase in MAPE from 0.747 at h = 1 to 3.255 at h = 12. The positive MPE values also show that the ARIMA-M3 forecasts are generally below the actual values, meaning that the model tends to under forecast electricity prices.

ARIMA Model 4 TSCV










```
##           ME  RMSE   MAE    MPE  MAPE   ACF1 Theil's U
## h=1    0.000 0.002 0.001 -0.011 0.723 -0.291     0.207
## h=3    0.000 0.002 0.002 -0.113 1.057  0.423     0.302
## h=6   -0.001 0.003 0.003 -0.435 1.380  0.606     0.386
## h=12  -0.001 0.004 0.003 -0.491 1.794  0.748     0.391
```

For ARIMA-M4, the time series cross validation results also show that forecast accuracy becomes

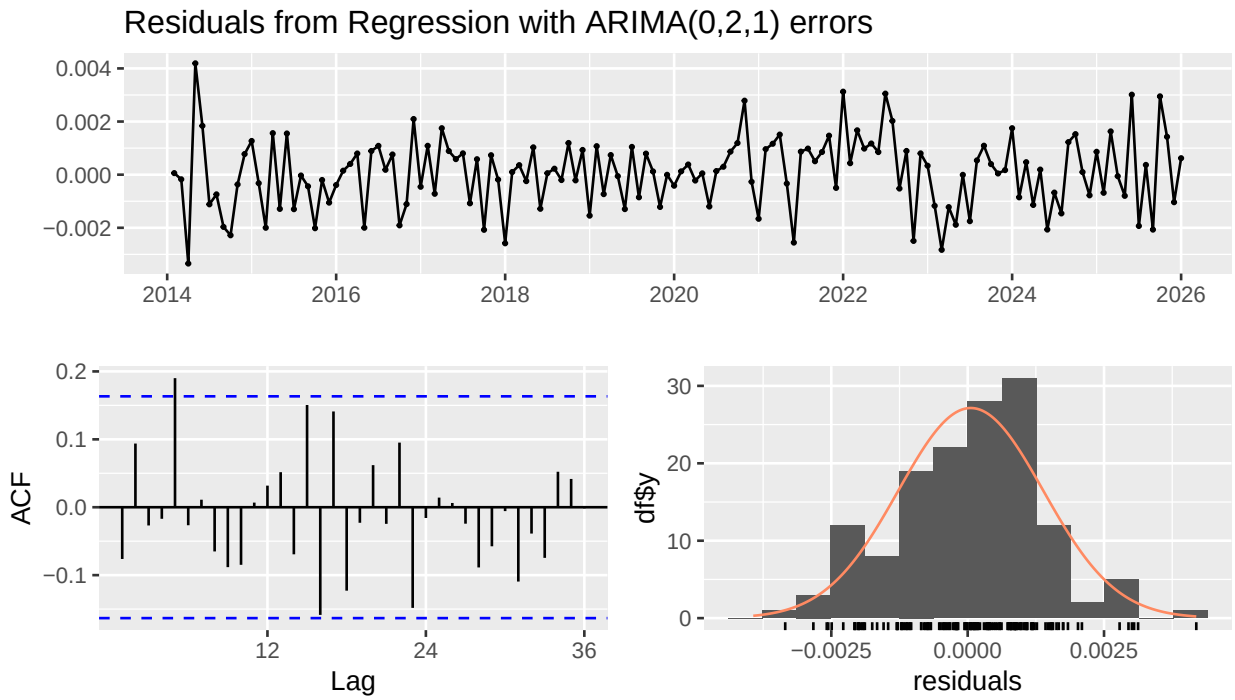
worse as the forecast horizon increases. At h = 1, the model has an MAPE = 0.723. At h = 12, these values increase to an MAPE = 1.794. This means ARIMA-M4 also performs better for short-term forecasts than for long-term forecasts.

The MPE values are negative for all four horizons: -0.011 at h = 1, -0.113 at h = 3, -0.435 at h = 6, and -0.491 at h = 12. Since MPE is negative, ARIMA-M4 tends to overforecast the actual electricity price. However, the MAPE values are lower than ARIMA-M3 at every horizon, which means ARIMA-M4 is more accurate overall even though it has a slight overforecasting bias.

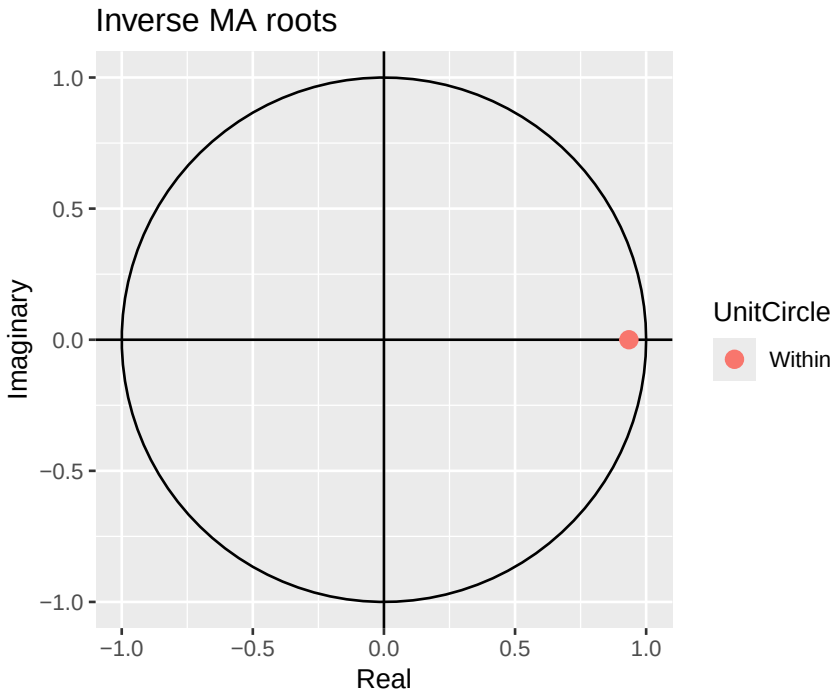
The overlay graph shows that ARIMA-M4 follows the actual electricity price series better than ARIMA-M3. This suggests that adding the level adjustment and trend adjustment improves the model's forecasting performance.

ARIMA Forecasts

```
## Series: window(price_clean, start = c(2014, 2))
## Regression with ARIMA(0,2,1) errors
##
## Coefficients:
##           ma1     Jan     Feb     Mar    Apr     May     Jun     Jul     Aug
##       -0.9343  0.0015  0.0015  0.0012  4e-04  0.0015  0.0057  0.0058  0.0056
## s.e.   0.0371  0.0004  0.0005  0.0006  6e-04  0.0007  0.0007  0.0007  0.0006
##          Sep     Oct     Nov
##       0.0049  0.0017  -1e-04
## s.e.  0.0006  0.0005   4e-04
##
## sigma^2 = 1.948e-06:  log likelihood = 737.33
## AIC=-1448.66   AICc=-1445.82   BIC=-1410.23
##
## Training set error measures:
##                       ME        RMSE         MAE      MPE     MAPE      MASE
## Training set 5.71946e-05 0.001325998 0.001047982 0.034044 0.704833 0.1845679
##                      ACF1
## Training set -0.07620039
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(0,2,1) errors
## Q* = 32.168, df = 23, p-value = 0.0967
##
## Model df: 1.   Total lags used: 24
```




```
## [1] 0.934
```

```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -3.345e-03 -8.105e-04  1.233e-04  5.719e-05  8.920e-04  4.192e-03
##
## No extreme outliers detected (3 × IQR rule).
```

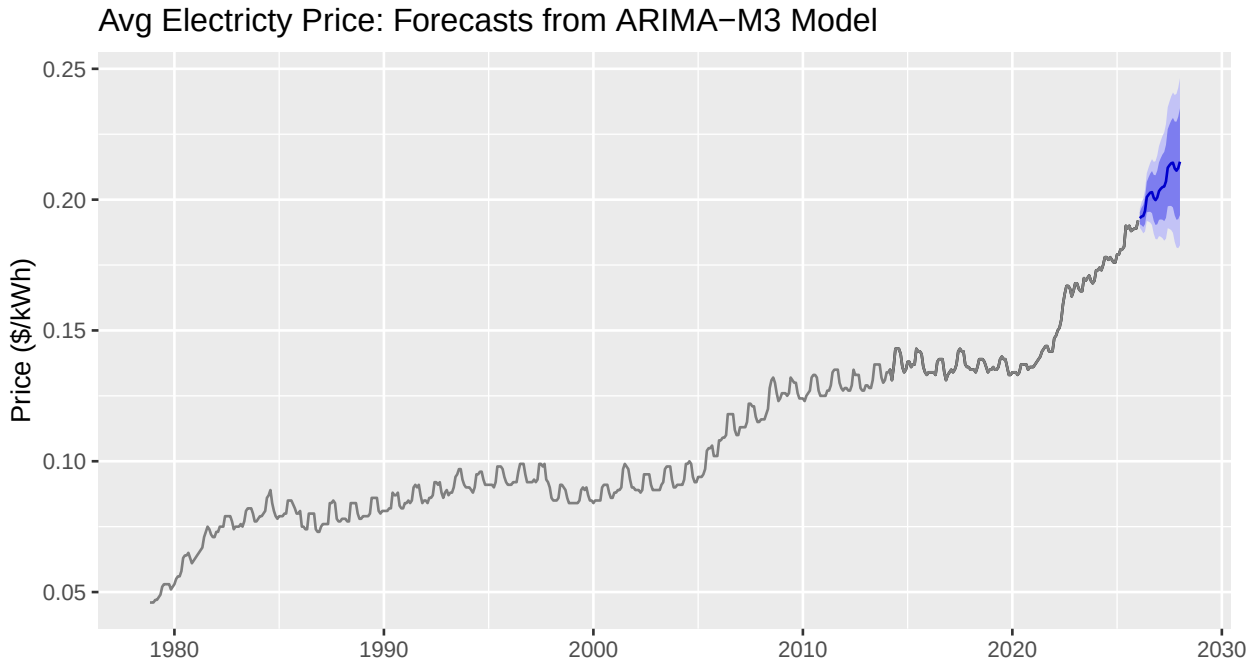
Function `auto.arima` selects a regression with ARIMA(0,2,1) errors for ARIMA-M3. This model includes seasonal dummy variables as exogenous regressors.

The diagnostic checks show that ARIMA-M3 does not fully pass the model diagnosis. The Ljung-Box test has a p-value of 0.0967, which is above the .05 threshold. Therefore, we can't reject the null hypothesis that the residuals are serially correlated. We tried a manual input of the differencing and the model still was not stable. We believe this may be cause by the shock from the war. Due to the war happening so late in our time series it's probable that the model is being influenced by the late change.

The residual ACF plot also shows at least one significant autocorrelation spike outside the 95% confidence bands. The inverse MA root is inside the unit circle but very close to the boundary, which suggests that the model is close to non-invertibility. However, there are no extreme outliers detected using the 3 × IQR rule.

Overall, ARIMA-M3 captures the seasonal pattern in electricity prices, but it does not fully pass the residual diagnostic checks.

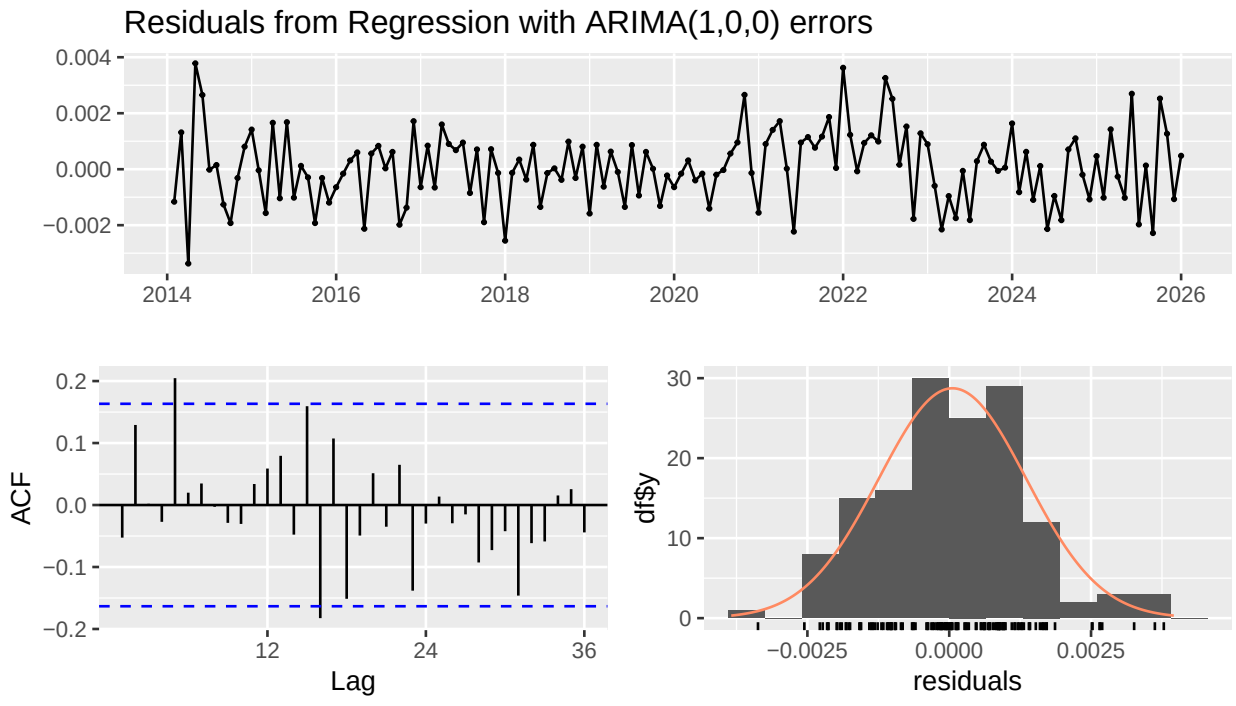
Confidence Intervals



	Point Forecast	Lo 90	Hi 90	Lo 99	Hi 99
Feb 2026	0.1929071	0.1906116	0.1952026	0.1893124	0.1965019
Mar 2026	0.1935485	0.1901938	0.1969032	0.1882951	0.1988019
Apr 2026	0.1937677	0.1895252	0.1980103	0.1871239	0.2004115
May 2026	0.1957315	0.1906768	0.2007863	0.1878158	0.2036472
Jun 2026	0.2009399	0.1951129	0.2067669	0.1918149	0.2100649
Jul 2026	0.2018928	0.1953161	0.2084696	0.1915937	0.2121920
Aug 2026	0.2026737	0.1953595	0.2099878	0.1912198	0.2141275
Sep 2026	0.2028657	0.1948204	0.2109110	0.1902668	0.2154646
Oct 2026	0.2006773	0.1919028	0.2094518	0.1869365	0.2144181
Nov 2026	0.1998168	0.1903123	0.2093213	0.1849328	0.2147008
Dec 2026	0.2008258	0.1905884	0.2110632	0.1847942	0.2168575
Jan 2027	0.2032461	0.1922715	0.2142207	0.1860600	0.2204323
Feb 2027	0.2041532	0.1924359	0.2158706	0.1858040	0.2225025
Mar 2027	0.2047946	0.1923282	0.2172610	0.1852723	0.2243169
Apr 2027	0.2050139	0.1917914	0.2182363	0.1843076	0.2257201
May 2027	0.2069777	0.1929917	0.2209636	0.1850758	0.2288795
Jun 2027	0.2121860	0.1974288	0.2269433	0.1890762	0.2352958
Jul 2027	0.2131390	0.1976022	0.2286757	0.1888085	0.2374695
Aug 2027	0.2139198	0.1975951	0.2302444	0.1883555	0.2394841
Sep 2027	0.2141118	0.1969907	0.2312329	0.1873003	0.2409233
Oct 2027	0.2119234	0.1939973	0.2298496	0.1838512	0.2399957
Nov 2027	0.2110629	0.1923229	0.2298030	0.1817161	0.2404097

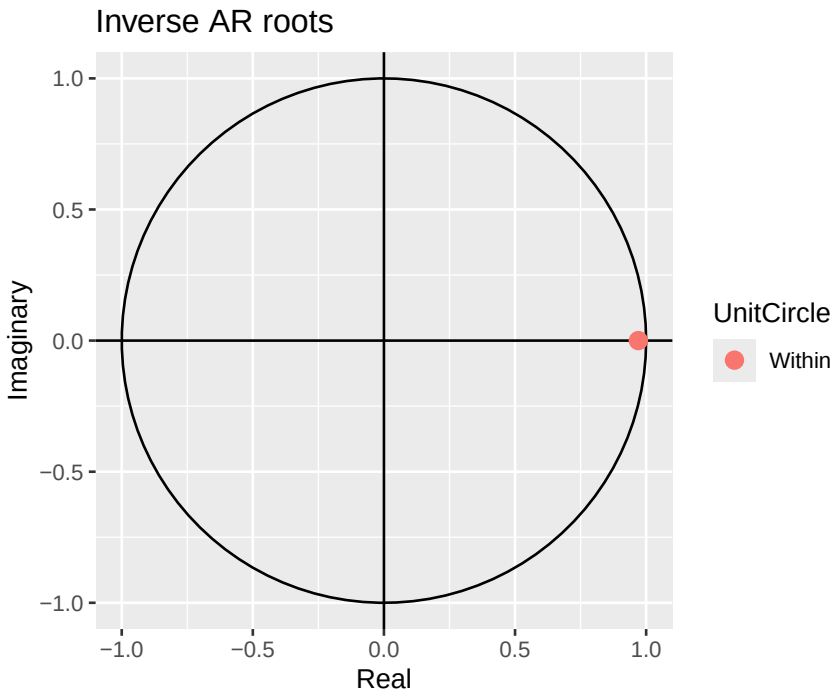
Dec 2027	0.2120719	0.1925092	0.2316347	0.1814368	0.2427071
Jan 2028	0.2144923	0.1940979	0.2348866	0.1825549	0.2464297

```
## Series: window(price_clean, start = c(2014, 2))
## Regression with ARIMA(1,0,0) errors
##
## Coefficients:
##          ar1  intercept     Jan     Feb     Mar     Apr     May     Jun     Jul
##       0.9703     0.1373  0.0015  0.0015  0.0011   3e-04  0.0014  0.0056  0.0057
## s.e.  0.0199     0.0035  0.0004  0.0005  0.0006   6e-04  0.0007  0.0007  0.0007
##          Aug     Sep     Oct     Nov  D_level  D_trend
##       0.0055  0.0048  0.0017  -1e-04   0.0018   0.0011
## s.e.  0.0006  0.0006  0.0005   4e-04   0.0014   0.0002
##
## sigma^2 = 1.877e-06:  log likelihood = 751.54
## AIC=-1471.07   AICc=-1466.79   BIC=-1423.55
##
## Training set error measures:
##                        ME        RMSE         MAE        MPE      MAPE
## Training set 5.758265e-05 0.001296888 0.001018203 0.03516739 0.6856985
##                    MASE        ACF1
## Training set 0.1793233 -0.05262511
```




```
##
##  Ljung-Box test
##
```

```
## data:  Residuals from Regression with ARIMA(1,0,0) errors
## Q* = 32.567, df = 23, p-value = 0.08892
##
## Model df: 1.   Total lags used: 24
```




```
## [1] 0.97
```

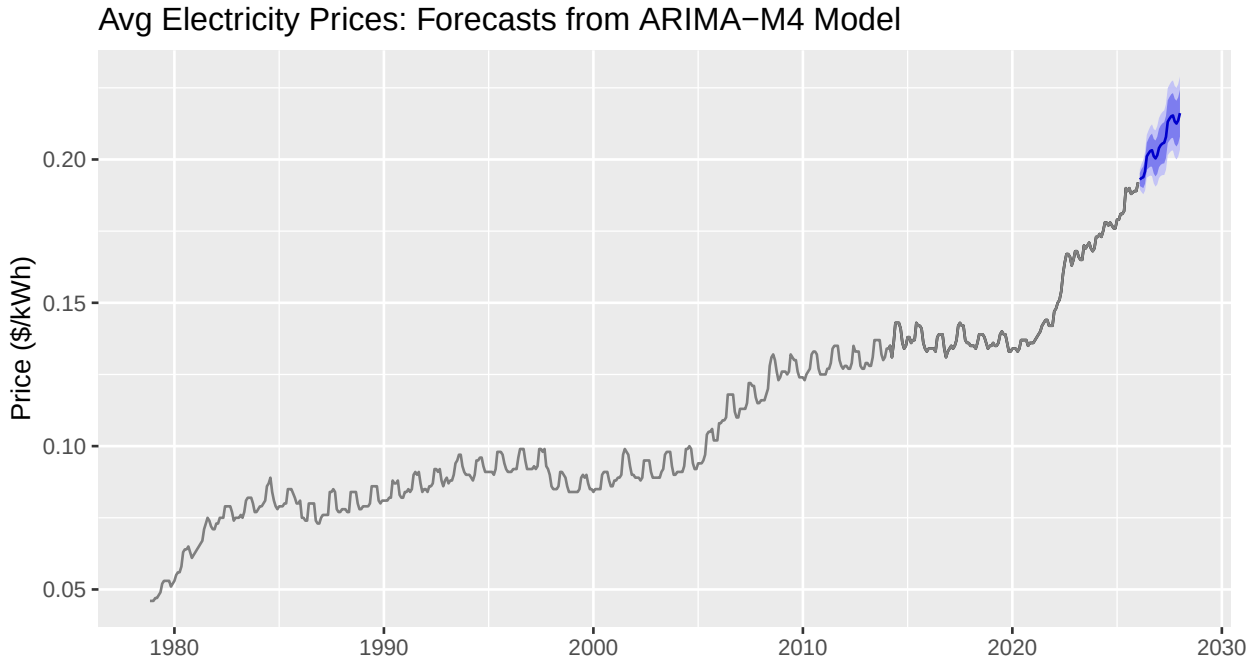
```
## Summary Statistics:
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -3.371e-03 -9.438e-04  2.522e-05  5.758e-05  8.786e-04  3.782e-03
##
## No extreme outliers detected (3 × IQR rule).
```

This model includes seasonal dummy variables, a level adjustment, and a trend adjustment as exogenous regressors. We found similar results to our initial ARIMA-M4 where the trend is significant and the level is not. This is consistent because the trend sees a noticeable jump but the level change takes a longer period which is likely why it's not captured in the model.Compared with ARIMA-M3, this model is designed to capture both the monthly seasonal pattern and the post-2022 level/trend change in average electricity prices.

The residual diagnostic plot shows that the residuals are centered around zero, but the ACF plot still has at least one autocorrelation spike outside the 95% confidence bands. This suggests that some autocorrelation may still remain in the residuals. However, the residuals appear roughly normally distributed based on the histogram.

The inverse AR root is inside the unit circle, but it is very close to the boundary. This suggests that the AR component is stable, but the model is close to the unit circle. Therefore, the model should be interpreted carefully.

Overall, ARIMA-M4 is still preferred over ARIMA-M3 because it performs better in the time series cross validation. ARIMA-M4 has lower MAPE values at every forecast horizon: 0.723 compared with 0.754 at h = 1, 1.057 compared with 1.352 at h = 3, 1.380 compared with 1.697 at h = 6, and 1.794 compared with 3.068 at h = 12. This means that even though the diagnostic plots show some remaining issues, ARIMA-M4 gives better out-of-sample forecast accuracy than ARIMA-M3.



	Point Forecast	Lo 90	Hi 90	Lo 99	Hi 99
Feb 2026	0.1930064	0.1907526	0.1952602	0.1894770	0.1965359
Mar 2026	0.1935410	0.1904006	0.1966813	0.1886232	0.1984587
Apr 2026	0.1938065	0.1900166	0.1975965	0.1878715	0.1997416
May 2026	0.1958234	0.1915103	0.2001364	0.1890692	0.2025776
Jun 2026	0.2010914	0.1963382	0.2058445	0.1936480	0.2085348
Jul 2026	0.2021105	0.1969774	0.2072437	0.1940721	0.2101490
Aug 2026	0.2029640	0.1974973	0.2084308	0.1944031	0.2115249
Sep 2026	0.2032351	0.1974718	0.2089983	0.1942099	0.2122603
Oct 2026	0.2011320	0.1951030	0.2071610	0.1916906	0.2105734
Nov 2026	0.2003630	0.1940940	0.2066320	0.1905458	0.2101802
Dec 2026	0.2014696	0.1949829	0.2079564	0.1913114	0.2116279
Jan 2027	0.2039935	0.1973082	0.2106789	0.1935244	0.2144627
Feb 2027	0.2050185	0.1981515	0.2118854	0.1942648	0.2157721
Mar 2027	0.2055710	0.1985372	0.2126047	0.1945562	0.2165858
Apr 2027	0.2058539	0.1986667	0.2130411	0.1945988	0.2171090
May 2027	0.2078877	0.2005590	0.2152164	0.1964110	0.2193644
Jun 2027	0.2131721	0.2057126	0.2206316	0.2014906	0.2248536
Jul 2027	0.2142072	0.2066266	0.2217877	0.2023360	0.2260783
Aug 2027	0.2150761	0.2073833	0.2227689	0.2030292	0.2271230

Sep 2027	0.2153621	0.2075651	0.2231591	0.2031521	0.2275721
Oct 2027	0.2132736	0.2053798	0.2211674	0.2009119	0.2256352
Nov 2027	0.2125187	0.2045348	0.2205026	0.2000159	0.2250214
Dec 2027	0.2136390	0.2055712	0.2217068	0.2010049	0.2262731
Jan 2028	0.2161762	0.2080302	0.2243221	0.2034196	0.2289327

Project VI

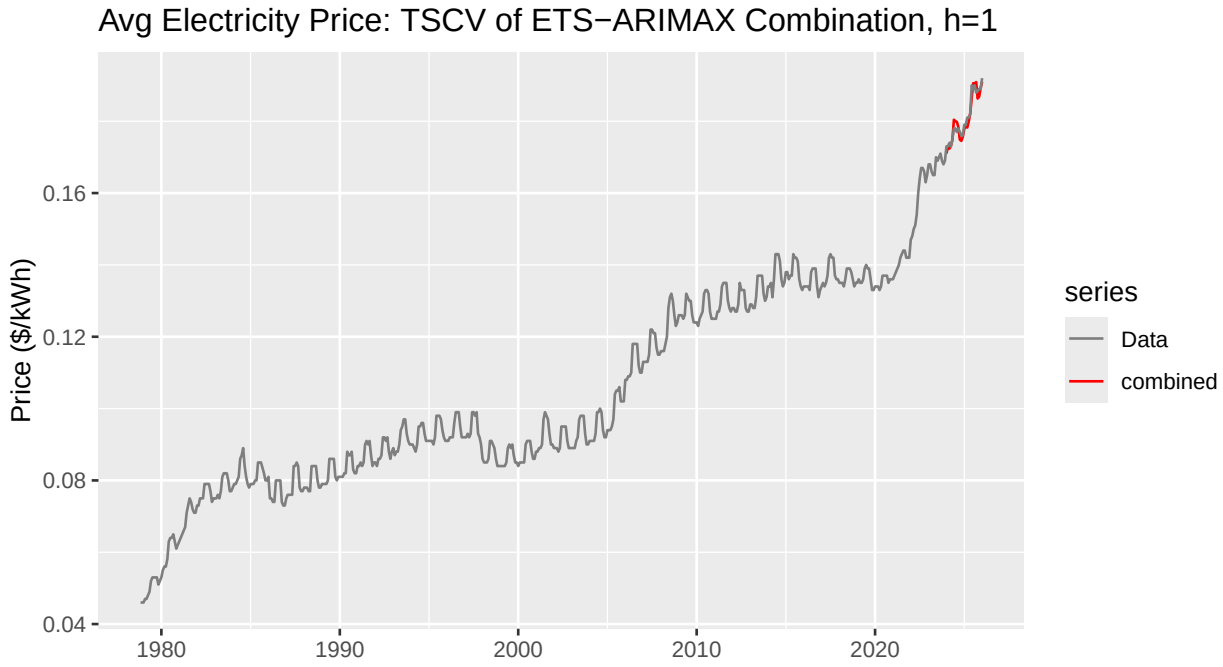
Forecast Comparison

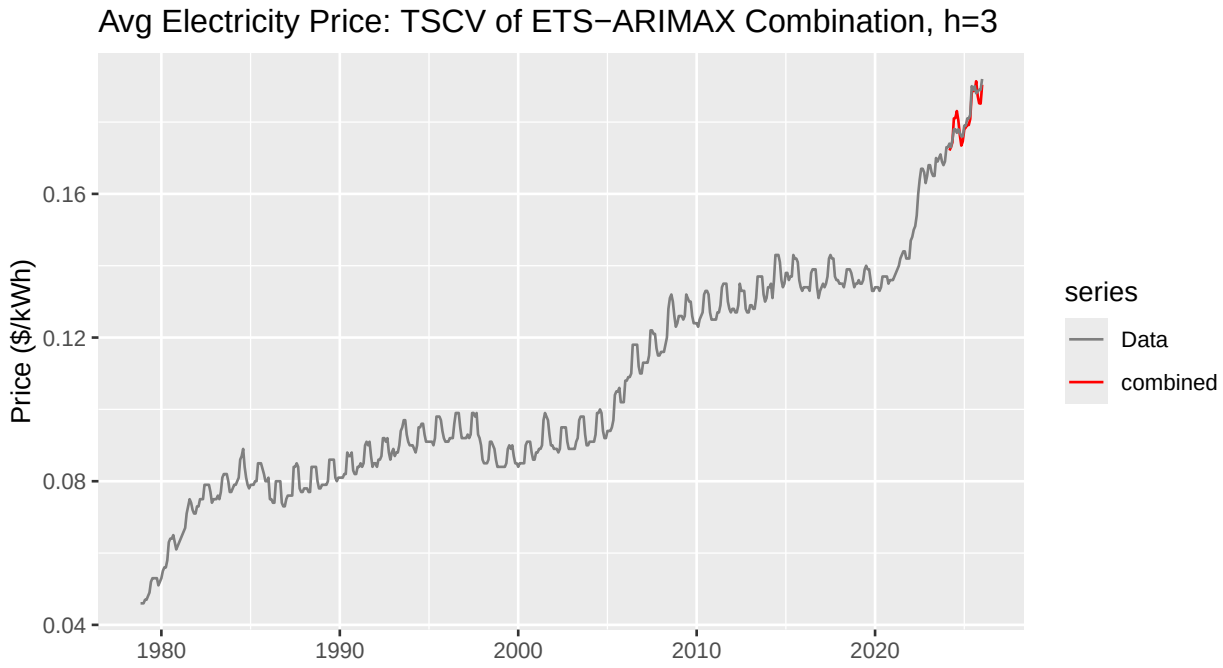
```
##      STL Benchmark ETS(M,A,M) ARIMA-M4
## h=1          0.441      0.388   -0.011
## h=3          0.916      0.517   -0.113
## h=6          1.507      0.321   -0.435
## h=12         3.190      1.806   -0.491
```

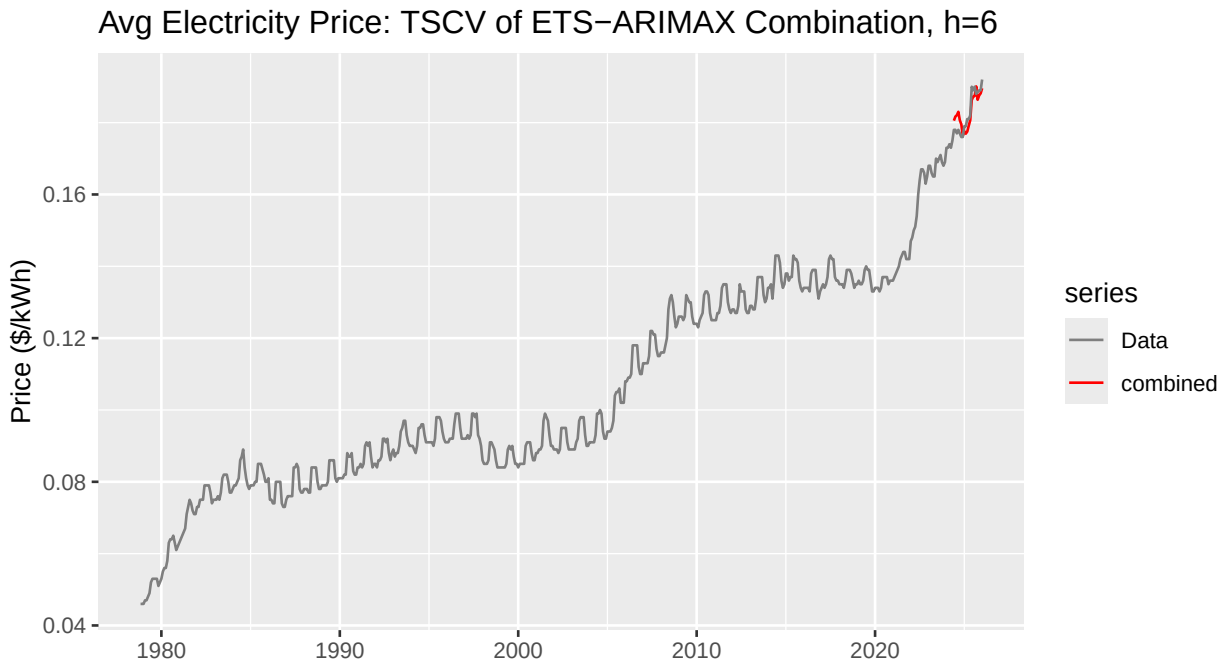
```
##      STL Benchmark ETS(M,A,M) ARIMA-M4
## h=1          0.721      1.132    0.723
## h=3          1.251      1.345    1.057
## h=6          1.677      1.614    1.380
## h=12         3.190      2.619    1.794
```

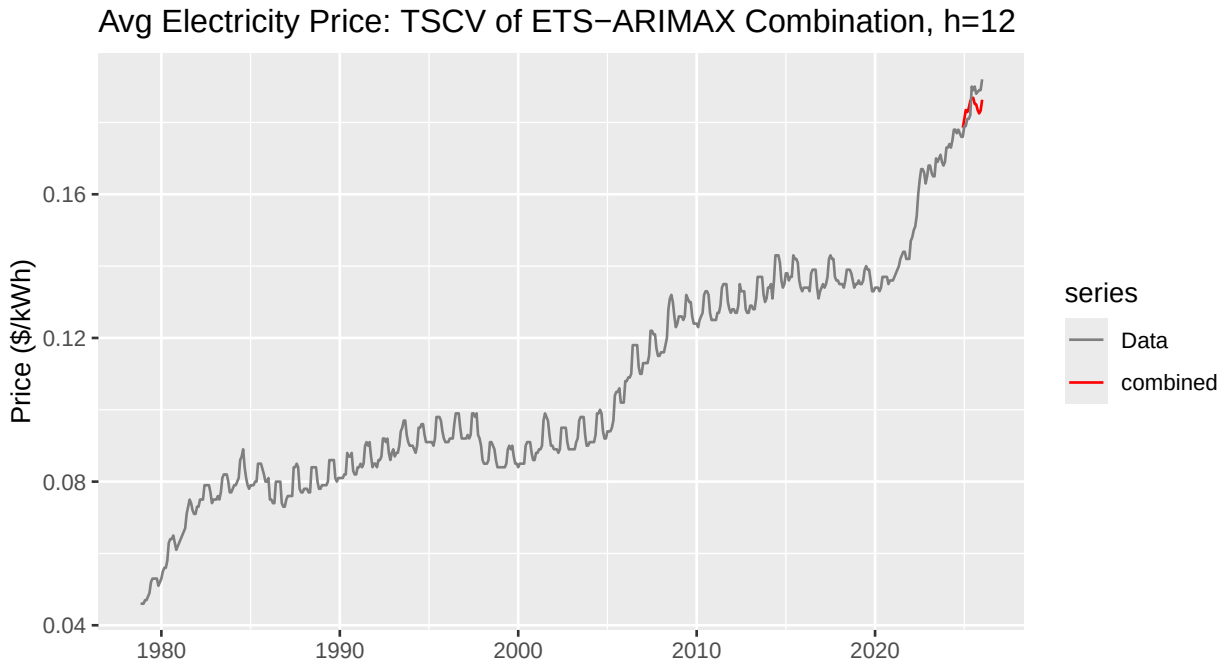
Based on the time series cross validation, The ARIMA-M4 model performs the best at most time intervals for the MAPE. It is best at h=3,h=6 and h=12 but the STL performs marginally better at the h=1.The ARIMA-M4 model performs the best for MPE at h=1, h=3, and h=12. However, The ETS (M,A,M) performs best at h=6. From these statistics we can conclude that the ARIMA model has the best performance since it has the highest accuracy and at all intervals other than h=6 it has the lowest bias.

Combination Forecasts









Evaluation of Combined model

We decided to combine the ARIMA-M4 and the ETS(M,A,M) because ARIMAX is better at forecasting in the short-run but ETS allows change in seasonality over time which should improve our model performance in the long run.

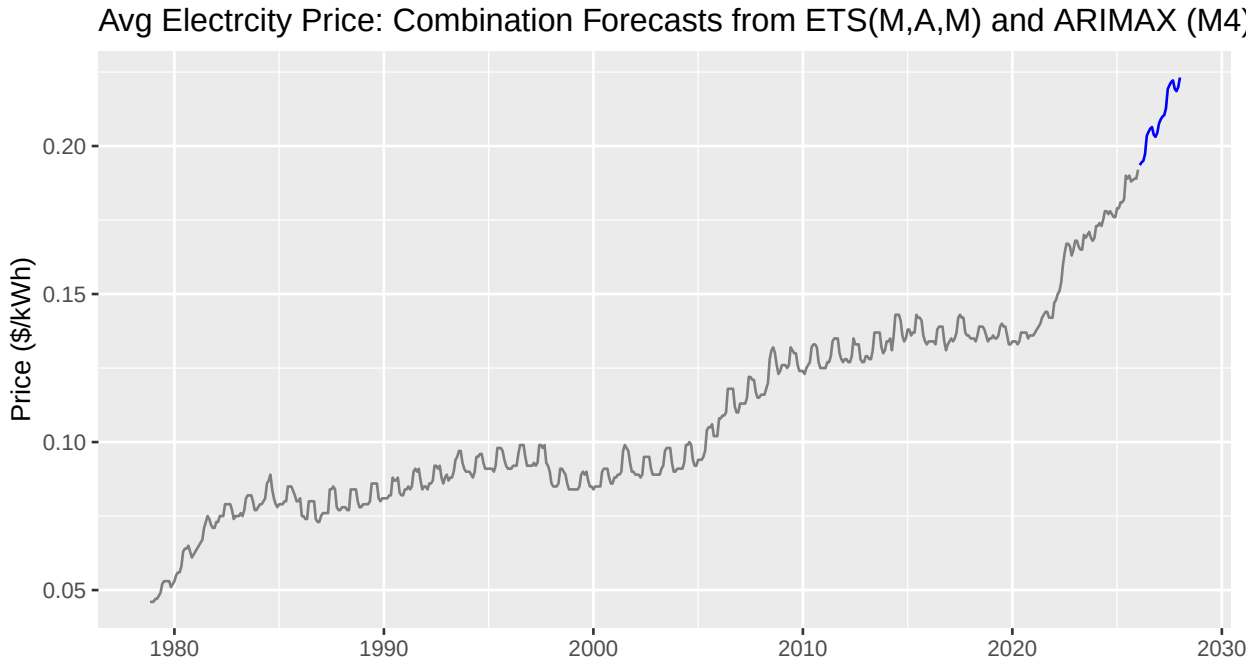
```
##          ME  RMSE   MAE    MPE  MAPE  ACF1 Theil's U
```

```
## h=1  0.000 0.002 0.001  0.188 0.776 0.136     0.234
## h=3  0.000 0.002 0.002  0.202 1.086 0.530     0.310
## h=6  0.000 0.003 0.003 -0.057 1.401 0.722     0.355
## h=12 0.001 0.004 0.004  0.657 2.063 0.719     0.425
```

###Base models vs Combo model

```
##       STL Benchmark ETS(M,A,M) ARIMA-M4 ETS-ARIMAX Combined
## h=1           0.441      0.388   -0.011                0.188
## h=3           0.916      0.517   -0.113                0.202
## h=6           1.507      0.321   -0.435               -0.057
## h=12          3.190      1.806   -0.491                0.657
```

```
##       STL Benchmark ETS(M,A,M) ARIMA-M4 ETS-ARIMAX Combined
## h=1           0.721      1.132    0.723                0.776
## h=3           1.251      1.345    1.057                1.086
## h=6           1.677      1.614    1.380                1.401
## h=12          3.190      2.619    1.794                2.063
```



year	month	STL	ETS.MAM	ARIMAX.M4	ETS.ARIMAX.Combined
2026	2	0.1922079	0.1941166	0.1930064	0.1935615
2026	3	0.1930208	0.1953907	0.1935410	0.1944658
2026	4	0.1927216	0.1961772	0.1938065	0.1949919
2026	5	0.1938803	0.1987112	0.1958234	0.1972673
2026	6	0.1980184	0.2058177	0.2010914	0.2034546

2026	7	0.1976821	0.2074666	0.2021105	0.2047886
2026	8	0.1978041	0.2087857	0.2029640	0.2058749
2026	9	0.1975375	0.2095998	0.2032351	0.2064174
2026	10	0.1956373	0.2064726	0.2011320	0.2038023
2026	11	0.1947355	0.2058578	0.2003630	0.2031104
2026	12	0.1949553	0.2073916	0.2014696	0.2044306
2027	1	0.1968144	0.2111833	0.2039935	0.2075884
2027	2	0.1970222	0.2130544	0.2050185	0.2090364
2027	3	0.1978352	0.2142991	0.2055710	0.2099351
2027	4	0.1975359	0.2150099	0.2058539	0.2104319
2027	5	0.1986946	0.2176358	0.2078877	0.2127617
2027	6	0.2028328	0.2252647	0.2131721	0.2192184
2027	7	0.2024965	0.2269162	0.2142072	0.2205617
2027	8	0.2026185	0.2282073	0.2150761	0.2216417
2027	9	0.2023519	0.2289471	0.2153621	0.2221546
2027	10	0.2004516	0.2253858	0.2132736	0.2193297
2027	11	0.1995499	0.2245718	0.2125187	0.2185453
2027	12	0.1997696	0.2261033	0.2136390	0.2198711
2028	1	0.2016287	0.2300950	0.2161762	0.2231356

```
##          ME  RMSE   MAE   MPE  MAPE  ACF1 Theil's U
## h=1  0.000 0.002 0.001 0.272 0.725 0.012     0.225
## h=3  0.001 0.002 0.002 0.440 1.135 0.478     0.301
## h=6  0.001 0.003 0.003 0.464 1.363 0.699     0.342
## h=12 0.003 0.004 0.004 1.502 1.943 0.718     0.458

##      STL Benchmark ETS(M,A,M) ARIMA-M4 3 Model Combination
## h=1          0.441      0.388   -0.011               0.272
## h=3          0.916      0.517   -0.113               0.440
## h=6          1.507      0.321   -0.435               0.464
## h=12         3.190      1.806   -0.491               1.502

##      STL Benchmark ETS(M,A,M) ARIMA-M4 3 Model Combination
## h=1          0.721      1.132    0.723               0.725
## h=3          1.251      1.345    1.057               1.135
## h=6          1.677      1.614    1.380               1.363
## h=12         3.190      2.619    1.794               1.943
```

Just as a test we decided to combine all 3 models to see if their was any improvement. Similar to our previous model combination the ARIMA-M4 produced the best results and is still our preferred model. e # Part VI

Reflection

The time series data used in this project contains both a positive long-term trend and strong seasonality. One of the most significant features of the data is the structural shock that occurs

around 2022 during the Russian invasion of Ukraine. This event caused a noticeable shift in the level of the series and created additional forecasting challenges. Aside from this disruption, the majority of the dataset follows a relatively stable pattern, with only the most recent years showing substantial deviations from the historical trend.

Over the course of the semester, we explored several forecasting approaches and compared their performance, strengths, and weaknesses. The methods used included:

- STL Benchmark Model
- ETS(A,A,A)
- ETS(M,A,M)
- Multiple ARIMA/ARIMAX models

One of the biggest lessons from this project was learning that no forecasting method is universally superior. Different models perform better depending on the structure of the data, the forecasting horizon, and the presence of outside shocks.

STL Benchmark Model

The STL model was the simplest and most interpretable method we used. By decomposing the series into trend, seasonal, and remainder components, it provided a useful baseline understanding of the data. Another advantage of the STL model is that it requires relatively few assumptions and parameters compared to more advanced forecasting methods.

However, despite its simplicity and interpretability, the STL model produced the weakest forecasting performance overall. While it captured the general trend and seasonal movement of the data, it struggled to adapt to sudden structural changes such as the Russia/Ukraine War shock. This demonstrated one of the major limitations of simpler decomposition methods: they may perform adequately under stable conditions but often fail when large external events affect the time series.

ETS Models

The ETS(A,A,A) model served as a strong baseline forecasting model. It was able to adapt to changes in trend over time while remaining relatively stable and easy to interpret. The forecasts generated by this model were fairly reliable, although the model displayed slightly more bias and lower accuracy than some of the more advanced approaches.

The ETS(M,A,M) model ultimately performed better because the multiplicative seasonal structure captured the increasing seasonal variation in the data more effectively. Compared to the additive ETS model, the multiplicative version adjusted more naturally as the magnitude of the series increased over time.

One major advantage of ETS methods overall is their stability. These models are generally less sensitive to small changes in the data and are easier to maintain over long forecasting periods. However, a major weakness is that ETS models struggle to explicitly account for outside events or exogenous shocks such as wars, policy changes, or economic disruptions. Because of this limitation, the ETS models were less effective at handling the sudden shift associated with the Russia/Ukraine War.

ARIMA and ARIMAX Models

The ARIMA and ARIMAX models were the most complex forecasting methods used in this project, but they also produced the strongest forecasting performance overall. The ARIMA-M4 model, which included seasonal dummy variables and non-seasonal differencing, a level control and trend control achieved the highest accuracy and lowest bias across most forecast horizons.

One of the major strengths of the ARIMAX framework is its ability to include external variables that may influence the series. This allowed us to better account for the effects of the Russia/Ukraine War shock compared to the STL and ETS approaches. By controlling for external influences, the ARIMAX models were able to separate temporary disruptions from the long-term underlying trend.

Despite their superior accuracy, the ARIMA and ARIMAX models introduced several additional challenges.

Model Instability and Forecasting Challenges

One of the most difficult issues we encountered was model instability over time. During the diagnostic process, the recommended differencing structure changed between the ARIMA-M3 and ARIMA-M4 models. Conceptually, this instability likely occurred because the underlying data itself became less stable after the geopolitical shock in 2022.

Structural breaks can significantly change autocorrelation patterns within a time series, causing yje previously optimal model to become less effective. This issue became especially noticeable during Time Series Cross Validation (TSCV), where models are repeatedly retrained on rolling subsets of the data. Small changes in the training sample often produced different differencing recommendations and parameter estimates.

To address this issue, we attempted to manually specify the differencing parameters in the ARIMA models. However, the models continued to return very similar outputs, suggesting that the instability was not simply caused by the optimization process, but rather by the changing structure of the data itself.

This part of the project highlighted one of the major real-world difficulties of forecasting: models are rarely static. As economic conditions, external shocks, and patterns evolve, forecasting models often require continual reevaluation and adjustment.

Data Preparation Challenges

Another major challenge involved the preprocessing and preparation of the dataset. The series contained two missing observations:

One missing point caused by the partial government shutdown at the end of 2025 and one unexplained missing observation from the 1980s. To address these gaps, we used interpolation with the approx method. This approach worked reasonably well for maintaining continuity in the dataset, but it also demonstrated how sensitive forecasting models can be to missing data and data quality issues.

This experience reinforced the importance of data cleaning and preprocessing in forecasting projects. Even highly sophisticated models can perform poorly if the underlying data contains inconsistencies, outliers, or missing observations.

Conclusion

As previously discussed the Russia-Ukraine war was a clear catalyst for higher energy prices in 2022 and therefore spilling over to electricity prices. However this only tells part of the story.

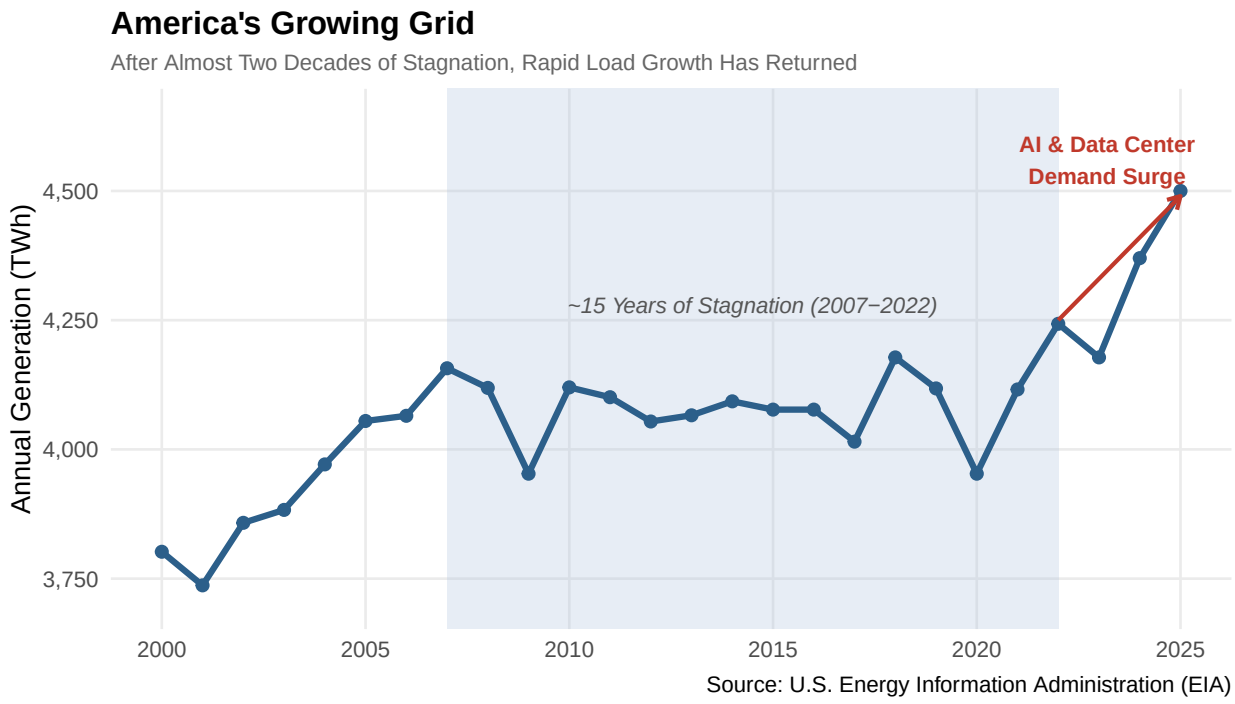


Figure 1: *The U.S. electricity grid has not grown in the past 15 years.*

Another part to the story is AI energy. AI energy is so constrained and so bottlenecked we have never seen something quite like this before. An estimate from Boston Consulting Group forecasts global AI data center power demand to grow 55% over the next 3 years, which will be undeniably led by the United States because we are the AI leader. As shown in the figure above, the United States has had flat electricity consumption for the past 15 years. The U.S. is expected to get into a 100% CAGR on energy for inference and remain at a 30% CAGR for training. We are seeing this demand from a country that has not grown its electricity grid in the past 15 years.

The U.S. grid is extremely stretched. We are hitting peak loads and electricity pricing is surging as a result from data center demand. However, even without peak loads and price surges it would still take a long time because big tech data centers are not located in dense urban areas so therefore the transmission process takes time. Data centers also need to on-board hundreds of megawatts (MWs) all at once. And our grid was not designed to handle this type of load, it was designed for more of a gradual increase from residential and commercial use. So this is another strong catalyst that is contributing to the surge in electricity prices.

The final part to this story is what is happening right now in the Strait of Hormuz. Roughly 1/4 of all internationally traded oil has been disrupted as a result of this war. And even if this war ends this second, it takes time for oil rigs to come back on. The U.S. doesn't generate electricity from oil however this supply shock affects electricity prices through the broad increase in energy prices. Natural gas prices surge since energy markets are correlated and since the U.S. uses natural gas to power electricity in many regions, electricity prices increase as a result.